

Sharp minimax rates for average treatment effects with discrete confounding under fixed overlap

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Abstract

This paper characterizes the fixed-interior minimax mean-squared-error risk for average treatment effect estimation in a finite-alphabet observational experiment with unrestricted category probabilities in the calibrated range stated below. For each fixed overlap parameter $0 < \epsilon < 1/2$, and for alphabet sizes d in the calibrated range $d \leq \rho_\epsilon n \log n$, the sharp risk is

$$R_{n,d,\epsilon} \asymp_\epsilon \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

The upper bound is attained by a computable split-sample hybrid estimator with universal numerical tuning. Pilot-heavy categories are estimated by empirical within-category treatment-control ratios, while pilot-light categories are estimated by a Chebyshev reciprocal polynomial and unbiased multinomial factorial moments. The matching lower bound follows by transferring the large-alphabet lower construction of [Zeng et al. \[2024\]](#) to a control-zero subclass on which the average treatment effect equals a treated-response functional. The characterization implies a parametric-risk region $d = O(\sqrt{n} \log n)$ and, along the stated calibrated range, consistency exactly when $d = o(n \log n)$. The same construction gives a deterministic upper envelope near the randomized endpoint, while a centered linear estimator has risk at most $n^{-1} + 4(1/2 - \epsilon)^2$ for every n, d ; at $\epsilon = 1/2$, the minimax risk is bracketed between $1/(100n)$ and $1/n$, uniformly over positive alphabet sizes.

1 Introduction

Average treatment effect estimation under unconfoundedness is usually presented as a problem of adjustment: estimate treatment propensities and outcome regressions well enough, then combine them through inverse-probability, regression, doubly robust, or targeted estimators. In regular low-dimensional models, the semiparametric theory gives the familiar $1/n$ mean-squared-error scale and identifies efficient first-order estimators [[Hahn, 1998](#), [Hirano et al., 2003](#), [Robins et al., 1994](#), [Bang and Robins, 2005](#), [van der Laan and Rubin, 2006](#)]. When the adjustment variable has many levels, however, the leading difficulty can come from the empirical scarcity of treatment and control observations inside many covariate categories. This paper studies that difficulty in a discrete benchmark where the covariate X , the adjustment variable, takes d unrestricted categories, treatment and outcome are binary, and overlap is fixed away from zero and one. Large alphabets arise routinely in practice: adjustment by geographic units, provider or product

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identifiers, or interacted demographic cells produces many sparsely populated categories, and K binary covariates encode as a single categorical variable with $d = 2^K$ levels, so even a moderate number of binary adjustments places d far beyond $\sqrt{n}$.

The target is the average treatment effect functional identified by the finite observed law under the usual consistency and conditional-exchangeability conditions. In category k , let p_k be the category mass, π_k the treatment probability, and μ_{ak} the conditional mean outcome in arm a . Under fixed overlap, $\epsilon \leq \pi_k \leq 1 - \epsilon$ for every positive-mass category. The resulting estimand is

$$\tau(P) = \sum_{k=1}^d p_k (\mu_{1k} - \mu_{0k}),$$

equivalently the four-cell functional in [Definition 7](#). The minimax risk $R_{n,d,\epsilon}$, the worst-case mean-squared error over the overlap-restricted iid experiment, is defined in [Definition 8](#).

The contributions are the following.

1. For each fixed interior overlap level $0 < \epsilon < 1/2$, the sharp minimax mean-squared-error rate $n^{-1} + d^2/\{n^2(\log n)^2\}$ for the average treatment effect over the unrestricted finite-alphabet class, throughout the calibrated range $d \leq \rho_\epsilon n \log n$ ([Theorem 1](#)). The result identifies the sharp large-alphabet term $d^2/\{n^2(\log n)^2\}$ in this fixed-interior calibrated range.
2. A computable split-sample ratio-polynomial hybrid estimator with universal numerical tuning that attains the upper bound: empirical ratios on pilot-certified heavy categories, and a Chebyshev reciprocal polynomial lifted by unbiased factorial moments on light categories ([Definition 35](#)).
3. The resulting regime characterization inside the calibrated range: the parametric scale $1/n$ throughout $d = O(\sqrt{n} \log n)$, and consistency exactly when $d = o(n \log n)$.
4. An endpoint analysis connecting the construction to randomized assignment: a deterministic risk envelope near $\epsilon = 1/2$ and a bracketing of the minimax risk at the parametric order at the endpoint itself, uniformly in d ([Theorem 2](#)).

The main result, [Theorem 1](#), establishes the sharp fixed-interior minimax rate. For every fixed $0 < \epsilon < 1/2$, there are constants $a_\epsilon, \rho_\epsilon, C_\epsilon > 0$ and $N_\epsilon < \infty$ such that, whenever $n \geq N_\epsilon$, $d > 0$, and $d \leq \rho_\epsilon n \log n$,

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq R_{n,d,\epsilon} \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

The same theorem identifies two immediate consequences inside this calibrated range. The parametric scale $1/n$ holds throughout $d = O(\sqrt{n} \log n)$. Along admissible sequences, consistency is equivalent to $d/(n \log n) \rightarrow 0$.

The upper bound is constructive. The estimator in [Definition 35](#) splits the sample deterministically. The pilot split classifies categories as heavy or light. Heavy categories use direct empirical treatment-control ratios, where the pilot mass floor gives denominator control. Light categories use a Chebyshev-derived polynomial approximation to the reciprocal ratio structure, and the polynomial is lifted to the sample by factorial moments of the split-one multinomial counts. This produces a finite arithmetic estimator with fixed numerical tuning $A_0 = 6$, $\lambda_0 = 256$, and

$b_0 = 4096$; the tuning is shared across fixed interior overlap classes, while the constants in the risk bound depend on ϵ .

The logarithmic improvement arises from treating sparse categories as a nonsmooth functional-estimation problem. The cell contribution is a homogeneous ratio functional: it multiplies category mass by the difference of two conditional outcome means. In low-mass cells, direct ratios have unstable denominators. Polynomial approximation replaces the reciprocal behavior by a degree of order $\log n$, and factorial moments estimate the resulting polynomial without plug-in bias. This is the same approximation-versus-variance mechanism that appears in large-alphabet functional estimation [Paninski, 2003, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2020], here applied to the causal four-cell contribution and aggregated over unrestricted confounding categories.

The closest comparison is Zeng et al. [2024]. Their discrete strong-overlap analysis provides the large-alphabet lower-bound input used below, with scale $d^2/(n^2 \log^2 n) + 1/n$ in the corresponding range. The present paper attains that lower scale in the fixed-interior regime by constructing the ratio-polynomial hybrid estimator. The lower side of Theorem 1 uses their published control-zero lower-bound conclusion and transfers it to the ATE risk because, on that subclass, the ATE functional equals the treated-response functional.

The endpoint analysis in Theorem 2 records how the construction connects to randomized assignment. A centered linear estimator satisfies the uniform bound

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P[(\hat{\tau}_{\text{ctr}} - \tau(P))^2] \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

for every $0 < \epsilon \leq 1/2$ and every positive n, d . At $\epsilon = 1/2$, where treatment is randomized within every positive-mass category, the minimax risk lies between $1/(100n)$ and $1/n$. For fixed interior ϵ , the selected estimator in Theorem 2 compares the hybrid envelope with the centered endpoint envelope and attains

$$\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\}$$

up to the displayed fixed- ϵ constant in the same calibrated range.

The paper also includes a two-category witness, Proposition 1, to separate the adjusted causal target from the marginal treated-minus-control contrast. The construction satisfies consistency, conditional exchangeability, and fixed overlap; its ATE equals $1/2$, while its naive observed contrast equals ϵ . This example fixes the causal interpretation before the minimax analysis turns to the many-category problem.

The formal statements and proofs are supported by a Lean 4 development, with the precise verification scope recorded in the verification appendix. The main text presents the assumptions, estimators, and risk statements in the usual mathematical form.

Section 2 reviews the related literature. Section 3 introduces the finite-alphabet observational experiment, the ATE functional, the minimax risk, the two-category witness, and the hybrid estimator. Section 4 states the sharp fixed-interior minimax theorem and the universal hybrid envelope. Section 5 discusses the rate, the estimator decomposition, endpoint behavior, and extensions. The appendices give the auxiliary lemmas and deferred proofs.

2 Related work

The paper builds on the potential-outcomes formulation of causal effects and the semiparametric theory of average treatment effect estimation. The potential-outcomes framework introduced by Rubin [1974] and the propensity-score analysis of Rosenbaum and Rubin [1983] give the observational model its target estimand and overlap structure. Semiparametric efficiency calculations and doubly robust constructions then identify the first-order behavior of average treatment effect estimators under regular low-dimensional conditions [Hahn, 1998, Hirano et al., 2003, Robins et al., 1994, Bang and Robins, 2005, van der Laan and Rubin, 2006]. Higher-order influence-function methods extend this line of work to settings where nuisance estimation and functional curvature affect attainable risk [Robins et al., 2008, 2009, 2017, Liu et al., 2017, Kennedy et al., 2024].

A second strand studies causal estimation when covariates are rich enough that classical fixed-dimensional approximations no longer describe the leading difficulty. Results on high-dimensional and weak-overlap causal inference emphasize how overlap, nuisance complexity, and target curvature interact in observational problems [D’Amour et al., 2021, Balakrishnan et al., 2023, Jin and Syrgkanis, 2025]. The present analysis isolates one discrete benchmark within that broader program: finite-alphabet confounding with unrestricted category probabilities and fixed interior overlap. In that benchmark, the alphabet size d and sample size n enter the minimax rate explicitly, so the problem connects causal identification with large-alphabet functional estimation.

The minimax perspective follows the classical decision-theoretic and nonparametric tradition in which rates are derived from least favorable local alternatives and matching constructive estimators [Bickel et al., 1993, van der Vaart, 1998, Tsybakov, 2009]. Adaptation and nonsmooth functional estimation provide the relevant methodological template: logarithmic factors can be intrinsic when the target functional has singular or boundary-sensitive structure, and polynomial approximation can convert that structure into an estimable object [Lepski et al., 1999, Cai and Low, 2011]. For the average treatment effect under discrete confounding, the ratio form of the cell contribution creates the nonsmooth feature that drives the large-alphabet term.

The technical construction is also close to the literature on discrete-distribution functionals. Classical species and support-size problems introduced polynomial and moment ideas for unseen mass and related functionals [Fisher et al., 1943, Good and Toulmin, 1956, Efron and Thisted, 1976]. Modern large-alphabet theory sharpened this approach through approximation-theoretic estimators, factorial moments, and matching lower bounds [Paninski, 2003, Valiant and Valiant, 2011, 2017, Orlitsky et al., 2016, Wu and Yang, 2019, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2015, 2020]. The hybrid estimator in this paper uses that same approximation principle in a causal cell problem: heavy categories are handled by ratio estimation, while light categories are handled by a factorial-moment polynomial approximation.

The closest comparison is Zeng et al. [2024]. Their strong-overlap discrete-covariate analysis gives an $O(d^2/n^2 + 1/n)$ upper bound for the standard plug-in, inverse-probability-weighted, and doubly robust estimator class, together with a minimax lower bound of order $d^2/(n^2 \log^2 n) + 1/n$. The result established here attains that lower scale in the fixed-interior overlap regime by constructing a computable balanced ratio-polynomial hybrid estimator, thereby identifying the sharp fixed-interior minimax mean-squared-error rate $n^{-1} + d^2/(n^2 \log^2 n)$ over the stated finite-alphabet experiment class.

3 Setup and assumptions

Throughout, constants denoted by C , c , and their indexed variants are positive and may change from line to line; dependence on the fixed overlap level ϵ is displayed when it matters. Asymptotic notation is along integer sequences with $n \rightarrow \infty$ and alphabet size $d = d_n$, and all category sums range over $k \in \{1, \dots, d\}$. Probability laws for one observed unit are denoted by P , expectations under P by E_P , and logarithms are natural.

3.1 Data, identification, and overlap

The observed-data model is a finite-alphabet observational experiment. The binary treatment and outcome coordinates are recorded first, followed by the iid sampling condition and the overlap restriction that defines the fixed-interior class.

Definition 1. [synth_7] (Binary observed treatment A) *We write $A \in \{0, 1\}$ for the binary observed treatment, with $A = 1$ denoting treated status and $A = 0$ denoting untreated status.*

Definition 2. [synth_8] (Binary observed outcome Y) *We say Y is the observed outcome when $Y \in \{0, 1\}$ is the binary outcome coordinate of a single observed unit drawn from the observed-data law P .*

The finite-alphabet iid observational world consists of observed units $O_i = (X_i, A_i, Y_i)$, where the covariate X takes finitely many values, A is the treatment coordinate, and Y is the observed outcome.

Assumption 1. [ass:iid-sampling] (IID observed-data model) *The observed data are $O_i = (X_i, A_i, Y_i)$, $i = 1, \dots, n$, with $O_1, \dots, O_n \stackrel{\text{iid}}{\sim} P$, where the covariate X takes values in $\{1, \dots, d\}$, the treatment A and outcome Y are binary. For each category $k \in \{1, \dots, d\}$ write $p_k = P(X = k)$ for the category mass, $\pi_k = P(A = 1 \mid X = k)$ for the treatment probability, and, for each treatment arm $a \in \{0, 1\}$, $\mu_{ak} = E_P[Y \mid A = a, X = k]$ for the conditional mean outcome.*

The iid observed-data model is the standard independent and identically distributed sampling condition for discrete-covariate ATE analysis [Zeng et al., 2024]. It fixes the single-observation law, the category masses p_k , the treatment probabilities π_k , and the outcome regressions μ_{ak} that enter the estimand.

Assumption 2. [ass:overlap] (Fixed overlap) *For every category k with $p_k > 0$, the treatment probability π_k satisfies*

$$\epsilon \leq \pi_k \leq 1 - \epsilon.$$

The fixed-overlap condition is the standard strong overlap requirement in this finite-alphabet setting [Zeng et al., 2024]. It keeps every positive-mass category inside a common treatment-probability interval, which makes within-category treated and control outcome means jointly identifiable from the observed law.

For the causal interpretation and the compact witness below, the observed-data model is paired with binary potential outcomes. The potential-outcome overlay follows the usual consistency and conditional exchangeability formulation.

Assumption 3. [ass:consistency] (Potential-outcome consistency) *The observed outcome agrees almost surely with the potential outcome realized under the observed treatment:*

$$Y = Y(A) \quad \text{almost surely.}$$

Potential-outcome consistency is the standard condition equating the realized observed outcome with the potential outcome selected by the observed treatment [Zeng et al., 2024]. In words, the potential-outcome overlay supplies binary potential outcomes $Y(0), Y(1)$ on the same probability space as (X, A, Y) .

Definition 3. [synth_9] (Control potential outcome $Y(0)$) *We say $Y(0)$ is the control potential outcome when $Y(0)$ is a $\{0, 1\}$ -valued random variable on the same probability space as (X, A, Y) whose value is the outcome under treatment level 0.*

Assumption 4. [ass:conditional-exchangeability] (Conditional exchangeability) *The binary potential outcomes are conditionally independent of treatment given covariates:*

$$(Y(0), Y(1)) \perp A \mid X.$$

Conditional exchangeability is the standard no-unmeasured-confounding condition after conditioning on the finite covariate X [Zeng et al., 2024]. Together, Assumptions 3 and 4 give the observed-law expression of the average treatment effect used below.

The statistical experiment is expressed directly at the observed-data level. The next definitions introduce the overlap-restricted product-law class, the four-cell cone, the cell contribution, the ATE functional, and the corresponding minimax risk.

3.2 The experiment class and the average treatment effect

Definition 4. [def:experiment-class] (Experiment class $\mathcal{E}_{n,d,\epsilon}$) *For $n, d \in \mathbb{N}$ and $0 < \epsilon \leq 1/2$, the overlap-restricted iid product-law experiment class associated with Assumptions 1 and 2 is*

$$\mathcal{E}_{n,d,\epsilon} := \{P^{\otimes n} : P \in \text{Prob}(\{1, \dots, d\} \times \{0, 1\}^2) \text{ and } P \text{ satisfies the overlap condition}\}.$$

The experiment class $\mathcal{E}_{n,d,\epsilon}$ collects iid product laws $P^{\otimes n}$ whose one-observation laws satisfy the fixed-overlap restriction in Assumption 2.

Definition 5. [def:overlap-cone] (Overlap cone $\mathcal{C}_\epsilon$) *The overlap-restricted cone of nonnegative four-cell vectors is*

$$\mathcal{C}_\epsilon := \{u = (u_{00}, u_{01}, u_{10}, u_{11}) \in \mathbb{R}_+^4 : \epsilon(u_{00} + u_{01} + u_{10} + u_{11}) \leq u_{10} + u_{11} \leq (1 - \epsilon)(u_{00} + u_{01} + u_{10} + u_{11})\}.$$

The cone $\mathcal{C}_\epsilon$ is the category-level version of overlap: its coordinates are the four treatment-outcome cell masses in a single covariate category.

Definition 6. [def:cell-functional] (Cell functional $\phi(u)$) *For $u = (u_{00}, u_{01}, u_{10}, u_{11}) \in \mathcal{C}_\epsilon$, with $\mathcal{C}_\epsilon$ as in Definition 5, define*

$$\phi(u) = (u_{00} + u_{01} + u_{10} + u_{11}) \left\{ \frac{u_{11}}{u_{10} + u_{11}} - \frac{u_{01}}{u_{00} + u_{01}} \right\}$$

when $u \neq (0, 0, 0, 0)$, and set

$$\phi(0, 0, 0, 0) = 0.$$

The homogeneous cell contribution ϕ multiplies the category mass by the treated-minus-control difference in conditional outcome means. Its ratio form is the nonsmooth feature handled by the estimator in Definition 35.

Definition 7. [def:ate-functional] (Average treatment effect $\tau(P)$) For a probability law P on $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$, with arbitrary masses on this finite observation alphabet, define, for $a, y \in \{0, 1\}$ and $k \in \{1, \dots, d\}$, the joint cell mass

$$q_{aky} = P(A = a, X = k, Y = y),$$

and set

$$q_k = (q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1}).$$

The average treatment effect functional is

$$\tau(P) = \sum_{k=1}^d \phi(q_k).$$

The joint atom q_{aky} and four-cell vector q_k put the ATE functional $\tau(P)$ in a form that depends only on the observed law. Under [Assumptions 3](#) and [4](#), this observed-law functional has the usual potential-outcome interpretation [[Rubin, 1974](#), [Rosenbaum and Rubin, 1983](#)].

3.3 Minimax risk and a two-category witness

Definition 8. [def:minimax-risk] (Minimax risk $R_{n,d,\epsilon}$) Define the minimax mean-squared-error risk by

$$R_{n,d,\epsilon} = \inf_{\hat{\tau}} \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P[(\hat{\tau} - \tau(P))^2],$$

where the infimum ranges over all measurable estimators $\hat{\tau}$ based on $O_1, \dots, O_n$.

The risk $R_{n,d,\epsilon}$ measures the best worst-case mean-squared error over the fixed-overlap experiment class. The main theorem in the next section characterizes this quantity over its stated sample-size and dimension range.

A two-category construction clarifies the causal target by displaying a law in which the ATE functional and the marginal observed contrast differ exactly. The construction fixes four outcome regressions and then couples them to binary potential outcomes.

Definition 9. [synth_15] (Control outcome regression in category 1) We write $\mu_{01} := E_P[Y \mid A = 0, X = 1]$ whenever $P(A = 0, X = 1) > 0$.

Definition 10. [synth_16] (Control outcome regression μ_{02}) For an observed-data law P with $P(A = 0, X = 2) > 0$, define $\mu_{02} := E_P[Y \mid A = 0, X = 2]$.

Definition 11. [synth_17] (Treated outcome regression in category 1) We write μ_{11} for the conditional mean $E_P[Y \mid A = 1, X = 1]$, whenever $P(A = 1, X = 1) > 0$.

Definition 12. [synth_18] (Treated outcome regression in category 2) In the two-category witness, μ_{12} is the treated outcome regression in the second covariate category, $\mu_{12} = E_P[Y \mid A = 1, X = 2]$, and this construction sets $\mu_{12} = 0$.

Definition 13. [def:two-category-witness] (Two-category witness law) For $d = 2$ and $0 \leq \epsilon \leq 1$, set

$$p_1 = p_2 = \frac{1}{2}, \quad \pi_1 = \epsilon, \quad \pi_2 = 1 - \epsilon, \quad \mu_{01} = \mu_{02} = 0, \quad \mu_{11} = 1, \quad \mu_{12} = 0.$$

Define the corresponding full-data extension by drawing

$$(Y(0), Y(1)) = (0, \mathbf{1}\{X = 1\})$$

setting $Y = Y(A)$, and then drawing $A \mid X = k \sim \text{Bernoulli}(\pi_k)$.

Proposition 1. [prop:two-category-confounding] (Two-category confounding witness) *For every real ϵ with $0 < \epsilon \leq 1/2$, let Q be the full-data law constructed in Definition 13, and let P be its observed marginal law for $O = (X, A, Y)$. Then:*

- *(Consistency.)* Q satisfies Assumption 3.
- *(Conditional exchangeability.)* Q satisfies Assumption 4.
- *(Overlap.)* P satisfies ϵ -overlap in the sense of Assumption 2.

The average treatment effect functional in Definition 7 and the naive observed contrast under P obey

$$\tau(P) = \frac{1}{2}, \quad E_P[Y \mid A = 1] - E_P[Y \mid A = 0] = \epsilon.$$

Proposition 1 records the exact separation: the full-data law satisfies the standard causal assumptions, the observed marginal satisfies overlap, and the ATE equals $1/2$ while the naive treated-minus-control contrast equals ϵ . This example anchors the estimand before the minimax analysis turns to unrestricted alphabet size.

The estimator used for the upper bound combines deterministic sample splitting, direct ratio estimation in pilot-heavy categories, and factorial-moment polynomial estimation in pilot-light categories. The following definitions fix the auxiliary notation and then state the estimator in full.

Definition 14. [synth_10] (Centered endpoint estimator $\hat{\tau}_{\text{ctr}}$) *For $n \in \mathbb{N}$ with $n > 0$ and observed units $O_i = (X_i, A_i, Y_i) \in \{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$, define the centered endpoint estimator by*

$$\hat{\tau}_{\text{ctr}}(O_1, \dots, O_n) = \frac{1}{n} \sum_{i=1}^n 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right).$$

Definition 15. [synth_13] (Sample law μ_n) *For a sample $O_1, \dots, O_n$ with single-observation law P on $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$, we write $\mu_n = P^{\otimes n}$ for the n -fold product law of the observed sample.*

3.4 Sample splits and the hybrid estimator

Definition 16. [def:sample-splits] (Sample splits I_0, I_1 and counts $N_{aky}^{(j)}$) *For a sample $O_1, \dots, O_n$ with $O_i = (X_i, A_i, Y_i) \in \{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$, define the deterministic half-sample splits, their sizes, and the logarithmic scale*

$$I_0 = \{1, \dots, \lfloor n/2 \rfloor\}, \quad I_1 = \{\lfloor n/2 \rfloor + 1, \dots, n\}, \quad m_j = |I_j|, \quad L = \log(en),$$

and, for each split $j \in \{0, 1\}$, treatment arm $a \in \{0, 1\}$, category $k \in \{1, \dots, d\}$, and outcome $y \in \{0, 1\}$, the split cell count

$$N_{aky}^{(j)} = \#\{i \in I_j : (A_i, X_i, Y_i) = (a, k, y)\}.$$

The deterministic split I_0, I_1 separates category selection from estimation, and the split cell counts $N_{aky}^{(j)}$ provide the multinomial factorial moments used by the light branch.

Definition 17. [synth_3] (Chebyshev polynomial T_M) For $M \in \mathbb{N}$, T_M denotes the degree- M Chebyshev polynomial of the first kind, characterized by $T_M(\cos \theta) = \cos(M\theta)$ for every $\theta \in \mathbb{R}$.

Definition 18. [synth_49] (Numerical cutoff N_0) We define N_0 to be the least $N \in \mathbb{N}$ such that, for every $n \geq N$, with $L = \log(en)$, $m_1 = |I_1|$, $M(n) = \max\{2, \lfloor \alpha_0 L \rfloor\}$, and $B(n) = b_0 L / m_1$, the inequalities $\alpha_0 L \geq 2$, $4M(n)^2 \leq m_1$, and $4M(n)/m_1 \leq 3B(n)/4$ all hold.

Definition 19. [synth_38] (Degree schedule $M(n)$) For $n \geq 2$, the logarithmic polynomial degree schedule is

$$M(n) = \max\{2, \lfloor \alpha_0 \log(en) \rfloor\},$$

where

$$\alpha_0 = \min \left\{ 1, \frac{1}{32 \log 6}, \frac{1}{256 \cdot 8 \log(27/4)} \right\}.$$

Definition 20. [synth_4] (Falling factorial split-cell count) For a split-one cell count $N_{aky}^{(1)}$ and multi-index coordinate $r_{ay} \in \mathbb{N}_0$, we write $(N_{aky}^{(1)})_{r_{ay}}$ for the falling factorial $N_{aky}^{(1)}(N_{aky}^{(1)} - 1) \cdots (N_{aky}^{(1)} - r_{ay} + 1)$ when $r_{ay} \geq 1$, with $(N_{aky}^{(1)})_0 = 1$.

Definition 21. [synth_5] (Four-cell monomial u^r) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$ and a multi-index $r = (r_{ay})_{a,y \in \{0,1\}} \in \mathbb{N}_0^4$, write $u^r = \prod_{a,y \in \{0,1\}} u_{ay}^{r_{ay}}$.

The multi-index $r = (r_{ay})_{a,y \in \{0,1\}}$ indexes monomials in the four observed cells. Falling factorials give unbiased polynomial-moment estimates for multinomial cell probabilities, with the scalar convention $(z)_r$ used analogously.

Definition 22. [synth_28] (Calibration constant b_0) The calibration constant b_0 is the fixed numerical value 4096.

Definition 23. [synth_29] (Hybrid degree constant α_0) The numerical hybrid degree constant is $\alpha_0 = \min \left\{ 1, \frac{1}{32 \log 6}, \frac{1}{2048 \log(27/4)} \right\}$.

Definition 24. [synth_30] (Total four-cell mass $s(u)$) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$, define $s(u) = u_{00} + u_{01} + u_{10} + u_{11}$.

Definition 25. [synth_31] (Control-cell mass $s_0(u)$) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$, define $s_0(u) = u_{00} + u_{01}$.

Definition 26. [synth_32] (Treated-arm mass $s_1(u)$) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$, ordered by $(A, Y) = (0, 0), (0, 1), (1, 0), (1, 1)$, define $s_1(u) := u_{10} + u_{11}$.

Definition 27. [synth_33] (Control outcome-one cell $t_0(u)$) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$, define $t_0(u) = u_{01}$.

Definition 28. [synth_34] (Treated outcome-one cell $t_1(u)$) For a four-cell vector $u = (u_{00}, u_{01}, u_{10}, u_{11})$, define $t_1(u) = u_{11}$.

Definition 29. [synth_40] (Coefficients c_r) For a nonnegative integer M and scale $B > 0$, c_r is the coefficient of u^r in the monomial expansion of the ratio-polynomial $P_{M,B}$:

$$P_{M,B}(u) = \sum_{|r| \leq M} c_r u^r.$$

Definition 30. [synth_41] (Split-size falling factorial $(m_1)_{|r|}$) For the estimation-split size m_1 and four-cell multi-index r , the denominator $(m_1)_{|r|}$ denotes the falling factorial $\prod_{\ell=0}^{|r|-1} (m_1 - \ell)$, with the empty product equal to 1.

Definition 31. [synth_48] (Light-cell approximation scale $B(n)$) For the calibrated hybrid estimator, the light-cell approximation scale is $B(n) = b_0 L / m_1$, with $L = \log(en)$, $m_1 = |I_1|$, and $b_0 = 4096$.

Definition 32. [synth_35] (Factorial-moment contribution $\hat{P}_k$) For $n \geq N_0$, set $M = M(n)$ and $B = B(n)$. If the ratio-polynomial has monomial expansion

$$P_{M,B}(u) = \sum_{|r| \leq M} c_r u^r,$$

then the factorial-moment polynomial contribution for category k is

$$\hat{P}_k = \sum_{|r| \leq M} c_r \frac{\prod_{a,y \in \{0,1\}} (N_{aky}^{(1)})^{r_{ay}}}{(m_1)_{|r|}}.$$

Definition 33. [synth_46] (Category count N_k) For a sample $O_i = (X_i, A_i, Y_i)$, $i \in I$, indexed by a finite set I , the category count for $k \in \{1, \dots, d\}$ is $N_k = \sum_{i \in I} \mathbf{1}\{X_i = k\}$.

Definition 34. [synth_47] (Arm-specific category count N_{ak}) For observations $O_i = (X_i, A_i, Y_i)$ indexed by a finite set I , category $k \in \{1, \dots, d\}$, and arm $a \in \{0, 1\}$, define $N_{ak} := \sum_{i \in I} \mathbf{1}\{X_i = k, A_i = a\}$.

The estimator layer uses the deterministic split, pilot categories, and four-cell factorial moments to balance ratio estimation and polynomial approximation. Heavy categories are selected from the pilot split; light categories use the polynomial $P_{M,B}$ through the factorial-moment contribution $\hat{P}_k$.

Definition 35. [def:hybrid-estimator-handle] (Hybrid estimator $\hat{\tau}_n^{\text{hyb}}$) Using the deterministic splits I_0, I_1 , split sizes m_j , logarithmic scale $L = \log(en)$, and split counts $N_{aky}^{(j)}$ from [Definition 16](#), set

$$A_0 = 6, \quad \lambda_0 = 256, \quad b_0 = 4096,$$

$$D_{A_0} = 8 \log(9A_0/8) = 8 \log(27/4),$$

and

$$\alpha_0 = \min \left\{ 1, \frac{1}{32 \log A_0}, \frac{1}{256 D_{A_0}} \right\}.$$

For $n \geq 2$, define

$$M(n) = \max\{2, \lfloor \alpha_0 L \rfloor\}, \quad B(n) = b_0 L / m_1.$$

Let N_0 be the least numerical cutoff such that, for every $n \geq N_0$,

$$\alpha_0 L \geq 2, \quad 4M(n)^2 \leq m_1, \quad 4M(n)/m_1 \leq 3B(n)/4.$$

For every n , define the pilot-certified heavy and light category sets

$$\hat{\mathcal{H}}_n = \left\{ k : \sum_{a,y} N_{aky}^{(0)} > \lambda_0 L \right\}, \quad \hat{\mathcal{L}}_n = \hat{\mathcal{H}}_n^c \quad (n \geq N_0), \quad \hat{\mathcal{L}}_n = \emptyset \quad (n < N_0),$$

so that for $n < N_0$ every category is treated by the ratio branch below. For $n \geq N_0$, abbreviate $M = M(n)$ and $B = B(n)$. Put

$$N_k^{(1)} = \sum_{a,y} N_{aky}^{(1)}, \quad N_{ak}^{(1)} = \sum_y N_{aky}^{(1)},$$

and define the heavy-category ratio contribution

$$\hat{h}_k = \frac{N_k^{(1)}}{m_1} \left\{ \frac{N_{1k1}^{(1)}}{N_{1k}^{(1)}} \mathbf{1}\{N_{1k}^{(1)} > 0\} - \frac{N_{0k1}^{(1)}}{N_{0k}^{(1)}} \mathbf{1}\{N_{0k}^{(1)} > 0\} \right\}.$$

Define

$$H_M(x) = \frac{1 - T_M(1 - 2x)}{2M^2}, \quad E_M(x) = \frac{H_M(x)}{x}, \quad G_M(x) = \frac{1 - E_M(x)}{x},$$

where E_M and G_M are interpreted by polynomial continuation at zero. For $u = (u_{00}, u_{01}, u_{10}, u_{11})$, set

$$s_a(u) = u_{a0} + u_{a1}, \quad s(u) = s_0(u) + s_1(u), \\ t_1(u) = u_{11}, \quad t_0(u) = u_{01},$$

and define the light-cell polynomial

$$P_{M,B}(u) = \frac{s(u)t_1(u)}{B} G_M\left(\frac{s_1(u)}{B}\right) - \frac{s(u)t_0(u)}{B} G_M\left(\frac{s_0(u)}{B}\right).$$

Expand

$$P_{M,B}(u) = \sum_{|r| \leq M} c_r u^r.$$

For each category k , define

$$\hat{P}_k = \sum_{|r| \leq M} c_r \frac{\prod_{a,y} (N_{aky}^{(1)})^{r_{ay}}}{(m_1)^{|r|}}.$$

The heavy and light sums are

$$\hat{T}_{\mathcal{H}} = \sum_{k \in \hat{\mathcal{H}}_n} \hat{h}_k, \quad \hat{T}_{\mathcal{L}} = \sum_{k \in \hat{\mathcal{L}}_n} \hat{P}_k.$$

The truncated balanced ratio-polynomial hybrid estimator is

$$\hat{\tau}_n^{\text{hyb}} = \max\{-1, \min(1, \hat{T}_{\mathcal{H}} + \hat{T}_{\mathcal{L}})\}.$$

The same calibrated estimator is used for every overlap level.

The numerical constants in [Definition 35](#) are chosen for proof convenience and are not optimized. In particular, $\alpha_0 = \min\{1, (32 \log A_0)^{-1}, (256 D_{A_0})^{-1}\} \approx 2.6 \times 10^{-4}$, so the cutoff N_0 determined by $\alpha_0 L \geq 2$ corresponds to an astronomically large sample size, and the calibration range $L \geq 240$ appearing in the auxiliary lemmas is of the same nature. The rate statements should therefore be read as sharp-order characterizations of the calibrated estimator's large-sample regime: the exponents and the logarithmic factor are intrinsic to the problem, while the explicit finite- n thresholds reflect the slack of the concrete numerical calibration rather than any barrier in the analysis. Below the cutoff, the estimator simply applies the ratio branch to every category, and the operation-count clause of [Theorem 2](#) is unaffected.

The pilot partition $\hat{\mathcal{H}}_n, \hat{\mathcal{L}}_n$ sends empirically large categories to the ratio branch and empirically small categories to the polynomial branch. The light sum $\hat{T}_{\mathcal{L}}$ aggregates factorial-polynomial contributions, and the truncated hybrid estimator $\hat{\tau}_n^{\text{hyb}}$ is the estimator analyzed in the main fixed-interior theorem.

The remaining definitions collect auxiliary notation used in the proof statements and appendices. They keep the sample-splitting, factorial-moment, and coefficient notation explicit at the point where the estimator is introduced.

Definition 36. [\[synth_1\]](#) (Pilot category count) *With the pilot-split cell counts $N_{aky}^{(0)}$ of [Definition 16](#), define the pilot category count $N_k^{(0)} = \sum_{a,y \in \{0,1\}} N_{aky}^{(0)}$.*

Definition 37. [\[synth_2\]](#) (Pilot split size m_0) *We write $m_0 := |I_0|$ for the cardinality of the deterministic pilot split I_0 .*

Definition 38. [\[synth_23\]](#) (Treatment-arm outcome count N_{ak1}) *For a finite index set I and observations $(O_i)_{i \in I}$ with $O_i = (X_i, A_i, Y_i)$, we write $N_{ak1} = \sum_{i \in I} \mathbf{1}\{A_i = a, X_i = k, Y_i = 1\}$ for the number of observations in treatment arm a , category k , and outcome level one.*

Definition 39. [\[synth_43\]](#) (Finite-index product law $P^{\otimes J}$) *For a finite index set J and a probability law P for one observed unit $O = (X, A, Y)$, we write $P^{\otimes J}$ for the product law on families $(O_j)_{j \in J}$ under which the coordinates are independent and each coordinate has law P .*

Definition 40. [\[synth_45\]](#) (Euler factor e) *We write $e = \exp(1)$ for the base of the natural logarithm.*

Definition 41. [\[synth_19\]](#) (Chebyshev reciprocal coefficients $g_{M,j}$) *For an integer $M \geq 2$, we say $g_{M,j}$, $0 \leq j \leq M-2$, are the Chebyshev reciprocal coefficients when they are the unique real coefficients satisfying*

$$\frac{1 - \frac{1 - T_M(1-2x)}{2M^2x}}{x} = \sum_{j=0}^{M-2} g_{M,j} x^j,$$

with the left-hand side interpreted by polynomial continuation at $x = 0$, where T_M is the degree- M Chebyshev polynomial of the first kind.

Definition 42. [\[synth_20\]](#) (Sparse factorial-polynomial mean $\bar{P}_k$) *Given a law P , a category $k \in \{1, \dots, d\}$, a degree M , and a scale B , define the sparse factorial-polynomial mean of category k by $\bar{P}_k = P_{M,B}(q_k)$, where $q_k = (q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1})$ is the four-cell mass vector for category k under P .*

Together, [Definitions 4, 7, 8](#) and [35](#) define the functional, risk, and estimator objects used in the rate statements. The next section states the fixed-interior minimax characterization for this experiment class.

4 Main results

The main theorem gives the fixed-interior minimax characterization for the risk in [Definition 8](#). The statement uses the same overlap-restricted experiment class, ATE functional, and hybrid estimator introduced in [Definitions 4, 7 and 35](#). For each fixed overlap parameter in the interior range $0 < \epsilon < 1/2$, it identifies the mean-squared-error scale

$$r_{n,d} = \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}$$

through matching lower and upper bounds in the displayed calibration range.

Theorem 1. [\[thm:sharp-minimax-fixed-interior\]](#) (Sharp fixed-interior minimax rate)¹ *For an overlap parameter ϵ with $0 < \epsilon < 1/2$, there exist constants $a_\epsilon, \rho_\epsilon, C_\epsilon > 0$ and an integer $N_\epsilon < \infty$ such that the following statements hold. Let*

$$r_{n,d} = \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

- **(Sharp sandwich.)** *For every n, d , every single-observation law P , and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ as in [Definition 4](#), if $d > 0$, $n \geq N_\epsilon$, and*

$$d \leq \rho_\epsilon n \log n,$$

then the minimax mean-squared-error risk $R_{n,d,\epsilon}$ from [Definition 8](#) and the hybrid estimator $\hat{\tau}_n^{\text{hyb}}$ from [Definition 35](#) satisfy

$$a_\epsilon r_{n,d} \leq R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[\left(\hat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] \leq C_\epsilon r_{n,d}.$$

- **(Parametric interior.)** *For every $K > 0$, there exists $C_K > 0$ such that, for every n, d , every single-observation law P , and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, if $d > 0$, $n \geq N_\epsilon$,*

$$d \leq K\sqrt{n} \log n, \quad d \leq \rho_\epsilon n \log n,$$

then

$$\frac{a_\epsilon}{n} \leq R_{n,d,\epsilon} \leq \frac{C_K}{n}.$$

- **(Consistency threshold.)** *For any sequences n_j, d_j with $n_j \rightarrow \infty$, with $d_j > 0$ eventually, and with*

$$d_j \leq \rho_\epsilon n_j \log n_j$$

eventually, one has

$$R_{n_j,d_j,\epsilon} \rightarrow 0 \iff \frac{d_j}{n_j \log n_j} \rightarrow 0.$$

¹**Formalization scope.** This result uses the published conclusion [\[Zeng et al., 2024\]](#) (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Theorem 1 has three immediate implications for the finite-alphabet observational problem. First, the large-alphabet term is $d^2/\{n^2(\log n)^2\}$, matching the logarithmic scale suggested by the lower-bound phenomenon in [Zeng et al. \[2024\]](#). Second, the ordinary $1/n$ parametric rate persists throughout the interior regime $d = O(\sqrt{n} \log n)$, subject to the theorem's fixed-overlap calibration range. Third, along admissible sequences in that range, consistency is characterized by $d = o(n \log n)$.

The intuition is that the risk separates into a sampling-variance component and a sparse-cell approximation component. Heavy categories contribute through stabilized empirical ratios, while light categories are handled by the factorial-moment polynomial branch in [Definition 35](#). The Chebyshev construction follows the approximation-theoretic approach used for nonsmooth large-alphabet functionals [[Jiao et al., 2015](#), [Wu and Yang, 2016](#), [Han et al., 2020](#), [Cai and Low, 2011](#)]; here it is applied to the cell contribution ϕ and summed across unrestricted covariate categories.

The second theorem records the operational and endpoint consequences of the same construction. It states that the hybrid estimator has a finite arithmetic implementation with fixed numerical tuning, gives its fixed-interior envelope, combines it with a centered estimator near the randomized endpoint, and brackets the endpoint risk.

Theorem 2. `[thm:overlap-adaptive-universal-hybrid]` (Universal hybrid envelope) *With $\hat{\tau}_n^{\text{hyb}}$ as in [Definition 35](#) and $R_{n,d,\epsilon}$ as in [Definition 8](#), the following conclusions hold.*

- **Computability.** *There is a constant $K \in \mathbb{N}$, $K > 0$, such that for every $n, d \in \mathbb{N}$ there is a finite real-arithmetic program which reads the split count vector, returns $\hat{\tau}_n^{\text{hyb}}$ exactly on every sample $O_1, \dots, O_n$, and uses at most $KdM(n)^4$ arithmetic and comparison operations.*
- **Fixed-interior calibration.** *For every overlap parameter ϵ with $0 < \epsilon < 1/2$, there are constants $C_\epsilon, \rho_\epsilon > 0$ and $N_\epsilon \in \mathbb{N}$ such that, for all $n, d \in \mathbb{N}$ with $n > 0$, $d > 0$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$,*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\hat{\tau}_n^{\text{hyb}} - \tau(P))^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

For the deterministic estimator $\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}}$ that equals $\hat{\tau}_n^{\text{hyb}}$ when

$$C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and equals $\hat{\tau}_{\text{ctr}}$ otherwise, the same n, d, ϵ satisfy

$$R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right],$$

and

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right] \leq \max\{C_\epsilon, 4\} \left[\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\} \right].$$

- **Centered randomization bound.** For every overlap parameter ϵ with $0 < \epsilon \leq 1/2$ and every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, the centered estimator

$$\hat{\tau}_{\text{ctr}} = \frac{1}{n} \sum_{i=1}^n 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right)$$

satisfies

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P[(\hat{\tau}_{\text{ctr}} - \tau(P))^2] \leq \frac{1}{n} + 4(1/2 - \epsilon)^2.$$

- **Endpoint bracket.** For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$,

$$\frac{1}{100n} \leq R_{n,d,1/2} \leq \frac{1}{n}.$$

[Theorem 2](#) makes the estimator in [Definition 35](#) an explicit finite-sample object: for each n and d , it is computed from split counts using a number of arithmetic and comparison operations of order $dM(n)^4$. The same fixed numerical tuning applies across fixed interior overlap classes; the constants in the risk envelope depend on ϵ , while the construction itself is the one already displayed in [Definition 35](#).

The centered randomization bound explains the endpoint behavior. As ϵ approaches $1/2$, treatment assignment is close to randomized within each category, and the centered linear estimator has worst-case risk bounded by $1/n + 4(1/2 - \epsilon)^2$. The selected estimator compares this bound with the fixed-interior hybrid envelope and obtains the combined rate

$$\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\}$$

up to the displayed constant. At the exact randomized endpoint $\epsilon = 1/2$, the minimax risk is bracketed at the parametric scale $1/n$, uniformly over positive d .

5 Discussion and extensions

[Theorem 1](#) identifies the fixed-interior minimax mean-squared-error rate for unrestricted finite-alphabet confounding under bounded overlap. The characterization separates the risk into the ordinary sampling term $1/n$ and the large-alphabet term $d^2/\{n^2(\log n)^2\}$. This scale gives a benchmark for observational problems in which the covariate distribution is unrestricted across many discrete categories, complementing classical semiparametric ATE theory for regular low-dimensional models [[Hahn, 1998](#), [Hirano et al., 2003](#), [Robins et al., 1994](#), [Bang and Robins, 2005](#), [van der Laan and Rubin, 2006](#)] and higher-order analyses of nonregular causal functionals [[Robins et al., 2008, 2009, 2017](#), [Liu et al., 2017](#), [Kennedy et al., 2024](#)].

The estimator in [Definition 35](#) explains how the rate is attained constructively. Heavy categories are estimated through empirical treatment-control ratios, where the relevant denominators are large enough for the ratio behavior to be controlled. Light categories are handled through a Chebyshev-derived polynomial approximation to the cell contribution in [Definition 6](#), with factorial moments converting the polynomial into an unbiased split-sample estimate of its population analogue. The resulting hybrid treats the singular ratio structure of ϕ as an approximation problem on sparse cells and as an ordinary ratio-estimation problem on sufficiently populated cells.

This decomposition also clarifies the logarithmic factor. In the finite-alphabet experiment class of [Definition 4](#), the alphabet size can create many categories whose population mass is too small for direct ratio estimation to behave parametrically. Polynomial approximation spreads the information in those sparse cells across a degree of order $\log n$, and the factorial-moment construction makes that approximation estimable from counts. The main theorem therefore turns the large-alphabet causal difficulty into the same approximation-versus-variance balance that appears in nonsmooth functional estimation, while preserving the adjusted ATE target in [Definition 7](#).

The endpoint statements in [Theorem 2](#) connect the fixed-interior result to the randomized boundary. When $\epsilon = 1/2$, treatment is randomized within every positive-mass category, and the minimax risk is bracketed at the parametric scale $1/n$. For interior values close to the endpoint, the centered estimator has the displayed risk bound $1/n + 4(1/2 - \epsilon)^2$. The selected estimator combines this endpoint behavior with the hybrid envelope, yielding the upper scale

$$\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\}$$

within the fixed-interior calibration range. Thus the same analysis gives both a sharp fixed-class rate and a deterministic upper envelope that reflects proximity to randomized assignment.

The results are also informative for high-dimensional causal inference. In many semiparametric and machine-learning approaches, structural conditions on nuisance functions, such as smoothness, sparsity, or predictive structure, shape attainable rates [[Balakrishnan et al., 2023](#), [Jin and Syrgkanis, 2025](#)]. Here the category probabilities and outcome regressions are unrestricted, so the rate isolates the cost of discrete confounding itself. The parametric interior regime $d = O(\sqrt{n} \log n)$ and the consistency criterion $d = o(n \log n)$, both under the stated calibration range of [Theorem 1](#), give concrete benchmarks against which additional structural restrictions can be compared.

Limitations and future work. A remaining open direction is the triangular-array lower-envelope problem in which the overlap level varies with sample size, $\epsilon = \epsilon_n$, especially near the randomized endpoint. The fixed-class theorem establishes the sharp rate for each fixed $0 < \epsilon < 1/2$, and the endpoint bracket establishes the parametric scale at $\epsilon = 1/2$. The combined upper envelope in [Theorem 2](#) supplies a deterministic attainable bound across the fixed-interior calibration range. A matching lower envelope for sequences ϵ_n would characterize the transition between the fixed-interior sparse-cell difficulty and the randomized endpoint behavior.

Appendices

A Proofs and auxiliary lemmas

This appendix collects the auxiliary statements used to assemble the fixed-interior lower bound, the heavy-cell and light-cell upper bounds, the endpoint bracket, and the combined estimator envelope. The statements are grouped by their mathematical role in the argument: pilot classification, lower-bound transfer, endpoint control, calibration, polynomial approximation, ratio residuals, factorial moments, and the two aggregate bounds for the hybrid estimator.

The first ingredient controls the pilot split. It shows that empirical heavy and light categories are aligned with population mass thresholds, with either an arbitrary polynomial tail after choosing the threshold large enough or the fixed numerical threshold used in [Definition 35](#).

Lemma 1. `[lem:pilot-sandwich]` (Pilot sandwich tail bound) *Let $L = \log(en)$, and let $m_0 = |I_0|$ be the pilot-split size from [Definition 16](#). For a threshold $t > 0$, define*

$$\widehat{\mathcal{H}} = \{k : N_k^{(0)} > tL\}, \quad \widehat{\mathcal{L}} = \widehat{\mathcal{H}}^c, \quad B_- = \frac{tL}{2m_0}, \quad B_+ = \frac{2tL}{m_0}.$$

The following bounds hold.

- **(Arbitrary polynomial tail.)** For every $K > 0$ and every $c > 0$, there are constants $t_0 > 0$, $C > 0$, and N such that, for every $t \geq t_0$, every $n \geq N$, every dimension d with $d \leq cnL$, every law P on $\{1, \dots, d\} \times \{0, 1\}^2$, and every sample law satisfying [Assumption 1](#),

$$\mathbb{P}_{\mu_n} \left(\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq B_-\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq B_+\} \right) \leq Cn^{-K}.$$

- **(Fixed threshold.)** For every $c > 0$, there are constants $C > 0$ and N such that, for every $n \geq N$, every dimension d with $d \leq cnL$, every law P on $\{1, \dots, d\} \times \{0, 1\}^2$, and every sample law satisfying [Assumption 1](#), the same sandwich event at threshold $t = 256$ obeys

$$\mathbb{P}_{\mu_n} \left(\widehat{\mathcal{H}} \not\subseteq \left\{k : p_k \geq \frac{256L}{2m_0}\right\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \left\{k : p_k \leq \frac{2 \cdot 256L}{m_0}\right\} \right) \leq Cn^{-4}.$$

Proof of [Lemma 1](#). Fix n, d, P, t , and let μ_n be a sample law satisfying [Assumption 1](#); thus $\mu_n = P^{\otimes n}$. Write $L = \log(en)$, $m_0 = |I_0|$, and $p_k = P(X = k)$. For a category k , set

$$Z_{ik} = \mathbf{1}\{X_i = k\}, \quad i \in I_0,$$

so that

$$N_k^{(0)} = \sum_{i \in I_0} Z_{ik}.$$

Under [Assumption 1](#), the one-observation mean of this indicator is p_k , and $N_k^{(0)}$ is the Bernoulli count obtained from the deterministic pilot split of size $m_0 = \lfloor n/2 \rfloor$.

The pilot heavy and light sets used in the bad event are

$$\widehat{\mathcal{H}} = \{k : \lfloor tL \rfloor < N_k^{(0)}\}, \quad \widehat{\mathcal{L}} = \widehat{\mathcal{H}}^c.$$

Thus

$$\mathcal{B}_t = \left\{ \widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq tL/(2m_0)\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq 2tL/m_0\} \right\}.$$

If the heavy-set condition fails, then for some k , $p_k < tL/(2m_0)$ and $\lfloor tL \rfloor < N_k^{(0)}$; since $N_k^{(0)}$ is integer-valued, the latter implies $tL < N_k^{(0)}$. If the light-set condition fails, then for some k , $2tL/m_0 < p_k$ and $k \notin \widehat{\mathcal{H}}$; equivalently $\lfloor tL \rfloor \geq N_k^{(0)}$, which gives $N_k^{(0)} \leq tL$. Hence

$$\mathcal{B}_t \subseteq \bigcup_{k=1}^d \left\{ p_k < \frac{tL}{2m_0}, \ tL < N_k^{(0)} \right\} \cup \bigcup_{k=1}^d \left\{ \frac{2tL}{m_0} < p_k, \ N_k^{(0)} \leq tL \right\}.$$

For any $\ell \in \mathbb{R}$, the Bernoulli-count upper tail used here gives

$$m_0 p_k < \ell/2 \implies \mu_n\{\ell < N_k^{(0)}\} \leq \exp\{-\ell(\log 2 - 1/2)\}.$$

Applying this with $\ell = tL$ bounds the upper-count branch whenever $p_k < tL/(2m_0)$.

Similarly, the lower-tail bound gives

$$2\ell < m_0 p_k \implies \mu_n\{N_k^{(0)} \leq \ell\} \leq \exp\{-m_0 p_k/8\} \leq \exp\{-\ell/4\}.$$

Applying this with $\ell = tL$ bounds the lower-count branch whenever $2tL/m_0 < p_k$.

Now fix a category k . For the first cellwise event in the displayed union, either $p_k < tL/(2m_0)$, in which case the event is contained in $\{tL < N_k^{(0)}\}$ and the upper-tail bound applies, or the mass inequality fails, in which case the event is empty. For the second cellwise event, either $2tL/m_0 < p_k$, in which case the event is contained in $\{N_k^{(0)} \leq tL\}$ and the lower-tail bound applies, or the mass inequality fails, in which case the event is empty. Taking the union bound over the d categories therefore yields, for $t \geq 0$, $n \geq 2$, and every μ_n satisfying [Assumption 1](#),

$$\mu_n(\mathcal{B}_t) \leq d [\exp\{-(tL)(\log 2 - 1/2)\} + \exp\{-(tL)/4\}].$$

Now fix $K > 0$ and $c > 0$. Since $\log 2 - 1/2 > 0$, set

$$t_0 = \max \left\{ \frac{K+4}{\log 2 - 1/2}, 4(K+4) \right\}, \quad C = 2c, \quad N = 2.$$

For $t \geq t_0$,

$$K+4 \leq t(\log 2 - 1/2), \quad K+4 \leq t/4.$$

Thus, whenever $n \geq N$, $d \leq cnL$, and μ_n satisfies [Assumption 1](#),

$$\begin{aligned} \mu_n(\mathcal{B}_t) &\leq d \left[e^{-(tL)(\log 2 - 1/2)} + e^{-(tL)/4} \right] \\ &\leq cnL \cdot 2e^{-(K+4)L} \\ &\leq 2cn^2 e^{-(K+4)L} \\ &= 2c e^{-(K+4)} n^{-K-2} \\ &\leq 2cn^{-K}. \end{aligned}$$

Here $L \leq n$ for $n \geq 1$, and

$$e^{-qL} = e^{-q} n^{-q}$$

has been used with $q = K+4$, together with $e^{-(K+4)} \leq 1$ and $n^{-K-2} \leq n^{-K}$ for $n \geq 1$. This proves the arbitrary-polynomial assertion.

For the fixed calibration, take $K = 4$ and $t = 256$. The numerical inequalities

$$8 \leq 256(\log 2 - 1/2), \quad 8 \leq 256/4$$

follow from $0.6931471803 < \log 2$ and elementary arithmetic. The same decay calculation gives, for every $c > 0$, the choices $C = 2c$ and $N = 2$, and hence

$$\mu_n(\mathcal{B}_{256}) \leq Cn^{-4}.$$

This is the stated fixed-calibration bound. □

[Lemma 1](#) is the selection device behind the hybrid decomposition. On its high-probability event, categories sent to the ratio branch have a population mass floor, while categories sent to the polynomial branch have a population mass ceiling at the scale L/m_0 . The fixed-threshold clause matches the numerical constant in [Definition 35](#).

The lower-bound component transfers a published large-alphabet functional lower bound to the ATE risk. On the subclass with zero control means, the ATE functional in [Definition 7](#) coincides with the treated response functional used by [Zeng et al. \[2024\]](#), so restricting the experiment class yields the following risk lower bound.

Lemma 2. `[lem:ate-lower-bound-transfer]` (ATE lower-bound transfer)² *Fix an overlap parameter ϵ with $0 < \epsilon < 1/2$, and let $R_{n,d,\epsilon}$ be the minimax mean-squared-error risk in [Definition 8](#). Then there exist constants $a_\epsilon, b_\epsilon > 0$ and $N_\epsilon \in \mathbb{N}$ such that, for every $n, d \in \mathbb{N}$, if $0 < d$, $N_\epsilon \leq n$, and*

$$d \leq b_\epsilon n \log n,$$

then

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq R_{n,d,\epsilon}.$$

Proof of [Lemma 2](#). With the cell-mass notation of [Definition 7](#), write

$$\mu_{ak}(P) = \frac{q_{ak1}}{q_{ak0} + q_{ak1}}$$

for every arm $a \in \{0, 1\}$ and category k , using the totalized real-ratio convention that the displayed value is 0 when $q_{ak0} + q_{ak1} = 0$. On positive arm cells this is the usual conditional mean $E_P[Y \mid A = a, X = k]$. Define the control-zero subclass by

$$\mathcal{D}_0^{(n,d)}(\epsilon) := \{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon} : \mu_{0k}(P) = 0 \text{ for every } k = 1, \dots, d\}.$$

For $P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)$, write the treated-arm functional as

$$\psi_1(P) = \sum_{k=1}^d p_k \mu_{1k}.$$

For this auxiliary subclass risk, the supremum over $\mathcal{D}_0^{(n,d)}(\epsilon)$ is the order-theoretic supremum over the displayed class, equal to 0 when that class is empty.

Since $0 < \epsilon < 1/2$, the fixed-sample control-zero lower bound of [\[Zeng et al., 2024, Theorem 2, Appendix C.5, Lemma 4, and Appendix D.2\]](#) gives constants $a_\epsilon, b_\epsilon > 0$ and $N_\epsilon \in \mathbb{N}$ such that, for every n, d with $0 < d$, $N_\epsilon \leq n$, and $d \leq b_\epsilon n \log n$,

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq \inf_{\hat{\psi}} \sup_{P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)} E_P \left[(\hat{\psi} - \psi_1(P))^2 \right],$$

where the infimum is over measurable estimators based on $O_1, \dots, O_n$.

²**Formalization scope.** This result uses the published conclusion [\[Zeng et al., 2024\]](#) (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

We next identify the ATE functional on this subclass from the displayed finite-cell definitions. For a fixed category k , set

$$s_{ak} = q_{ak0} + q_{ak1}, \quad p_k = s_{0k} + s_{1k}.$$

If $q_k = (0, 0, 0, 0)$, then $\phi(q_k) = 0$, $p_k = 0$, and hence

$$\phi(q_k) = p_k(\mu_{1k} - \mu_{0k}).$$

If $q_k \neq (0, 0, 0, 0)$, [Definitions 6](#) and [7](#) give

$$\phi(q_k) = p_k \left\{ \frac{q_{1k1}}{s_{1k}} - \frac{q_{0k1}}{s_{0k}} \right\} = p_k(\mu_{1k} - \mu_{0k}),$$

where the second equality is the totalized definition of μ_{ak} . Summing over categories yields

$$\tau(P) = \sum_{k=1}^d p_k(\mu_{1k} - \mu_{0k}).$$

For $P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)$, the defining control-zero condition gives $\mu_{0k} = 0$ for every k , so

$$\tau(P) = \sum_{k=1}^d p_k(\mu_{1k} - \mu_{0k}) = \sum_{k=1}^d p_k \mu_{1k} = \psi_1(P).$$

The class suprema used below are bounded above on inhabited classes. For any overlap law P , the same finite-cell identity gives

$$\tau(P) = \sum_{k=1}^d p_k(\mu_{1k} - \mu_{0k}).$$

The category masses satisfy $p_k \geq 0$ and $\sum_{k=1}^d p_k = 1$, and the totalized binary outcome means satisfy $0 \leq \mu_{ak} \leq 1$. Hence $|\mu_{1k} - \mu_{0k}| \leq 1$ for each k , so

$$|\tau(P)| \leq \sum_{k=1}^d p_k |\mu_{1k} - \mu_{0k}| \leq \sum_{k=1}^d p_k = 1.$$

For any estimator $\hat{\tau}$, pointwise domination on the finite sample alphabet gives

$$|\hat{\tau}(o_1, \dots, o_n)| \leq \sum_{o'_1, \dots, o'_n} |\hat{\tau}(o'_1, \dots, o'_n)|.$$

Combining this inequality with $|\tau(P)| \leq 1$ yields

$$|\hat{\tau}(o_1, \dots, o_n) - \tau(P)| \leq \sum_{o'_1, \dots, o'_n} |\hat{\tau}(o'_1, \dots, o'_n)| + 1.$$

After squaring and integrating under $P^{\otimes n}$,

$$E_P[(\hat{\tau} - \tau(P))^2] \leq \left\{ \sum_{o_1, \dots, o_n} |\hat{\tau}(o_1, \dots, o_n)| + 1 \right\}^2.$$

For $P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)$, the identity $\tau(P) = \psi_1(P)$ gives the same bound with $\psi_1(P)$ in place of $\tau(P)$. Each displayed mean squared error is also nonnegative because it is the integral of a square.

Fix a measurable estimator $\hat{\tau}$. If $\mathcal{D}_0^{(n,d)}(\epsilon)$ is empty, its subclass supremum is 0. The ambient supremum over $\mathcal{E}_{n,d,\epsilon}$ is 0 when the ambient class is empty, and otherwise it is at least one nonnegative mean squared error. Hence the estimatorwise comparison holds in the empty-subclass case.

If $\mathcal{D}_0^{(n,d)}(\epsilon)$ is inhabited, take $P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)$. By the definition of the subclass, the same product law is in $\mathcal{E}_{n,d,\epsilon}$, and the target identity gives

$$E_P[(\hat{\tau} - \psi_1(P))^2] = E_P[(\hat{\tau} - \tau(P))^2] \leq \sup_{Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_Q[(\hat{\tau} - \tau(Q))^2],$$

where the upper boundedness needed for the final supremum is supplied by the preceding envelope. Taking the supremum over $P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)$ gives, for every measurable $\hat{\tau}$,

$$\sup_{P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)} E_P[(\hat{\tau} - \psi_1(P))^2] \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P[(\hat{\tau} - \tau(P))^2].$$

The same empty-class convention and nonnegativity show that every subclass worst-case risk is bounded below by 0. Taking the infimum over the common measurable estimator class therefore yields

$$\inf_{\hat{\psi}} \sup_{P^{\otimes n} \in \mathcal{D}_0^{(n,d)}(\epsilon)} E_P[(\hat{\psi} - \psi_1(P))^2] \leq R_{n,d,\epsilon}.$$

Combining the cited lower bound with this restriction comparison yields

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq R_{n,d,\epsilon}$$

for every n, d in the stated range. $\square$

Lemma 2 supplies the lower side of the sharp sandwich in [Theorem 1](#). The constants depend on the fixed interior overlap level, and the dimension range has the same $n \log n$ scale as the upper-bound construction.

The endpoint and near-endpoint controls are based on a centered linear estimator. These statements give the parametric endpoint bracket and the deterministic comparison term used by the selected estimator.

Lemma 3. [[lem:near-randomization-linear-upper](#)] (Centered linear upper bound) *For every $n, d \in \mathbb{N}$ with $n > 0$, every overlap parameter ϵ , every single-observation law P , and every sample law μ_n such that $\mu_n \in \mathcal{E}_{n,d,\epsilon}$ for P in the sense of [Definition 4](#), define*

$$\hat{\tau}_{\text{ctr}} = \frac{1}{n} \sum_{i=1}^n 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right).$$

Then the mean squared error of $\hat{\tau}_{\text{ctr}}$ for the average treatment effect functional $\tau(P)$ in [Definition 7](#) satisfies

$$\int (\hat{\tau}_{\text{ctr}}(o_1, \dots, o_n) - \tau(P))^2 d\mu_n(o_1, \dots, o_n) \leq \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2.$$

Proof of Lemma 3. For one observation $z = (x, a, y)$, define

$$W(z) = 2(2a - 1) \left(y - \frac{1}{2} \right),$$

with $a, y \in \{0, 1\}$. Then

$$\hat{\tau}_{\text{ctr}} = \frac{1}{n} \sum_{i=1}^n W(O_i), \quad W(z)^2 = 1 \quad \text{for every } z.$$

Writing $q_{aky} = P(A = a, X = k, Y = y)$, direct summation gives

$$E_P[W(O)] = \sum_{k=1}^d \{q_{1k1} - q_{1k0} + q_{0k0} - q_{0k1}\}.$$

For the sums below, set

$$p_k = \sum_{a=0}^1 \sum_{y=0}^1 q_{aky}, \quad \pi_k = \frac{q_{1k0} + q_{1k1}}{p_k}, \quad \mu_{ak} = \frac{q_{ak1}}{q_{ak0} + q_{ak1}},$$

using ordinary real division, so a quotient with zero denominator is assigned the value zero. If $p_k > 0$, overlap and $\epsilon > 0$ make both arm masses positive, and the definitions give

$$q_{1k1} = \mu_{1k} \pi_k p_k, \quad q_{1k0} = (1 - \mu_{1k}) \pi_k p_k,$$

and

$$q_{0k1} = \mu_{0k} (1 - \pi_k) p_k, \quad q_{0k0} = (1 - \mu_{0k}) (1 - \pi_k) p_k.$$

Hence

$$q_{1k1} - q_{1k0} + q_{0k0} - q_{0k1} = p_k (\mu_{1k} - \mu_{0k}) + 2 \left(\pi_k - \frac{1}{2} \right) p_k (\mu_{1k} + \mu_{0k} - 1).$$

If $p_k = 0$, all four joint masses in category k are zero; with the totalized definitions above, both weighted terms also vanish because they are multiplied by p_k . Thus the same identity holds with zero contribution in that category.

By Definition 7, $\tau(P) = \sum_{k=1}^d \phi(q_k)$, with $q_k = (q_{0k0}, q_{0k1}, q_{1k0}, q_{1k1})$. For a category with $p_k = 0$, the preceding zero-mass argument gives $q_k = 0$, so the zero-cell branch in Definition 6 gives $\phi(q_k) = 0 = p_k (\mu_{1k} - \mu_{0k})$. For a category with $p_k > 0$, overlap and $\epsilon > 0$ give positive treated and control arm masses, and the nonzero-cell branch of Definition 6 gives

$$\phi(q_k) = p_k \left\{ \frac{q_{1k1}}{q_{1k0} + q_{1k1}} - \frac{q_{0k1}}{q_{0k0} + q_{0k1}} \right\} = p_k (\mu_{1k} - \mu_{0k}).$$

Summing over categories yields

$$\tau(P) = \sum_{k=1}^d p_k (\mu_{1k} - \mu_{0k}).$$

Combining this identity with the preceding category identity gives the exact bias identity

$$E_P[W(O)] - \tau(P) = \sum_{k=1}^d 2 \left(\pi_k - \frac{1}{2} \right) p_k (\mu_{1k} + \mu_{0k} - 1).$$

The class assumptions give $0 < \epsilon \leq 1/2$ and, on every category with $p_k > 0$, $\epsilon \leq \pi_k \leq 1 - \epsilon$. The totalized outcome means satisfy $0 \leq \mu_{ak} \leq 1$ for every category and arm. Hence, for every positive-mass category,

$$\left| \pi_k - \frac{1}{2} \right| \leq \frac{1}{2} - \epsilon, \quad |\mu_{1k} + \mu_{0k} - 1| \leq 1,$$

and therefore

$$\left| 2 \left(\pi_k - \frac{1}{2} \right) p_k (\mu_{1k} + \mu_{0k} - 1) \right| \leq 2 \left(\frac{1}{2} - \epsilon \right) p_k.$$

For a zero-mass category, $p_k = 0$ makes both sides of this last bound equal to zero. Applying the triangle inequality to the bias identity and using $\sum_{k=1}^d p_k = 1$ gives

$$|E_P[W(O)] - \tau(P)| \leq \sum_{k=1}^d 2 \left(\frac{1}{2} - \epsilon \right) p_k = 2 \left(\frac{1}{2} - \epsilon \right).$$

The sample law in the class is the product law, and $n > 0$, so

$$E_{\mu_n}[\hat{\tau}_{\text{ctr}}] = E_P[W(O)].$$

Moreover, independence gives

$$\text{Var}_{\mu_n}(\hat{\tau}_{\text{ctr}}) = \frac{1}{n} \text{Var}_P(W(O)) \leq \frac{1}{n} E_P[W(O)^2] = \frac{1}{n}.$$

Finally, the bias-variance identity gives

$$\int (\hat{\tau}_{\text{ctr}} - \tau(P))^2 d\mu_n = \text{Var}_{\mu_n}(\hat{\tau}_{\text{ctr}}) + (E_{\mu_n}[\hat{\tau}_{\text{ctr}}] - \tau(P))^2.$$

Combining the preceding variance and bias bounds yields

$$\int (\hat{\tau}_{\text{ctr}} - \tau(P))^2 d\mu_n \leq \frac{1}{n} + \left\{ 2 \left(\frac{1}{2} - \epsilon \right) \right\}^2 = \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2.$$

□

Lemma 4. `[lem:one-category-bernoulli-lower]` (Bernoulli randomization lower bound) *For all positive integers n and d , the minimax mean-squared-error risk $R_{n,d,\epsilon}$ of [Definition 8](#), evaluated at $\epsilon = 1/2$, satisfies*

$$\frac{1}{100n} \leq R_{n,d,1/2}.$$

Proof of [Lemma 4](#). Since $d > 0$, the category space is nonempty. Set

$$\delta = \frac{2/5}{\sqrt{n}}.$$

Since $n > 0$, $\sqrt{n} \geq 1$, and hence $\delta \geq 0$, $\delta \leq 2/5$, and

$$\delta^2 = \frac{4}{25n}.$$

Consider two laws P_0 and P_1 on the d -category observation space. Under both laws, X is uniform on the category space and $A \mid X \sim \text{Bernoulli}(1/2)$. Under P_0 , both treatment-arm outcome means equal $1/2$. Under P_1 , the control outcome mean equals $1/2$ and the treated outcome mean equals $1/2 + \delta$. The bounds above give $0 \leq 1/2 + \delta \leq 1$, so these Bernoulli probabilities are valid. For each P_j , membership of the product sample law in $\mathcal{E}_{n,d,1/2}$ is obtained from $0 < 1/2$, $1/2 \leq 1/2$, the defining identity of the iid product law $P_j^{\otimes n}$, and the overlap fact that every positive-mass category has treatment probability $1/2$. Thus the two product sample laws belong to $\mathcal{E}_{n,d,1/2}$.

For these two laws,

$$\tau(P_0) = 0, \quad \tau(P_1) = \delta, \quad |\tau(P_1) - \tau(P_0)| = \delta.$$

We next bound the total variation distance between the product laws. Write $p_{j,kay} = P_j(X = k, A = a, Y = y)$. The null law assigns positive mass to every singleton of the finite observation space, so $P_1 \ll P_0$, and hence $P_1^{\otimes n} \ll P_0^{\otimes n}$. The finite chi-squared formula gives

$$1 + \chi^2(P_1 \| P_0) = \sum_{k=1}^d \sum_{a,y \in \{0,1\}} \frac{p_{1,kay}^2}{p_{0,kay}}.$$

For a fixed category k , the two control cells agree with the null and contribute $1/(2d)$. The two treated cells contribute

$$\frac{\{d^{-1}(1/2)(1/2 - \delta)\}^2}{d^{-1}(1/2)(1/2)} + \frac{\{d^{-1}(1/2)(1/2 + \delta)\}^2}{d^{-1}(1/2)(1/2)} = \frac{1}{2d} \left(1 + \frac{\delta^2}{(1/2)(1 - 1/2)} \right).$$

Summing over the d categories yields

$$\chi^2(P_1 \| P_0) = \frac{(1/2)\delta^2}{(1/2)(1 - 1/2)} = 2\delta^2.$$

By tensorization of chi-squared divergence over the iid product law,

$$1 + \chi^2(P_1^{\otimes n} \| P_0^{\otimes n}) = \left(1 + \frac{(1/2)\delta^2}{(1/2)(1 - 1/2)} \right)^n.$$

The single-observation chi-squared value is nonnegative, so $(1 + x)^n \leq \exp(nx)$ gives

$$1 + \chi^2(P_1^{\otimes n} \| P_0^{\otimes n}) \leq \exp\left(n \frac{(1/2)\delta^2}{(1/2)(1 - 1/2)}\right).$$

Using $\delta^2 = 4/(25n)$,

$$n \frac{(1/2)\delta^2}{(1/2)(1 - 1/2)} = \frac{8}{25} \leq \log 2,$$

and therefore

$$\chi^2(P_1^{\otimes n} \| P_0^{\otimes n}) \leq 1.$$

The chi-squared-to-total-variation comparison on the finite product space, together with symmetry of total variation, yields

$$\|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}} = \|P_1^{\otimes n} - P_0^{\otimes n}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\chi^2(P_1^{\otimes n} \| P_0^{\otimes n})} \leq \frac{1}{2}.$$

Now fix any measurable estimator $\hat{\tau}$. With $s = \delta/2$, the separation above and the total-variation bound imply

$$\max_{j=0,1} P_j^{\otimes n} \left(|\hat{\tau} - \tau(P_j)| \geq \frac{\delta}{2} \right) \geq \frac{1}{4}.$$

On each of the two laws,

$$\left(\frac{\delta}{2} \right)^2 P_j^{\otimes n} \left(|\hat{\tau} - \tau(P_j)| \geq \frac{\delta}{2} \right) \leq E_{P_j^{\otimes n}} [(\hat{\tau} - \tau(P_j))^2].$$

Therefore

$$\max_{j=0,1} E_{P_j^{\otimes n}} [(\hat{\tau} - \tau(P_j))^2] \geq \frac{\delta^2}{16} = \frac{1}{100n}.$$

It remains to compare these two risks with the supremum in [Definition 8](#). A point of that supremum is a single-observation law P together with the class membership assertion $P^{\otimes n} \in \mathcal{E}_{n,d,1/2}$. For any such P , the overlap part of the membership assertion gives $|\tau(P)| \leq 1$. Since the sample alphabet is finite, every sample point ω satisfies

$$|\hat{\tau}(\omega)| \leq \sum_{\omega' \in (\{1, \dots, d\} \times \{0,1\}^2)^n} |\hat{\tau}(\omega')|.$$

Thus, under the product law generated by any such single-observation law P ,

$$E_{P^{\otimes n}} [(\hat{\tau} - \tau(P))^2] \leq \left(\sum_{\omega' \in (\{1, \dots, d\} \times \{0,1\}^2)^n} |\hat{\tau}(\omega')| + 1 \right)^2.$$

This finite upper bound applies uniformly over the indexed family in the supremum of [Definition 8](#). Since $P_0^{\otimes n}$ and $P_1^{\otimes n}$ are members of $\mathcal{E}_{n,d,1/2}$, that supremum is an upper bound for the two displayed risks. Hence the worst-case risk of the fixed estimator $\hat{\tau}$ is at least $1/(100n)$. The class of measurable estimators is nonempty, for instance by the constant-zero estimator, so taking the infimum over estimators in [Definition 8](#) gives

$$\frac{1}{100n} \leq R_{n,d,1/2}.$$

□

Lemma 5. [[lem:randomized-endpoint-minimax](#)] (Randomized endpoint risk) *For every $n, d \in \mathbb{N}$ with $0 < n$ and $0 < d$, the minimax mean-squared-error risk $R_{n,d,\epsilon}$ defined in [Definition 8](#) satisfies, at $\epsilon = 1/2$,*

$$\frac{1}{100n} \leq R_{n,d,1/2} \leq \frac{1}{n}.$$

Proof of [Lemma 5](#). [Lemma 4](#) gives the lower bound

$$\frac{1}{100n} \leq R_{n,d,1/2}.$$

For the upper bound, use the centered estimator

$$\hat{\tau}_{\text{ctr}} = \frac{1}{n} \sum_{i=1}^n 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right).$$

It is measurable because the sample space is finite. Write, for a measurable estimator $\hat{\tau}$,

$$W(\hat{\tau}) = \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,1/2}} E_P[(\hat{\tau} - \tau(P))^2].$$

The real-valued range $\{W(\hat{\tau}) : \hat{\tau} \text{ measurable}\}$ is bounded below by 0. Indeed, if $\mathcal{E}_{n,d,1/2}$ is empty, the defining supremum has the empty index set and equals 0. If the class is nonempty, then in the bounded-supremum case $W(\hat{\tau})$ dominates the mean squared error at any class member, and that mean squared error is an integral of a square; in the remaining real-supremum convention branch the value is again 0. Hence the defining infimum for the minimax risk may be compared with the admissible estimator $\hat{\tau}_{\text{ctr}}$:

$$R_{n,d,1/2} \leq W(\hat{\tau}_{\text{ctr}}).$$

It remains to bound this worst-case risk. If $\mathcal{E}_{n,d,1/2}$ is empty, then $W(\hat{\tau}_{\text{ctr}}) = 0 \leq 1/n$. If the class is nonempty, then for each $P^{\otimes n} \in \mathcal{E}_{n,d,1/2}$, [Lemma 3](#) gives

$$E_P[(\hat{\tau}_{\text{ctr}} - \tau(P))^2] \leq \frac{1}{n} + 4 \left(\frac{1}{2} - \frac{1}{2} \right)^2 = \frac{1}{n}.$$

Taking the supremum over the nonempty class therefore gives

$$W(\hat{\tau}_{\text{ctr}}) \leq \frac{1}{n}.$$

Consequently $R_{n,d,1/2} \leq 1/n$, and combining this with the lower bound yields

$$\frac{1}{100n} \leq R_{n,d,1/2} \leq \frac{1}{n}.$$

□

[Lemma 3](#) bounds the centered estimator by a variance term and a squared distance from the randomized endpoint. [Lemmas 4](#) and [5](#) then place the exact endpoint risk at the parametric scale, uniformly over positive alphabet sizes.

The following selector lemma combines the fixed-interior hybrid envelope with the centered endpoint envelope. Its estimator choice is deterministic once n, d, ϵ , and the constants in the fixed-interior upper bound are specified.

Lemma 6. [[lem:combined-upper-envelope](#)] (Combined envelope selector) *Fix constants $\epsilon, C_\epsilon, \rho_\epsilon \in \mathbb{R}$ and $N_\epsilon \in \mathbb{N}$. Assume:*

- (*Interior overlap.*) $0 < \epsilon < 1/2$.
- (*Positive constants.*) $C_\epsilon > 0$ and $\rho_\epsilon > 0$.
- (*Hybrid bound.*) *For every $n, d \in \mathbb{N}$, if $0 < n$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$, then the hybrid estimator $\hat{\tau}_n^{\text{hyb}}$ of [Definition 35](#) satisfies*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[\left(\hat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Then for every $n, d \in \mathbb{N}$ with $0 < n$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$, let $K_\epsilon = \max\{C_\epsilon, 4\}$. Define the C_ϵ, ϵ -dependent selected estimator $\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}}$ to be $\hat{\tau}_n^{\text{hyb}}$ when

$$C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and to be $\hat{\tau}_{\text{ctr}}$ otherwise. This selected estimator satisfies

$$R_{n, d, \epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n, d, \epsilon}} \mathbb{E}_{P^{\otimes n}} \left[\left(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P) \right)^2 \right] \leq K_\epsilon \left[\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\} \right],$$

where $R_{n, d, \epsilon}$ is the minimax risk of [Definition 8](#).

Proof of [Lemma 6](#). Fix n, d satisfying the displayed hypotheses, and set

$$K_\epsilon = \max\{C_\epsilon, 4\}.$$

Throughout this proof, the quotient $d^2/(n^2(\log n)^2)$ is interpreted with the paper's rate convention: it is the real-division term appearing in the definition of the rate, and equals 0 when the denominator is 0.

The centered estimator satisfies the uniform bound

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n, d, \epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\hat{\tau}_{\text{ctr}} - \tau(P))^2 \right] \leq \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2.$$

Indeed, when the experiment class is inhabited, [Lemma 3](#) applies to each product law in the class and the supremum preserves the same upper bound. When the class is empty, the empty worst-case supremum is the real number 0, and the same bound follows from nonnegativity of the displayed right-hand side.

We compare the two branches in the deterministic definition of $\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}}$. If

$$C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right) \leq \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2,$$

then the selector returns $\hat{\tau}_n^{\text{hyb}}$, and the assumed hybrid bound gives

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n, d, \epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right] \leq C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right) = \min \left\{ C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right), \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \right\}.$$

If the displayed inequality fails, then

$$\frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \leq C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right),$$

the selector returns $\hat{\tau}_{\text{ctr}}$, and the centered bound gives the same conclusion with the second branch:

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n, d, \epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right] \leq \min \left\{ C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right), \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \right\}.$$

Thus, in all cases,

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right] \leq \min \left\{ C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right), \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \right\}.$$

The sample space is finite, so the selected estimator is measurable. For every measurable estimator, the worst-case mean squared error is bounded below by 0. To see this, if $\mathcal{E}_{n,d,\epsilon}$ is empty, the empty worst-case supremum is 0. If $\mathcal{E}_{n,d,\epsilon}$ is inhabited and the real-valued family of mean squared errors over $\mathcal{E}_{n,d,\epsilon}$ is bounded above, then a nonnegative squared-loss integral lies below the supremum. If that family is unbounded above, the totalized real supremum of the indexed family is, by convention, $\sup \emptyset = 0$, so the same lower bound holds. Hence the range over which the infimum defining $R_{n,d,\epsilon}$ is taken is bounded below, and evaluating that infimum at the selected estimator yields

$$R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P))^2 \right].$$

It remains to simplify the deterministic envelope. Since $K_\epsilon \geq C_\epsilon$, $K_\epsilon \geq 4$, and the scalar terms $1/n$, $d^2/(n^2(\log n)^2)$, and $(1/2 - \epsilon)^2$ are nonnegative,

$$C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right) \leq K_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right)$$

and

$$\frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \leq K_\epsilon \left(\frac{1}{n} + \left(\frac{1}{2} - \epsilon \right)^2 \right).$$

If

$$\frac{d^2}{n^2(\log n)^2} \leq \left(\frac{1}{2} - \epsilon \right)^2,$$

then the first of these deterministic bounds gives

$$\min \left\{ C_\epsilon \left(\frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right), \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2 \right\} \leq K_\epsilon \left\{ \frac{1}{n} + \min \left(\frac{d^2}{n^2(\log n)^2}, \left(\frac{1}{2} - \epsilon \right)^2 \right) \right\}.$$

In the remaining case,

$$\left(\frac{1}{2} - \epsilon \right)^2 \leq \frac{d^2}{n^2(\log n)^2},$$

the same conclusion follows from the second deterministic bound. Combining the selector bound with this deterministic comparison proves the stated upper envelope with $K_\epsilon = \max\{C_\epsilon, 4\}$. $\square$

The selector in [Lemma 6](#) is the algebraic step behind the combined envelope in [Theorem 2](#). It converts two available upper bounds into the smaller of the sparse-cell and near-randomization scales, up to $K_\epsilon = \max\{C_\epsilon, 4\}$.

The next block records the deterministic calibration and polynomial approximation facts used by the light branch. The Chebyshev construction follows the same approximation-theoretic principle used for nonsmooth discrete-distribution functionals [[Jiao et al., 2015](#), [Wu and Yang, 2016](#), [Han et al., 2020](#), [Cai and Low, 2011](#)].

Lemma 7. [lem:calibration-consequences] (Calibration consequences) *Let $L = \log(en)$, let $\ell = \log n$, and let $m_1 = |I_1|$ be the second split size from [Definition 16](#). Let $M = M(n)$ and $B = B(n)$ be the polynomial degree and light-cell approximation scale from [Definition 35](#). Suppose that*

- *(Calibration cutoff.) $n \geq N_0$.*
- *(Large logarithmic scale.) $L \geq 240$.*

Then

$$0 < \ell, \quad \ell \leq L, \quad L \leq 2\ell, \quad 0 < m_1, \quad n \leq 2m_1, \quad M \leq L, \quad B \leq \frac{8192L}{n}, \quad (6^{2M})^2 L^6 \leq n.$$

Proof of Lemma 7. Because $n \geq N_0$, the calibration cutoff gives

$$\alpha_0 L \geq 2, \quad 4M(n)^2 \leq m_1, \quad \frac{4M(n)}{m_1} \leq \frac{3B(n)}{4}.$$

In particular $L > 0$. The identity $L = \log(en) = 1 + \log n = 1 + \ell$ is valid for this positive n . Since $L \geq 240$, we have

$$\ell = L - 1 > 0, \quad \ell \leq L, \quad L = 1 + \ell \leq 2\ell.$$

The definition $M(n) = \max\{2, \lfloor \alpha_0 L \rfloor\}$ gives $M \geq 2$. Hence $4M^2 \leq m_1$ implies $0 < m_1$. For the deterministic second split,

$$m_1 = n - \lfloor n/2 \rfloor,$$

and therefore $n \leq 2m_1$.

Since $\alpha_0 L \geq 2$, the maximum in the definition of $M(n)$ is attained by the floor term:

$$M = \lfloor \alpha_0 L \rfloor.$$

Using $\alpha_0 \leq 1$ and $L > 0$,

$$M \leq \alpha_0 L \leq L.$$

By the definition $B(n) = 4096L/m_1$ and the bound $n \leq 2m_1$,

$$B = \frac{4096L}{m_1} \leq \frac{8192L}{n}.$$

Finally we verify the calibrated growth inequality. Since $\alpha_0 \leq (32 \log 6)^{-1}$ and $M \leq \alpha_0 L$,

$$4M \log 6 \leq \frac{L}{8}, \quad \text{so} \quad 6^{4M} \leq \exp(L/8).$$

Also $L \geq 240$ gives

$$\log L = \log 12 + \log(L/12) \leq 11 + (L/12 - 1) \leq L/8,$$

and hence

$$L^6 = \exp(6 \log L) \leq \exp(3L/4).$$

Multiplying,

$$6^{4M} L^6 \leq \exp(7L/8) \leq \exp(L - 1) = n.$$

Since $(6^{2M})^2 = 6^{4M}$, this is exactly

$$(6^{2M})^2 L^6 \leq n.$$

The displayed conclusions follow. □

Lemma 8. `[lem:factorial-moment-mean]` (Exact factorial mean) *Let $d, n \in \mathbb{N}$, let P be a single-observation law on $\{1, \dots, d\} \times \{0, 1\}^2$, and let k be a category. Suppose that $n \geq N_0$, where N_0 is the calibration cutoff in [Definition 35](#). With $M = M(n)$, $B = B(n)$, and $\widehat{P}_k$ the factorial-moment polynomial estimate defined in [Definition 35](#), its exact mean under the iid product law generated by P is*

$$\int \widehat{P}_k(O_1, \dots, O_n) dP^{\otimes n} = P_{M(n), B(n)}(q_k).$$

Proof of [Lemma 8](#). Write $M = M(n)$, $B = B(n)$, and let $q = q_k$ be the four-cell vector for category k . Set

$$g_{M,j} = (-1)^j \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4}.$$

The polynomial whose mean is evaluated is the finite polynomial

$$P_{M,B}(u) = B^{-1}s(u)t_1(u) \sum_{j=0}^{M-2} g_{M,j} \{B^{-1}s_1(u)\}^j - B^{-1}s(u)t_0(u) \sum_{j=0}^{M-2} g_{M,j} \{B^{-1}s_0(u)\}^j.$$

We compute the factorial lift in its sparse arm expansion. The factorial-moment contribution is

$$\widehat{P}_k = \widehat{A}_{1,k} - \widehat{A}_{0,k},$$

where, for $a \in \{0, 1\}$,

$$\widehat{A}_{a,k} = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} (B^{-1})^j \binom{j}{t} \frac{\prod_{b,y} (N_{bky}^{(1)})_{r_{by}^{a,c,j,t}}}{(m_1)_{|r^{a,c,j,t}|}}.$$

The merged exponent vector $r^{a,c,j,t}$ is defined by

$$u^{r^{a,c,j,t}} = u_c u_{a1} u_{a0}^t u_{a1}^{j-t}.$$

Thus $\widehat{P}_k$ is the difference of the two displayed sparse arm sums.

The cutoff condition $n \geq N_0$ includes $4M^2 \leq m_1$; since $M > 0$, this implies $M \leq m_1$. In every displayed summand, $0 \leq t \leq j \leq M-2$, and the merged exponent vector has total degree

$$|r^{a,c,j,t}| = j+2 \leq M \leq m_1.$$

For each such multi-index r , the factorial-moment identity for the second-split multinomial cell counts gives

$$\mathbb{E}_{P^{\otimes \infty}} \left[\frac{\prod_{b,y} (N_{bky}^{(1)})_{r_{by}}}{(m_1)_{|r|}} \right] = \prod_{b,y} q_{by}^{r_{by}}.$$

Because the arm sums are finite sums of integrable factorial monomials, termwise integration yields, for each arm a ,

$$\mathbb{E}_{P^{\otimes \infty}}[\widehat{A}_{a,k}] = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} (B^{-1})^j \binom{j}{t} q^{r^{a,c,j,t}}.$$

For fixed a and j , the inner finite sum collapses as

$$\sum_{t=0}^j \sum_{c \in \{0,1\}^2} \binom{j}{t} q^{r^{a,c,j,t}} = \left(\sum_{b,y} q_{by} \right) q_{a1} (q_{a0} + q_{a1})^j.$$

Indeed, the merged-exponent monomial identity writes each summand as $q_c q_{a1} q_{a0}^t q_{a1}^{j-t}$; summing over c gives the total mass $\sum_{b,y} q_{by}$, and summing over t gives the binomial theorem.

Writing $s(q) = \sum_{b,y} q_{by}$, $s_a(q) = q_{a0} + q_{a1}$, and $t_a(q) = q_{a1}$, we obtain

$$\mathbb{E}_{P^{\otimes \infty}}[\hat{A}_{a,k}] = B^{-1} s(q) t_a(q) \sum_{j=0}^{M-2} g_{M,j} \{B^{-1} s_a(q)\}^j.$$

Substituting this identity for $a = 1$ and $a = 0$ into the sparse arm representation of $\hat{P}_k$ gives

$$\mathbb{E}_{P^{\otimes \infty}}[\hat{P}_k] = B^{-1} s(q) t_1(q) \sum_{j=0}^{M-2} g_{M,j} \{B^{-1} s_1(q)\}^j - B^{-1} s(q) t_0(q) \sum_{j=0}^{M-2} g_{M,j} \{B^{-1} s_0(q)\}^j.$$

By the displayed definition of $P_{M,B}$, the right-hand side is $P_{M(n),B(n)}(q_k)$. $\square$

Lemma 9. [lem:coefficient-envelope] (Absolute coefficient envelope) *With $P_{M,B}$, $s(\cdot)$, c_r , and u^r as in Definition 35, write*

$$P_{M,B}(u) = \sum_r c_r u^r.$$

For every nonnegative integer M , every scale $B > 0$, and every nonnegative four-cell vector v ,

$$\sum_{r \in \text{supp}(P_{M,B})} |c_r| v^r \leq B 6^M \max \left\{ \left(\frac{s(v)}{B} \right)^2, \left(\frac{s(v)}{B} \right)^M \right\}.$$

Proof of Lemma 9. For a polynomial $Q(u) = \sum_r q_r u^r$, write its absolute coefficient envelope at the nonnegative vector v as

$$\mathcal{A}_v(Q) = \sum_{r \in \text{supp}(Q)} |q_r| \prod_i v_i^{r_i}.$$

For $Q = P_{M,B}$, this is the left-hand side of the asserted inequality. If $M = 0$ or $M = 1$, the defining range $j \in \{0, \dots, M-2\}$ is empty, so $P_{M,B} = 0$; the right-hand side is nonnegative because $B > 0$ and $v \geq 0$.

Assume now that $M \geq 2$. Write the polynomial continuation as

$$G_M(x) = \sum_{j=0}^{M-2} g_{M,j} x^j, \quad g_{M,j} = (-1)^j \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4}.$$

Set

$$R = s(v), \quad z = \frac{R}{B}, \quad \Gamma_M(z) = \sum_{j=0}^{M-2} |g_{M,j}| z^j.$$

Then $R \geq 0$ and $z \geq 0$. The coefficient envelope satisfies, for nonnegative v ,

$$\mathcal{A}_v(Q_1 - Q_0) \leq \mathcal{A}_v(Q_1) + \mathcal{A}_v(Q_0), \quad \mathcal{A}_v(Q_1 Q_2) \leq \mathcal{A}_v(Q_1) \mathcal{A}_v(Q_2),$$

and iterating the product bound gives the corresponding four-factor bound. For the linear mass polynomials,

$$\mathcal{A}_v(s) \leq R, \quad \mathcal{A}_v(s_a) \leq s_a(v) \leq R, \quad \mathcal{A}_v(u_{a1}) = v_{a1}.$$

Thus, for each arm a , the inner polynomial obeys

$$\begin{aligned} \mathcal{A}_v \left(\sum_{j=0}^{M-2} g_{M,j} \{B^{-1}s_a(u)\}^j \right) &\leq \sum_{j=0}^{M-2} |g_{M,j}| \{B^{-1}\mathcal{A}_v(s_a)\}^j \\ &\leq \sum_{j=0}^{M-2} |g_{M,j}| (R/B)^j = \Gamma_M(z). \end{aligned}$$

Combining this bound with the four-factor envelope inequality gives

$$\mathcal{A}_v \left(B^{-1}s(u)u_{a1} \sum_{j=0}^{M-2} g_{M,j} \{B^{-1}s_a(u)\}^j \right) \leq B^{-1}R v_{a1} \Gamma_M(z),$$

and applying the subtraction bound to the treated and control terms in $P_{M,B}$ yields

$$\mathcal{A}_v(P_{M,B}) \leq B^{-1}R\{v_{11} + v_{01}\}\Gamma_M(z) \leq B^{-1}R^2\Gamma_M(z),$$

because $v_{11} + v_{01} \leq s(v) = R$.

It remains to bound Γ_M . Let T_M be the Chebyshev polynomial of the first kind. The shifted-Chebyshev expansion is

$$T_M(1-2x) = 1 + \sum_{j=1}^M (-1)^j \frac{M}{M+j} \binom{M+j}{2j} 4^j x^j.$$

Separating the $j=1$ term and reindexing the remaining terms by $j=k+2$ gives

$$T_M(1-2x) = 1 - 2M^2x + 2M^2x^2 \sum_{k=0}^{M-2} g_{M,k}x^k = 1 - 2M^2x + 2M^2x^2G_M(x).$$

Since $M > 0$, the explicit coefficient formula gives

$$|g_{M,j}| = (-1)^j g_{M,j}, \quad \Gamma_M(t) = G_M(-t) \quad (t \geq 0).$$

Putting $x = -1$ in the shifted identity therefore yields

$$T_M(3) = 1 + 2M^2 + 2M^2\Gamma_M(1).$$

Because $M \geq 1$ and $\Gamma_M(1) \geq 0$, this implies

$$\Gamma_M(1) \leq T_M(3).$$

The standard inequality $T_m(x) \geq 1$ for $x \geq 1$ gives $T_m(3) \geq 0$. Together with the recurrence $T_{m+2}(3) = 6T_{m+1}(3) - T_m(3)$ and the initial values $T_0(3) = 1$, $T_1(3) = 3$, induction gives

$$T_M(3) \leq 6^M.$$

Therefore $\Gamma_M(1) \leq 6^M$. If $0 \leq z \leq 1$, termwise comparison in the nonnegative-power sum gives $\Gamma_M(z) \leq \Gamma_M(1) \leq 6^M$. If $z \geq 1$, then $z^j \leq z^{M-2}$ for every $0 \leq j \leq M-2$, hence

$$\Gamma_M(z) \leq z^{M-2} \Gamma_M(1) \leq 6^M z^{M-2}.$$

Equivalently,

$$\Gamma_M(z) \leq 6^M \max\{1, z^{M-2}\} \quad (z \geq 0).$$

Combining the coefficient-envelope bound with this estimate,

$$\mathcal{A}_v(P_{M,B}) \leq B^{-1} R^2 6^M \max\{1, z^{M-2}\}.$$

Since $R = Bz$, $B > 0$, and $M \geq 2$,

$$B^{-1} R^2 6^M \max\{1, z^{M-2}\} = B 6^M \max\{z^2, z^M\}.$$

Substituting $z = s(v)/B$ gives the claimed coefficient envelope. $\square$

Lemma 10. [lem:chebyshev-reciprocal-certificate] (Polynomial approximation certificate) *With the notation for $\mathcal{C}_\epsilon$, $s(u)$, $P_{M,B}$, and $\phi(u)$ from Definitions 5, 6 and 35, let M , B , the overlap level ϵ , and u satisfy:*

- (*Degree.*) $M \geq 1$.
- (*Overlap.*) $\epsilon > 0$.
- (*Cone membership.*) $u \in \mathcal{C}_\epsilon$.
- (*Mass bound.*) $s(u) \leq B$.

Then

$$|P_{M,B}(u) - \phi(u)| \leq \frac{2B}{\epsilon M^2}.$$

Proof of Lemma 10. Let $G_M(x) = \sum_{j=0}^{M-2} g_{M,j} x^j$. The shifted-Chebyshev expansion gives, for every real x ,

$$T_M(1 - 2x) = 1 - 2M^2 x + 2M^2 x^2 G_M(x).$$

If $0 \leq x \leq 1$, then $|T_M(1 - 2x)| \leq 1$, and the displayed identity implies

$$x |1 - x G_M(x)| \leq \frac{1}{M^2}.$$

First consider $u = 0$. Then $s(u) = s_0(u) = s_1(u) = u_{01} = u_{11} = 0$, so both $P_{M,B}(u)$ and $\phi(u)$ equal 0. Since $s(u) \leq B$, this case also gives $B \geq 0$, and hence

$$|P_{M,B}(u) - \phi(u)| = 0 \leq \frac{2B}{\epsilon M^2}.$$

Now suppose $u \neq 0$. Nonnegativity of the four coordinates gives $s(u) \geq 0$, and $u \neq 0$ gives $s(u) > 0$. Since $s(u) \leq B$, we have $B > 0$. The overlap inequalities give

$$s_1(u) \geq \epsilon s(u) > 0, \quad s_0(u) = s(u) - s_1(u) \geq \epsilon s(u) > 0.$$

Also $s_a(u) \leq s(u) \leq B$, so

$$x_a := \frac{s_a(u)}{B} \in [0, 1], \quad a \in \{0, 1\}.$$

For each arm a , the preceding Chebyshev certificate gives

$$\left| 1 - \frac{s_a(u)}{B} G_M\left(\frac{s_a(u)}{B}\right) \right| \leq \frac{B}{s_a(u)M^2}.$$

Using $0 \leq u_{a1} \leq s_a(u)$, this yields

$$\left| B^{-1} s(u) u_{a1} G_M\left(\frac{s_a(u)}{B}\right) - s(u) \frac{u_{a1}}{s_a(u)} \right| \leq \frac{B}{\epsilon M^2}.$$

Indeed, the left-hand side equals

$$\frac{s(u) u_{a1}}{s_a(u)} \left| 1 - \frac{s_a(u)}{B} G_M\left(\frac{s_a(u)}{B}\right) \right|,$$

and the factor $s(u)/s_a(u)$ is at most $1/\epsilon$.

Finally,

$$P_{M,B}(u) - \phi(u) = \left[B^{-1} s(u) u_{11} G_M\left(\frac{s_1(u)}{B}\right) - s(u) \frac{u_{11}}{s_1(u)} \right] - \left[B^{-1} s(u) u_{01} G_M\left(\frac{s_0(u)}{B}\right) - s(u) \frac{u_{01}}{s_0(u)} \right].$$

The triangle inequality and the two arm bounds give

$$|P_{M,B}(u) - \phi(u)| \leq \frac{B}{\epsilon M^2} + \frac{B}{\epsilon M^2} = \frac{2B}{\epsilon M^2}.$$

□

Lemma 7 turns the numerical choices in **Definition 35** into inequalities at the scales used in the variance and bias calculations. **Lemma 8** identifies the population target of each factorial-polynomial term exactly, while **Lemmas 9** and **10** control the coefficient growth and approximation error of $P_{M,B}$.

The heavy branch is controlled through residual and missing-arm bounds for deterministic category sets. These auxiliary statements quantify the two finite-sample effects that remain after conditioning on the pilot split: centered binomial variation within observed arms and the contribution of categories whose estimation split misses one treatment arm.

Lemma 11. [lem:heavy-residual-bound] (Fixed-set residual variance) *Let P be a law on $\{1, \dots, d\} \times \{0, 1\}^2$, and let $O_i = (X_i, A_i, Y_i)$, $i \in I$, be an iid P -sample indexed by a finite set I . Put $m = |I|$, $p_k = P(X = k)$, and $\mu_{ak} = E_P[Y \mid A = a, X = k]$. For a deterministic set $H \subseteq \{1, \dots, d\}$ and an arm $a \in \{0, 1\}$, define*

$$\mathcal{R}_a(H) = \sum_{k \in H} N_k \frac{\mathbf{1}\{N_{ak} > 0\}}{N_{ak}} \{N_{ak1} - \mu_{ak} N_{ak}\}.$$

Assume:

- **(Positive overlap.)** The overlap level satisfies $\epsilon > 0$, and every category with positive mass satisfies

$$\epsilon \leq P(A = 1 \mid X = k) \leq 1 - \epsilon.$$

- **(Positive mass on H .)** For every $k \in H$, $p_k > 0$.

Then

$$E_P[\mathcal{R}_a(H)^2] \leq \frac{2m}{\epsilon} \sum_{k \in H} p_k.$$

Proof of Lemma 11. For an observation array $z = (O_i)_{i \in I}$, with $O_i = (X_i, A_i, Y_i)$, write

$$N_k(z) = \sum_{i \in I} \mathbf{1}\{X_i = k\}, \quad N_{ak}(z) = \sum_{i \in I} \mathbf{1}\{X_i = k, A_i = a\},$$

and

$$N_{ak1}(z) = \sum_{i \in I} \mathbf{1}\{X_i = k, A_i = a, Y_i = 1\}.$$

Let $m = |I|$, and put

$$C_{ak}(z) = N_k(z) \frac{\mathbf{1}\{N_{ak}(z) > 0\}}{N_{ak}(z)}, \quad \xi_{ak,i}(z) = \mathbf{1}\{X_i = k, A_i = a\} \{Y_i - \mu_{ak}\}.$$

With the convention that the displayed ratio is zero when $N_{ak}(z) = 0$, the residual can be written as

$$\mathcal{R}_a(H) = \sum_{k \in H} C_{ak}(z) \sum_{i \in I} \xi_{ak,i}(z),$$

because $\sum_i \xi_{ak,i} = N_{ak1}(z) - \mu_{ak} N_{ak}(z)$.

Expand the square as a double sum over pairs (k, i) and (l, j) . If $i \neq j$, then the factor $C_{ak}(z)C_{al}(z)\xi_{al,j}(z)$ is unchanged when only the outcome coordinate of observation i is replaced; conditioning on all other coordinates and using the centered arm-category residual gives integral zero. If $i = j$ and $k \neq l$, then the two category indicators cannot both hold for the same observation, so the product of residuals is identically zero. Hence only the diagonal terms remain:

$$\mathbb{E}_{P^{\otimes I}}[\mathcal{R}_a(H)^2] = \sum_{k \in H} \sum_{i \in I} \mathbb{E}_{P^{\otimes I}}[C_{ak}(z)^2 \xi_{ak,i}(z)^2].$$

For each fixed k and i , the multiplier $C_{ak}(z)^2$ is nonnegative and depends only on the category and treatment coordinates of the array. The one-coordinate conditional Bernoulli variance bound gives

$$\mathbb{E}_{P^{\otimes I}}[C_{ak}(z)^2 \xi_{ak,i}(z)^2] \leq \mathbb{E}_{P^{\otimes I}}[C_{ak}(z)^2 \mathbf{1}\{X_i = k, A_i = a\}].$$

Therefore, for each fixed k ,

$$\begin{aligned} \sum_{i \in I} \mathbb{E}_{P^{\otimes I}}[C_{ak}(z)^2 \xi_{ak,i}(z)^2] &\leq \mathbb{E}_{P^{\otimes I}} \left[\sum_{i \in I} C_{ak}(z)^2 \mathbf{1}\{X_i = k, A_i = a\} \right] \\ &= \mathbb{E}_{P^{\otimes I}} \left[N_k(z)^2 \frac{\mathbf{1}\{N_{ak}(z) > 0\}}{N_{ak}(z)} \right]. \end{aligned}$$

Let $\rho_{ak} = P(A = a \mid X = k)$, so $\rho_{1k} = \pi_k$ and $\rho_{0k} = 1 - \pi_k$. For $k \in H$, the positive-mass hypothesis gives $p_k > 0$. Overlap gives $\rho_{ak} \geq \epsilon$, and $\rho_{ak} \leq 1$ because it is an arm probability. Put

$$C_k = \{X = k\}, \quad R_{ak} = \{X = k, A = a\}.$$

Then $R_{ak} \subset C_k$, and under the one-observation law

$$P(C_k) = p_k, \quad P(R_{ak}) = p_k \rho_{ak}, \quad P(C_k \setminus R_{ak}) = p_k(1 - \rho_{ak}), \quad P(C_k^c) = 1 - p_k.$$

Partition the product sample space according to

$$U = \{i : O_i \in C_k\}, \quad V = \{i : O_i \in R_{ak}\}.$$

The nested relation forces $V \subset U$. For fixed U, V with $V \subset U$, independence gives the product probability

$$(p_k \rho_{ak})^{|V|} \{p_k(1 - \rho_{ak})\}^{|U| - |V|} (1 - p_k)^{m - |U|},$$

and the integrand equals

$$|U|^2 \frac{\mathbf{1}\{|V| > 0\}}{|V|}.$$

Summing over all such pairs and then collecting first by $t = |U|$ and then by $s = |V|$ gives

$$\begin{aligned} & \mathbb{E}_{P^{\otimes I}} \left[N_k^2 \frac{\mathbf{1}\{N_{ak} > 0\}}{N_{ak}} \right] \\ &= \sum_{t=0}^m \binom{m}{t} p_k^t (1 - p_k)^{m-t} t^2 \sum_{s=0}^t \binom{t}{s} \rho_{ak}^s (1 - \rho_{ak})^{t-s} \frac{\mathbf{1}\{s > 0\}}{s}. \end{aligned}$$

It remains to bound the inner reciprocal-count sum. For every integer $s \geq 0$, with the left side interpreted as zero at $s = 0$,

$$\frac{\mathbf{1}\{s > 0\}}{s} \leq \frac{2}{s+1}.$$

Since $0 < \rho_{ak} \leq 1$, the binomial weights are nonnegative, and therefore

$$\begin{aligned} & \sum_{s=0}^t \binom{t}{s} \rho_{ak}^s (1 - \rho_{ak})^{t-s} \frac{\mathbf{1}\{s > 0\}}{s} \\ & \leq 2 \sum_{s=0}^t \binom{t}{s} \rho_{ak}^s (1 - \rho_{ak})^{t-s} \frac{1}{s+1} \\ &= \frac{2}{(t+1)\rho_{ak}} \sum_{s=0}^t \binom{t+1}{s+1} \rho_{ak}^{s+1} (1 - \rho_{ak})^{t-s} \\ &= \frac{2\{1 - (1 - \rho_{ak})^{t+1}\}}{(t+1)\rho_{ak}} \leq \frac{2}{(t+1)\rho_{ak}}. \end{aligned}$$

Here the middle equality uses $\binom{t}{s}/(s+1) = \binom{t+1}{s+1}/(t+1)$, and the final inequality uses $(1 - \rho_{ak})^{t+1} \geq 0$. Hence, using $t^2/(t+1) \leq t$ and the binomial first moment,

$$\begin{aligned} \mathbb{E}_{P^{\otimes I}} \left[N_k^2 \frac{\mathbf{1}\{N_{ak} > 0\}}{N_{ak}} \right] & \leq \sum_{t=0}^m \binom{m}{t} p_k^t (1 - p_k)^{m-t} \frac{2t}{\rho_{ak}} \\ &= \frac{2mp_k}{\rho_{ak}} \leq \frac{2mp_k}{\epsilon}. \end{aligned}$$

Summing over $k \in H$ gives

$$\mathbb{E}_{P^{\otimes I}}[\mathcal{R}_a(H)^2] \leq \sum_{k \in H} \frac{2mp_k}{\epsilon} = \frac{2m}{\epsilon} \sum_{k \in H} p_k.$$

□

Lemma 12. [lem:missing-arm-envelope] (Missing-arm second moment) *Let P be a probability law for one observed unit $O = (X, A, Y)$, with $X \in \{1, \dots, d\}$ and $A, Y \in \{0, 1\}$. Let J be a finite index set and write $m = |J|$. For an iid P -sample $(O_j)_{j \in J}$, a deterministic category set $H \subseteq \{1, \dots, d\}$, and an arm $a \in \{0, 1\}$, set*

$$Z_{ak} = N_k \mathbf{1}\{N_{ak} = 0\}, \quad N_k = \sum_{j \in J} \mathbf{1}\{X_j = k\}, \quad N_{ak} = \sum_{j \in J} \mathbf{1}\{X_j = k, A_j = a\}.$$

Assume:

- **(Sample size.)** $m \geq 3$.
- **(Overlap.)** P satisfies overlap at level ϵ , the overlap parameter, and $\epsilon > 0$.
- **(Mass floor.)** $B > 0$ and $p_k = P(X = k) \geq B$ for every $k \in H$.

Then

$$E_{P^{\otimes J}} \left[\left(\sum_{k \in H} Z_{ak} \right)^2 \right] \leq m + m^2 \left[\frac{|H|}{\{((m-2)/2)\epsilon\}^2 B} \right]^2.$$

Proof of Lemma 12. For a sample $z = (O_1, \dots, O_m)$, let $Z_{ak}(z)$ be the missing-arm category count

$$Z_{ak}(z) = \begin{cases} \#\{i : X_i = k\}, & \#\{i : X_i = k, A_i = a\} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus $\sum_{k \in H} Z_{ak}(z)$ is the fixed missing-arm count over H for arm a . Let

$$\rho_{ak} = P(A = a \mid X = k), \quad u = \frac{m-2}{2}\epsilon, \quad f_k = p_k \exp(-up_k).$$

The assumptions $m \geq 3$, $\epsilon > 0$, and $B > 0$ give $u > 0$, and the mass floor gives $p_k \geq B > 0$ for every $k \in H$. Hence ρ_{ak} is the ordinary conditional arm probability on H , with $0 \leq \rho_{ak} \leq 1$, and overlap gives $\epsilon \leq \rho_{ak}$, equivalently $\epsilon p_k \leq p_k \rho_{ak}$.

We first record the exact second moment. For a fixed category k , expand Z_{ak}^2 by separating one selected index from two ordered distinct selected indices. The one-index event requires O_i to lie in category k outside arm a , while the whole sample avoids the arm- a subcell in category k . Its probability is

$$p_k(1 - \rho_{ak})(1 - p_k \rho_{ak})^{m-1}.$$

For an ordered pair $i \neq j$, the corresponding two-index event has probability

$$\{p_k(1 - \rho_{ak})\}^2(1 - p_k \rho_{ak})^{m-2}.$$

Summing over the m one-index choices and the $(m)_2 = m(m-1)$ ordered pairs gives the diagonal contribution

$$m p_k(1 - \rho_{ak})(1 - p_k \rho_{ak})^{m-1} + (m)_2 \{p_k(1 - \rho_{ak})\}^2 (1 - p_k \rho_{ak})^{m-2}.$$

For distinct k, l , the same-index part in $Z_{ak}Z_{al}$ is zero because the category events are disjoint. The remaining ordered-pair event places one selected observation in category k outside arm a , the other in category l outside arm a , and the whole sample outside the two forbidden arm- a subcells. Since those forbidden subcells are disjoint, its probability is

$$p_k(1 - \rho_{ak})p_l(1 - \rho_{al})(1 - p_k \rho_{ak} - p_l \rho_{al})^{m-2}.$$

Therefore

$$\begin{aligned} \int \left(\sum_{k \in H} Z_{ak}(z) \right)^2 dP^{\otimes m}(z) &= \sum_{k \in H} \left[m p_k(1 - \rho_{ak})(1 - p_k \rho_{ak})^{m-1} \right. \\ &\quad \left. + (m)_2 \{p_k(1 - \rho_{ak})\}^2 (1 - p_k \rho_{ak})^{m-2} \right] \\ &\quad + \sum_{k \in H} \sum_{l \in H \setminus \{k\}} (m)_2 p_k(1 - \rho_{ak})p_l(1 - \rho_{al})(1 - p_k \rho_{ak} - p_l \rho_{al})^{m-2}. \end{aligned}$$

For the diagonal terms, $0 \leq p_k \leq 1$, $0 \leq \rho_{ak} \leq 1$, $(m)_2 \leq m^2$, and $\epsilon p_k \leq p_k \rho_{ak}$ imply

$$\begin{aligned} m p_k(1 - \rho_{ak})(1 - p_k \rho_{ak})^{m-1} + (m)_2 \{p_k(1 - \rho_{ak})\}^2 (1 - p_k \rho_{ak})^{m-2} \\ \leq m p_k + m^2 f_k^2. \end{aligned}$$

Indeed, the first summand is at most $m p_k$. For the second summand,

$$(1 - p_k \rho_{ak})^{m-2} \leq \exp\{-(m-2)\epsilon p_k\} = \exp(-u p_k)^2,$$

and $p_k(1 - \rho_{ak}) \leq p_k$.

For $k \neq l$, disjointness of the two category events gives $p_k + p_l \leq 1$, and hence $p_k \rho_{ak} + p_l \rho_{al} \leq 1$. Together with the two overlap inequalities,

$$\begin{aligned} (1 - p_k \rho_{ak} - p_l \rho_{al})^{m-2} &\leq \exp\{-(m-2)(p_k \rho_{ak} + p_l \rho_{al})\} \\ &\leq \exp\{-(m-2)\epsilon(p_k + p_l)\} \\ &\leq \exp(-u p_k) \exp(-u p_l). \end{aligned}$$

Using also $p_k(1 - \rho_{ak}) \leq p_k$, $p_l(1 - \rho_{al}) \leq p_l$, and $(m)_2 \leq m^2$, we obtain

$$(m)_2 p_k(1 - \rho_{ak})p_l(1 - \rho_{al})(1 - p_k \rho_{ak} - p_l \rho_{al})^{m-2} \leq m^2 f_k f_l.$$

Substituting the diagonal and cross envelopes into the exact expansion and collecting the square of the exponential sum gives

$$\int \left(\sum_{k \in H} Z_{ak}(z) \right)^2 dP^{\otimes m}(z) \leq m \sum_{k \in H} p_k + m^2 \left(\sum_{k \in H} f_k \right)^2.$$

The category masses over H sum to at most one, so the first term is at most m .

It remains to bound the exponential sum. For $t > 0$,

$$\exp(-t) \leq t^{-2}.$$

Applying this with $t = up_k$, and using $p_k \geq B$, gives

$$p_k \exp(-up_k) \leq \frac{1}{u^2 p_k} \leq \frac{1}{u^2 B}.$$

Therefore

$$\sum_{k \in H} f_k \leq \frac{|H|}{u^2 B}.$$

Since each $f_k \geq 0$, combining the last two bounds and substituting $u = ((m-2)/2)\epsilon$ yields

$$\int \left(\sum_{k \in H} Z_{ak}(z) \right)^2 dP^{\otimes m}(z) \leq m + m^2 \left[\frac{|H|}{\left(\left(\frac{m-2}{2} \epsilon \right)^2 B \right)} \right]^2.$$

□

Together, [Lemmas 11](#) and [12](#) provide the finite-set estimates needed after the pilot sandwich supplies the mass floor for heavy categories. The residual term scales with the total population mass of the selected set, and the missing-arm term is controlled by overlap and the pilot-certified mass floor.

The factorial-moment product and coefficient-sum statements support the second-moment analysis of the light branch. They are the count-level analogues of the approximation bounds above and are applied to the split-one multinomial cell counts.

Lemma 13. `[lem:factorial-product-moment]` (Factorial product moment) *Fix $n, d \in \mathbb{N}$, $P \in \text{Prob}(\{1, \dots, d\} \times \{0, 1\}^2)$, a category $k \in \{1, \dots, d\}$, and four-cell multi-indices r, s . Using the split counts and split size in [Definition 16](#), let*

$$\mathbf{m}_{k,r} = \frac{\prod_{(a,y)} (N_{aky}^{(1)})_{r_{ay}}}{(m_1)_{|r|}},$$

where $(z)_t$ denotes the falling factorial and $|r| = \sum_{(a,y)} r_{ay}$. Let q_{aky} denote the treatment-outcome cell mass in category k , as in [Definition 7](#). Assume:

- **(Nonempty estimation split.)** $m_1 > 0$.
- **(Degree ordering.)** $|r| \leq |s|$.
- **(Split-size condition.)** $4|s|^2 \leq m_1$.

Then, under the iid product law generated by P ,

$$\mathbb{E}_{P^{\otimes n}}[\mathbf{m}_{k,r} \mathbf{m}_{k,s}] \leq e \prod_{(a,y)} q_{aky}^{r_{ay}} \prod_{(a,y)} \left(q_{aky} + \frac{|s|}{m_1} \right)^{s_{ay}}.$$

Proof of Lemma 13. Let $\mathcal{C} = \{0, 1\} \times \{0, 1\}$, let $m = m_1$, and write

$$q_{ay} = q_{aky}, \quad r_{ay} = r(a, y), \quad s_{ay} = s(a, y), \quad |r| = \sum_{(a,y) \in \mathcal{C}} r_{ay}, \quad |s| = \sum_{(a,y) \in \mathcal{C}} s_{ay}.$$

Adjoin one extra label $\emptyset$ for observations outside category k . For the j -th observation in the estimation split define

$$Z_j = \begin{cases} (a, y), & (X_j, A_j, Y_j) = (k, a, y), \\ \emptyset, & X_j \neq k, \end{cases} \quad \alpha_{ay} = q_{ay}, \quad \alpha_{\emptyset} = 1 - \sum_{(a,y) \in \mathcal{C}} q_{ay}.$$

The variables $Z_1, \dots, Z_m$ are iid on $\mathcal{C} \cup \{\emptyset\}$ with masses α . Extend r, s by setting $r_{\emptyset} = s_{\emptyset} = 0$. If $N_b = \#\{j : Z_j = b\}$, then

$$\mathbf{m}_{k,r} = \frac{\prod_{b \in \mathcal{C} \cup \{\emptyset\}} (N_b)_{r_b}}{(m)_{|r|}}, \quad \mathbf{m}_{k,s} = \frac{\prod_{b \in \mathcal{C} \cup \{\emptyset\}} (N_b)_{s_b}}{(m)_{|s|}}.$$

This is just the split-count notation of Definition 16, with the outside-category exponent equal to zero.

For an overlap choice

$$H = (H_b)_{b \in \mathcal{C} \cup \{\emptyset\}}, \quad 0 \leq H_b \leq \min\{r_b, s_b\},$$

set

$$|H| = \sum_b H_b, \quad t_b(H) = r_b + s_b - H_b, \quad A_H = \prod_b \binom{r_b}{H_b} \binom{s_b}{H_b} H_b!.$$

The cellwise falling-factorial product identity gives

$$\prod_b (N_b)_{r_b} \prod_b (N_b)_{s_b} = \sum_H A_H \prod_b (N_b)_{t_b(H)}.$$

Taking expectations and using the factorial moment formula for multinomial counts,

$$\mathbb{E}[\mathbf{m}_{k,r} \mathbf{m}_{k,s}] = \sum_H A_H \frac{(m)_{|r|+|s|-|H|}}{(m)_{|r|} (m)_{|s|}} \prod_b \alpha_b^{r_b+s_b-H_b}.$$

The normalization ratio satisfies, for every such H ,

$$\frac{(m)_{|r|+|s|-|H|}}{(m)_{|r|} (m)_{|s|}} \leq \frac{e}{m^{|H|}}.$$

Indeed, if $|s| = 0$ then also $|r| = 0$, and the bound is immediate. Otherwise put $u = m - |s|$. The assumptions $4|s|^2 \leq m$ and $|r| \leq |s|$ imply $u > 0$ and $2|s|^2 \leq u$. Also

$$(m)_{|r|+|s|-|H|} \leq m^{|r|+|s|-|H|}, \quad (m)_{|r|} \geq u^{|r|}, \quad (m)_{|s|} \geq u^{|s|}.$$

Hence the ratio is at most

$$\frac{m^{|r|+|s|-|H|}}{u^{|r|+|s|}} = \frac{(m/u)^{|r|+|s|}}{m^{|H|}}.$$

Since $m/u = 1 + |s|/u$,

$$(|r| + |s|) \log(m/u) \leq (|r| + |s|) \frac{|s|}{u} \leq 1,$$

and therefore $(m/u)^{|r|+|s|} \leq e$.

Because all masses α_b are nonnegative,

$$\mathbb{E}[\mathbf{m}_{k,r} \mathbf{m}_{k,s}] \leq e \sum_H A_H m^{-|H|} \prod_b \alpha_b^{r_b+s_b-H_b}.$$

The remaining overlap sum factors by coordinates:

$$\sum_H A_H m^{-|H|} \prod_b \alpha_b^{r_b+s_b-H_b} = \prod_b \sum_{h=0}^{\min(r_b, s_b)} \binom{r_b}{h} \binom{s_b}{h} h! m^{-h} \alpha_b^{r_b+s_b-h}.$$

For each b and each $0 \leq h \leq \min\{r_b, s_b\}$, one has $h \leq s_b$ and

$$\binom{r_b}{h} h! = (r_b)_h \leq r_b^h \leq |r|^h \leq |s|^h.$$

The assumption $m_1 > 0$ gives $m > 0$, and $\alpha_b \geq 0$. After rewriting $\alpha_b^{r_b+s_b-h} = \alpha_b^{r_b} \alpha_b^{s_b-h}$, the preceding coefficient bound gives

$$\binom{r_b}{h} \binom{s_b}{h} h! m^{-h} \alpha_b^{r_b+s_b-h} \leq \alpha_b^{r_b} \binom{s_b}{h} \left(\frac{|s|}{m}\right)^h \alpha_b^{s_b-h}.$$

The right-hand summands are nonnegative for every $0 \leq h \leq s_b$, so enlarging the summation range and applying the binomial theorem yields

$$\sum_{h=0}^{\min(r_b, s_b)} \binom{r_b}{h} \binom{s_b}{h} h! m^{-h} \alpha_b^{r_b+s_b-h} \leq \alpha_b^{r_b} \sum_{h=0}^{s_b} \binom{s_b}{h} \left(\frac{|s|}{m}\right)^h \alpha_b^{s_b-h} = \alpha_b^{r_b} \left(\alpha_b + \frac{|s|}{m}\right)^{s_b}.$$

Multiplying these coordinatewise bounds gives

$$\mathbb{E}[\mathbf{m}_{k,r} \mathbf{m}_{k,s}] \leq e \prod_b \alpha_b^{r_b} \prod_b \left(\alpha_b + \frac{|s|}{m}\right)^{s_b}.$$

The $\emptyset$ -coordinate contributes 1, since $r_\emptyset = s_\emptyset = 0$. Replacing b by $(a, y) \in \mathcal{C}$ and m by m_1 yields

$$\mathbb{E}_{P^{\otimes n}}[\mathbf{m}_{k,r} \mathbf{m}_{k,s}] \leq e \prod_{(a,y)} q_{aky}^{r_{ay}} \prod_{(a,y)} \left(q_{aky} + \frac{|s|}{m_1}\right)^{s_{ay}}.$$

□

Lemma 14. [lem:coefficient-sum-certificate] (Coefficient sum certificate) *Let M be a positive integer and let $z \geq 0$. With $g_{M,j}$ denoting the coefficients of the polynomial continuation G_M in Definition 35, define*

$$g_M^+(z) := \sum_{0 \leq j < M-1} |g_{M,j}| z^j.$$

Then

$$g_M^+(z) \leq 6^M \max\{1, z^{M-2}\}.$$

Proof of Lemma 14. All powers below have nonnegative integer exponents; in particular, the exponent $M - 2$ in the statement is interpreted as natural-number subtraction, so it equals 0 when $M = 1$. Let T_M be the Chebyshev polynomial and let G_M be the polynomial continuation used in Definition 35. The continuation has coefficients $g_{M,j}$ satisfying

$$T_M(1 - 2x) = 1 - 2M^2x + 2M^2x^2G_M(x).$$

For these coefficients, the closed form is

$$g_{M,j} = (-1)^j \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4} \quad (j \geq 0).$$

Because $M > 0$, the factor after $(-1)^j$ is nonnegative. Hence

$$|g_{M,j}| = \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4} = (-1)^j g_{M,j},$$

and therefore, for every z ,

$$g_M^+(z) = \sum_{0 \leq j < M-1} |g_{M,j}| z^j = \sum_{0 \leq j < M-1} g_{M,j} (-z)^j = G_M(-z).$$

Taking $z = 1$ in this identity and substituting $x = -1$ in the Chebyshev identity gives

$$T_M(3) = 1 + 2M^2 + 2M^2 g_M^+(1).$$

The sum defining $g_M^+(1)$ is nonnegative. Since $M \geq 1$, we have $M^2 \geq 1$, and the displayed identity implies

$$g_M^+(1) \leq T_M(3).$$

The Chebyshev recurrence and the standard positivity bound $T_m(x) \geq 1$ for $x \geq 1$ give, at $x = 3$,

$$T_0(3) = 1, \quad T_1(3) = 3, \quad T_{m+2}(3) = 6T_{m+1}(3) - T_m(3), \quad T_m(3) \geq 0.$$

Induction using these facts yields

$$T_M(3) \leq 6^M.$$

Consequently,

$$g_M^+(1) \leq 6^M.$$

Now split according to z . If $0 \leq z \leq 1$, then $z^j \leq 1$ for every index in the sum, so

$$g_M^+(z) = \sum_{0 \leq j < M-1} |g_{M,j}| z^j \leq \sum_{0 \leq j < M-1} |g_{M,j}| = g_M^+(1) \leq 6^M \leq 6^M \max\{1, z^{M-2}\}.$$

If $1 \leq z$, then every index in the finite sum satisfies $j \leq M - 2$ with the same natural-number subtraction convention. Hence $z^j \leq z^{M-2}$, and

$$g_M^+(z) \leq z^{M-2} \sum_{0 \leq j < M-1} |g_{M,j}| = z^{M-2} g_M^+(1) \leq 6^M z^{M-2} \leq 6^M \max\{1, z^{M-2}\}.$$

The two cases prove the claimed bound. □

[Lemma 13](#) gives a uniform product-moment bound for normalized falling-factorial monomials, and [Lemma 14](#) supplies the one-arm coefficient growth bound inherited by the polynomial continuation. These two estimates feed directly into the variance and covariance controls for light cells.

The aggregate heavy-cell statement packages the pilot, residual, and missing-arm controls into the rate needed for the hybrid estimator. Its target is the pilot-heavy contribution to the ATE decomposition.

Lemma 15. [lem:universal-heavy-cell-rate] (Heavy-cell aggregate rate) *For every overlap parameter ϵ with $0 < \epsilon < 1/2$, there exist constants $C_\epsilon > 0$ and $\rho_\epsilon > 0$, and a sample-size cutoff $N_\epsilon < \infty$, such that the following holds. For every n, d , every single-observation law $P \in \text{Prob}(\{1, \dots, d\} \times \{0, 1\}^2)$, and every sample law μ_n for $O_1, \dots, O_n$, suppose that*

- **(Sample size.)** $n \geq N_\epsilon$.
- **(Dimension range.)** $d \leq \rho_\epsilon n \log n$.
- **(Experiment class.)** $\mu_n \in \mathcal{E}_{n,d,\epsilon}$ for P , as in [Definition 4](#).

Then the mean squared error, under μ_n , of the heavy-cell aggregate ratio branch $\hat{T}_\mathcal{H} = \sum_{k \in \hat{\mathcal{H}}_n} \hat{h}_k$ around the pilot-heavy target $\sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k)$ satisfies

$$\int \left(\sum_{k \in \hat{\mathcal{H}}_n} \hat{h}_k - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right)^2 d\mu_n \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Proof of [Lemma 15](#). Write

$$L = \log(en), \quad m_j = |I_j| \quad (j = 0, 1),$$

and let N_0 denote the calibration cutoff in [Definition 35](#). For $n \geq N_0$, the implemented heavy set $\hat{\mathcal{H}}_n$ is the pilot set at threshold 256:

$$\hat{\mathcal{H}} = \{k : \lfloor 256L \rfloor < N_k^{(0)}\}.$$

The corresponding pilot-bad event is

$$\mathcal{B} = \left\{ \neg \left(\forall k \in \hat{\mathcal{H}}, \frac{256L}{2m_0} \leq p_k \right) \text{ or } \neg \left(\forall k \notin \hat{\mathcal{H}}, p_k \leq \frac{2 \cdot 256L}{m_0} \right) \right\}.$$

For every sample, the empirical arm ratios in $\hat{h}_k$ use totalized division, so the ratio is 0 when the corresponding arm count is zero; with this convention each such ratio lies in $[0, 1]$. Hence the absolute empirical ratio contribution of category k is at most $N_k^{(1)}/m_1$, and summing over the selected categories gives

$$|\hat{T}_\mathcal{H}| \leq \sum_{k \in \hat{\mathcal{H}}_n} \frac{N_k^{(1)}}{m_1} \leq \sum_k \frac{N_k^{(1)}}{m_1} = \frac{m_1}{m_1} \leq 1.$$

For an overlap law P , the category vector q_k from [Definition 7](#) lies in the overlap cone of [Definition 5](#). If $p_k = 0$, then $q_k = 0$ and [Definition 6](#) sets $\phi(q_k) = 0$. If $p_k > 0$, the arm masses in q_k are positive and the displayed formula in [Definition 6](#) gives

$$\phi(q_k) = p_k(\mu_{1k} - \mu_{0k}),$$

where the binary-outcome regressions μ_{0k} and μ_{1k} lie in $[0, 1]$. Thus, in both cases, $|\phi(q_k)| \leq p_k$. Since the category masses sum to one,

$$\left| \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right| \leq \sum_{k \in \hat{\mathcal{H}}_n} p_k \leq \sum_k p_k = 1.$$

Thus the squared heavy-cell error is bounded by 4 pointwise for the overlap laws considered below. The fixed-threshold pilot sandwich from [Lemma 1](#), applied with dimension constant 1, gives constants $C_0 > 0$ and N_{bad} such that, whenever $n \geq N_{\text{bad}}$ and $d \leq nL$,

$$P^{\otimes n}(\mathcal{B}) \leq C_0 n^{-4}.$$

With $C_{\text{bad}} = 4C_0$, the range bound gives

$$\int_{\mathcal{B}} \left(\hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right)^2 dP^{\otimes n} \leq C_{\text{bad}} n^{-4}.$$

Set

$$K_{\epsilon}^{\mathcal{H}} = \frac{16}{\epsilon} + 12 + \frac{1}{\epsilon^4}, \quad C_{\epsilon} = 4K_{\epsilon}^{\mathcal{H}} + C_{\text{bad}}, \quad \rho_{\epsilon} = 1,$$

and take

$$N_{\epsilon} = \max\{8, N_0, N_{\text{bad}}\}.$$

These constants satisfy the required positivity conditions.

Assume $n \geq N_{\epsilon}$, $d \leq \rho_{\epsilon} n \log n$, and $\mu_n \in \mathcal{E}_{n,d,\epsilon}$. The experiment-class assumption gives $\mu_n = P^{\otimes n}$ and ϵ -overlap for P . Also $n \geq 8$, $n \geq N_0$, $n \geq N_{\text{bad}}$, and, since $\log n \leq L$,

$$d \leq nL.$$

Let

$$B_{-} = \frac{256L}{2m_0}.$$

On $\mathcal{B}^c$, the first half of the pilot sandwich and the equality $\hat{\mathcal{H}}_n = \hat{\mathcal{H}}$ give

$$k \in \hat{\mathcal{H}}_n \implies B_{-} \leq p_k.$$

Fix a deterministic category set H satisfying $p_k \geq B_{-}$ for every $k \in H$, and fix $a \in \{0, 1\}$. Define

$$S_a(H) = \sum_{k \in H} \frac{N_k^{(1)}}{m_1} \frac{N_{ak1}^{(1)}}{N_{ak}^{(1)}} \mathbf{1}\{N_{ak}^{(1)} > 0\}, \quad \theta_a(H) = \sum_{k \in H} p_k \mu_{ak},$$

where the displayed ratio is taken to be 0 when $N_{ak}^{(1)} = 0$. The empirical category-mass centering is

$$\sum_{k \in H} \frac{N_k^{(1)}}{m_1} \mu_{ak}.$$

The pointwise decomposition about this centering is

$$S_a(H) - \sum_{k \in H} \frac{N_k^{(1)}}{m_1} \mu_{ak} = \frac{1}{m_1} \left[\sum_{k \in H} N_k^{(1)} \frac{\mathbf{1}\{N_{ak}^{(1)} > 0\}}{N_{ak}^{(1)}} \{N_{ak1}^{(1)} - \mu_{ak} N_{ak}^{(1)}\} - \sum_{k \in H} \mu_{ak} N_k^{(1)} \mathbf{1}\{N_{ak}^{(1)} = 0\} \right].$$

For the first bracket, write $N_{ak1}^{(1)} - \mu_{ak}N_{ak}^{(1)}$ as the sum over the second split of the centered outcome residual in category k and arm a . Expanding the square, all off-diagonal terms vanish: terms from distinct categories are pointwise disjoint when they use the same observation, and terms using different observations integrate to zero by conditioning on one outcome coordinate and using the centered residual. The remaining diagonal terms are bounded by replacing the squared residual with the category-arm indicator. For each k , the resulting nested-count integral is

$$\int (N_k^{(1)})^2 \frac{\mathbf{1}\{N_{ak}^{(1)} > 0\}}{N_{ak}^{(1)}} dP^{\otimes n} \leq \frac{2m_1 p_k}{\epsilon},$$

where overlap supplies the lower bound by ϵ on the arm probability inside category k . Summing over $k \in H$ yields

$$\int \left[\sum_{k \in H} N_k^{(1)} \frac{\mathbf{1}\{N_{ak}^{(1)} > 0\}}{N_{ak}^{(1)}} \{N_{ak1}^{(1)} - \mu_{ak}N_{ak}^{(1)}\} \right]^2 dP^{\otimes n} \leq \frac{2m_1}{\epsilon} \sum_{k \in H} p_k.$$

For the second bracket, $0 \leq \mu_{ak} \leq 1$, so its square is bounded by the square of the missing-arm count

$$\sum_{k \in H} N_k^{(1)} \mathbf{1}\{N_{ak}^{(1)} = 0\}.$$

Since $n \geq 8$, $m_1 \geq 3$. The exact multinomial expansion of this count has diagonal terms

$$\begin{aligned} & m_1 p_k \{1 - P(A = a \mid X = k)\} \{1 - p_k P(A = a \mid X = k)\}^{m_1 - 1} \\ & + (m_1)_2 [p_k \{1 - P(A = a \mid X = k)\}]^2 \{1 - p_k P(A = a \mid X = k)\}^{m_1 - 2} \end{aligned}$$

and cross terms, for $k \neq \ell$,

$$(m_1)_2 p_k \{1 - P(A = a \mid X = k)\} p_\ell \{1 - P(A = a \mid X = \ell)\} \{1 - p_k P(A = a \mid X = k) - p_\ell P(A = a \mid X = \ell)\}^{m_1 - 2}.$$

Overlap gives $p_k P(A = a \mid X = k) \geq \epsilon p_k$. With $u = ((m_1 - 2)/2)\epsilon$, the diagonal display is bounded by

$$m_1 p_k + m_1^2 \{p_k e^{-up_k}\}^2,$$

and each cross term is bounded by

$$m_1^2 \{p_k e^{-up_k}\} \{p_\ell e^{-up_\ell}\}.$$

The masses of categories in H are at least B_- , so the elementary envelope

$$\sum_{k \in H} p_k e^{-up_k} \leq \frac{|H|}{u^2 B_-}$$

applies

$$\int \left[\sum_{k \in H} \mu_{ak} N_k^{(1)} \mathbf{1}\{N_{ak}^{(1)} = 0\} \right]^2 dP^{\otimes n} \leq m_1 + m_1^2 \left[\frac{|H|}{\{((m_1 - 2)/2)\epsilon\}^2 B_-} \right]^2.$$

Together with $(x - y)^2 \leq 2x^2 + 2y^2$, the two bracket bounds give

$$\int \left(S_a(H) - \sum_{k \in H} \frac{N_k^{(1)}}{m_1} \mu_{ak} \right)^2 dP^{\otimes n} \leq \frac{4 \sum_{k \in H} p_k}{m_1 \epsilon} + \frac{2}{m_1} + 2 \left[\frac{|H|}{\{((m_1 - 2)/2)\epsilon\}^2 B_-} \right]^2.$$

For the empirical category-mass fluctuation, use the one-observation score

$$g_{a,H}(O) = \mu_{aX} \mathbf{1}\{X \in H\}.$$

Its second-split average is the empirical category-mass centering, its expectation is $\theta_a(H)$, and $0 \leq g_{a,H} \leq 1$. The iid bounded-score variance bound therefore gives

$$\int \left(\sum_{k \in H} \frac{N_k^{(1)}}{m_1} \mu_{ak} - \theta_a(H) \right)^2 dP^{\otimes n} \leq \frac{1}{m_1}.$$

Splitting $S_a(H) - \theta_a(H)$ at the empirical category-mass centering yields

$$\int \{S_a(H) - \theta_a(H)\}^2 dP^{\otimes n} \leq \frac{8 \sum_{k \in H} p_k}{m_1 \epsilon} + \frac{6}{m_1} + 4 \left[\frac{|H|}{\{((m_1 - 2)/2)\epsilon\}^2 B_-} \right]^2.$$

The probability-mass identities give

$$\sum_{k \in H} p_k \leq \sum_k p_k = 1,$$

and the finite-set cardinality bound gives $|H| \leq d$. The split-size calibration for $n \geq 8$ gives

$$\frac{n}{2} \leq m_1, \quad \frac{n}{8} \leq \frac{m_1 - 2}{2}, \quad \frac{256L}{n} \leq B_-,$$

and $\log n \leq L$. Hence

$$\frac{8 \sum_{k \in H} p_k}{m_1 \epsilon} + \frac{6}{m_1} + 4 \left[\frac{|H|}{\{((m_1 - 2)/2)\epsilon\}^2 B_-} \right]^2 \leq K_\epsilon^{\mathcal{H}} \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Let $V = K_\epsilon^{\mathcal{H}} \{n^{-1} + d^2/[n^2 (\log n)^2]\}$. The preceding deterministic bound applies uniformly to every fixed H whose elements have mass at least B_- . To pass to the random pilot-heavy set on $\mathcal{B}^c$, partition the pilot sample space into the fibers on which $\hat{\mathcal{H}}_n = H$. On each contributing fiber, the good-pilot implication gives $p_k \geq B_-$ for every $k \in H$. The fixed- H arm error is a function of the estimation split, while the fiber is determined by the pilot split, so the product law factors the restricted integral into the fiber probability times the fixed- H product-law integral. Summing over the eligible fibers and using that their probabilities sum to at most one gives, for each $a \in \{0, 1\}$,

$$\int_{\mathcal{B}^c} \{S_a(\hat{\mathcal{H}}_n) - \theta_a(\hat{\mathcal{H}}_n)\}^2 dP^{\otimes n} \leq V.$$

The heavy contribution and target decompose into their two arm components:

$$\hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) = \{S_1(\hat{\mathcal{H}}_n) - \theta_1(\hat{\mathcal{H}}_n)\} - \{S_0(\hat{\mathcal{H}}_n) - \theta_0(\hat{\mathcal{H}}_n)\}.$$

Thus $(x - y)^2 \leq 2x^2 + 2y^2$ gives

$$\int_{\mathcal{B}^c} \left(\hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right)^2 dP^{\otimes n} \leq 4K_\epsilon^{\mathcal{H}} \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

The bad-event contribution satisfies

$$\int_{\mathcal{B}} \left(\hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right)^2 dP^{\otimes n} \leq C_{\text{bad}} n^{-4} \leq C_{\text{bad}} \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\},$$

because $n^{-4} \leq n^{-1}$ for $n \geq 8$ and $n^{-1} \leq n^{-1} + d^2/[n^2(\log n)^2]$. Adding the integrals over $\mathcal{B}$ and $\mathcal{B}^c$, and using $\mu_n = P^{\otimes n}$, yields

$$\int \left(\hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right)^2 d\mu_n \leq (4K_{\epsilon}^{\mathcal{H}} + C_{\text{bad}}) \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\},$$

which is the claimed bound with the chosen C_{ϵ} . $\square$

Lemma 15 gives the heavy branch the same rate as the full fixed-interior theorem. Its conditions match the interior overlap and dimension range used in [Theorem 1](#).

The light-branch analysis separates within-category variance, cross-category covariance, false-light selection, and the final aggregate polynomial rate. The following statements give those components in the calibrated regime.

Lemma 16. [lem:factorial-moment-variance] (Factorial-polynomial variance bound) *Let P be a law on the observed-data alphabet and let k be a category. With the sample splits and the calibration $M(n)$ and $B(n)$ defined in [Definitions 16](#) and [35](#), write $\hat{P}_k$ for the factorial-moment polynomial estimate in [Definition 35](#) and write $\bar{P}_k$ for its sparse factorial-polynomial mean.*

Assume that

- **(Nonempty second split.)** $m_1 > 0$.
- **(Positive degree and scale.)** $M(n) > 0$ and $B(n) > 0$.
- **(Split-size calibration.)** $4M(n)^2 \leq m_1$.
- **(Light-cell mass bounds.)** *The category mass $p_k = P(X = k)$ satisfies*

$$p_k + \frac{4M(n)}{m_1} \leq B(n) \quad \text{and} \quad p_k \leq B(n).$$

Then

$$E_{P^{\otimes n}} \left[\left(\hat{P}_k - \bar{P}_k \right)^2 \right] \leq 8(e+1) \left\{ B(n) 6^{M(n)} \right\}^2.$$

Proof of Lemma 16. All expectations below are taken under the iid sequence law generated by P . Write $M = M(n)$, $B = B(n)$, $m_1 = |I_1|$, and $q = q_k$. The coefficients in the sparse expansion are the coefficients of the polynomial continuation G_M :

$$g_{M,j} = (-1)^j \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4}, \quad G_M(x) = \sum_{j=0}^{M-2} g_{M,j} x^j.$$

For $a \in \{0,1\}$, $c \in \{0,1\}^2$, and $0 \leq t \leq j \leq M-2$, let $r^{a,c,j,t}$ be the sparse multi-index determined by

$$u^{r^{a,c,j,t}} = u_c u_{a1} u_{a0}^t u_{a1}^{j-t}.$$

The sparse arm mean is

$$\bar{A}_{a,k} = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} B^{-j} \binom{j}{t} q^{r^{a,c,j,t}},$$

and the sparse polynomial mean is

$$\bar{P}_k = \bar{A}_{1,k} - \bar{A}_{0,k}.$$

For a nonnegative four-cell vector v , set

$$\mathcal{E}_a(v) = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} \left| B^{-1} g_{M,j} B^{-j} \binom{j}{t} \right| v^{r^{a,c,j,t}}.$$

The triangle inequality gives $|\bar{A}_{a,k}| \leq \mathcal{E}_a(q)$. We first record the envelope bound used for both q and its shifted version. If $v \geq 0$, $\sum_c v_c \leq B$, $M > 0$, and $B > 0$, then summing first over c and then using the binomial theorem gives

$$\mathcal{E}_a(v) = B^{-1} \left(\sum_c v_c \right) v_{a1} \sum_{j=0}^{M-2} |g_{M,j}| \left(\frac{v_{a0} + v_{a1}}{B} \right)^j.$$

The absolute-coefficient series satisfies, for every $z \geq 0$,

$$\sum_{j=0}^{M-2} |g_{M,j}| z^j \leq 6^M \max\{1, z^{M-2}\}.$$

Indeed, the displayed coefficient formula gives $|g_{M,j}| = (-1)^j g_{M,j}$, hence $\sum_j |g_{M,j}| = G_M(-1)$. The shifted-Chebyshev identity

$$T_M(1-2x) = 1 - 2M^2x + 2M^2x^2G_M(x)$$

at $x = -1$ gives $T_M(3) = 1 + 2M^2 + 2M^2G_M(-1)$, and since $G_M(-1) \geq 0$ and $M > 0$, this bounds $\sum_j |g_{M,j}|$ by $T_M(3)$. The Chebyshev recurrence at 3, together with nonnegativity of the preceding values, gives $T_M(3) \leq 6^M$. For $0 \leq z \leq 1$, monotonicity of the nonnegative terms gives the bound from the value at 1; for $z \geq 1$, the inequality $z^j \leq z^{M-2}$ for $j \leq M-2$ gives the displayed maximum factor.

In the envelope application, $z = (v_{a0} + v_{a1})/B$ lies in $[0, 1]$. Since $0 \leq v_{a1} \leq \sum_c v_c \leq B$,

$$\mathcal{E}_a(v) \leq B^{-1} \left(\sum_c v_c \right)^2 6^M \leq B6^M.$$

Applying this with $v = q$, using $q \geq 0$ and $\sum_c q_c = p_k \leq B$, gives

$$|\bar{P}_k| \leq |\bar{A}_{1,k}| + |\bar{A}_{0,k}| \leq 2B6^M, \quad \bar{P}_k^2 \leq 4(B6^M)^2.$$

The sample arm contribution has the parallel factorial-moment expansion

$$\hat{A}_{a,k} = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} B^{-j} \binom{j}{t} \frac{\prod_{b,y} (N_{bky}^{(1)})_{r_{by}^{a,c,j,t}}}{(m_1)_{|r^{a,c,j,t}|}}, \quad \hat{P}_k = \hat{A}_{1,k} - \hat{A}_{0,k}.$$

For a multi-index r , write

$$F_r = \frac{\prod_{b,y} (N_{bky}^{(1)})_{r_{by}}}{(m_1)_{|r|}}.$$

For two displayed terms with multi-indices r and s , the range restrictions give $|r| = j + 2 \leq M$ and $|s| = j' + 2 \leq M$. Suppose first that $|r| \leq |s|$. Expanding the product of factorial counts by the cellwise overlap vector η gives

$$\mathbb{E}[F_r F_s] = \sum_{\eta} C_{r,s}(\eta) \frac{(m_1)_{|r|+|s|-|\eta|}}{(m_1)_{|r|}(m_1)_{|s|}} \prod_{b,y} q_{by}^{r_{by}+s_{by}-\eta_{by}},$$

where the sum is over all four-cell vectors $\eta = (\eta_{by})$ with integer coordinates satisfying $0 \leq \eta_{by} \leq \min\{r_{by}, s_{by}\}$, where $|\eta| = \sum_{b,y} \eta_{by}$, and where

$$C_{r,s}(\eta) = \prod_{b,y} \binom{r_{by}}{\eta_{by}} \binom{s_{by}}{\eta_{by}} \eta_{by}!.$$

The falling-factorial ratio in this display obeys

$$\frac{(m_1)_{|r|+|s|-|\eta|}}{(m_1)_{|r|}(m_1)_{|s|}} \leq \frac{e}{m_1^{|\eta|}}.$$

If $|s| = 0$, then $|r| = |\eta| = 0$ and the bound is immediate. Otherwise, $4|s|^2 \leq m_1$ gives $0 < |s| < m_1$. The elementary bounds

$$(m_1)_{|r|+|s|-|\eta|} \leq m_1^{|r|+|s|-|\eta|}, \quad (m_1 - |s|)^{|r|} \leq (m_1)_{|r|}, \quad (m_1 - |s|)^{|s|} \leq (m_1)_{|s|}$$

give

$$\frac{(m_1)_{|r|+|s|-|\eta|}}{(m_1)_{|r|}(m_1)_{|s|}} \leq \frac{m_1^{|r|+|s|-|\eta|}}{(m_1 - |s|)^{|r|+|s|}} = \frac{\{m_1/(m_1 - |s|)\}^{|r|+|s|}}{m_1^{|\eta|}}.$$

Finally,

$$(|r| + |s|) \log \left(\frac{m_1}{m_1 - |s|} \right) \leq \frac{(|r| + |s|)|s|}{m_1 - |s|} \leq \frac{2|s|^2}{m_1 - |s|} \leq 1,$$

where the first inequality uses $\log x \leq x - 1$, the second uses $|r| \leq |s|$, and the last follows from $4|s|^2 \leq m_1$ together with $|s| \leq 2|s|^2$. Exponentiating gives the displayed ratio bound.

The coefficient bound $\binom{r_{by}}{\eta_{by}} \binom{s_{by}}{\eta_{by}} \eta_{by}! \leq \binom{s_{by}}{\eta_{by}} |s|^{\eta_{by}}$, applied cell by cell and summed with the binomial theorem, yields

$$\mathbb{E}[F_r F_s] \leq e q^r \left(q + \frac{|s|}{m_1} \mathbf{1} \right)^s.$$

When $|s| \leq |r|$, the same argument with r and s interchanged applies. Since $q \leq q + M\mathbf{1}/m_1$ coordinatewise and $|r|, |s| \leq M$, monotonicity of nonnegative monomials gives, in either case,

$$\mathbb{E}[F_r F_s] \leq e \left(q + \frac{M}{m_1} \mathbf{1} \right)^r \left(q + \frac{M}{m_1} \mathbf{1} \right)^s,$$

using the size condition $4M^2 \leq m_1$. The unweighted factorial monomial product has nonnegative expectation, and the product of the two displayed coefficients is bounded above by the product of their absolute values; hence

$$\begin{aligned} & \mathbb{E} \left[\left(B^{-1} g_{M,j} B^{-j} \binom{j}{t} F_{r^{a,c,j,t}} \right) \left(B^{-1} g_{M,j'} B^{-j'} \binom{j'}{t'} F_{r^{a,c',j',t'}} \right) \right] \\ & \leq e \left| B^{-1} g_{M,j} B^{-j} \binom{j}{t} \right| \left(q + \frac{M}{m_1} \mathbf{1} \right)^{r^{a,c,j,t}} \left| B^{-1} g_{M,j'} B^{-j'} \binom{j'}{t'} \right| \left(q + \frac{M}{m_1} \mathbf{1} \right)^{r^{a,c',j',t'}}. \end{aligned}$$

Expanding $\widehat{A}_{a,k}^2$ as the finite double sum, applying this estimate term by term, and reassembling the absolute-coefficient sums gives

$$\mathbb{E}[\widehat{A}_{a,k}^2] \leq e \mathcal{E}_a \left(q + \frac{M}{m_1} \mathbf{1} \right)^2.$$

The shifted vector is nonnegative and has total mass

$$\sum_{b,y} \left(q_{by} + \frac{M}{m_1} \right) = p_k + \frac{4M}{m_1} \leq B,$$

by the shifted light-cell mass condition. Applying the envelope bound to the shifted vector gives

$$\mathbb{E}[\widehat{A}_{a,k}^2] \leq e (B6^M)^2 \quad (a = 0, 1).$$

Using $(u - v)^2 \leq 2u^2 + 2v^2$ with $u = \widehat{A}_{1,k}$ and $v = \widehat{A}_{0,k}$,

$$\mathbb{E}[\widehat{P}_k^2] \leq 2\mathbb{E}[\widehat{A}_{1,k}^2] + 2\mathbb{E}[\widehat{A}_{0,k}^2] \leq 4e (B6^M)^2.$$

Finally, pointwise,

$$(\widehat{P}_k - \bar{P}_k)^2 \leq 2\widehat{P}_k^2 + 2\bar{P}_k^2,$$

because $(\widehat{P}_k + \bar{P}_k)^2 \geq 0$. Integrating and substituting the two bounds above gives

$$\mathbb{E}[(\widehat{P}_k - \bar{P}_k)^2] \leq 2 \cdot 4e (B6^M)^2 + 2 \cdot 4(B6^M)^2 = 8(e + 1)(B6^M)^2,$$

as required. $\square$

Lemma 17. [lem:factorial-moment-covariance] (Cross-cell factorial covariance) *Let P be a law on the finite observed-data alphabet, and let k and l be categories. Define $\bar{P}_k$ and $\bar{P}_l$ as the sparse factorial-polynomial means of the light-cell factorial estimates $\widehat{P}_k$ and $\widehat{P}_l$ from Definition 35. Suppose that*

- (*Distinct categories.*) $k \neq l$.
- (*Degree and scale.*) $M(n) > 0$ and $B(n) > 0$, with $M(n)$, $B(n)$, and m_1 as in Definitions 16 and 35.
- (*Split size.*) $4M(n)^2 \leq m_1$.
- (*Light-cell masses.*) $p_k \leq B(n)$ and $p_l \leq B(n)$.

Then

$$E_{P^{\otimes n}} \left[\left(\hat{P}_k - \bar{P}_k \right) \left(\hat{P}_l - \bar{P}_l \right) \right] \leq \frac{8M(n)^2}{m_1} \left\{ B(n) 6^{M(n)} \right\}^2.$$

Proof of Lemma 17. Let μ be the iid product law generated by P on the sample sequence, and abbreviate

$$M = M(n), \quad B = B(n), \quad m = m_1, \quad R = B 6^M, \quad \eta = \frac{2M^2}{m}.$$

Write the polynomial continuation as

$$G_M(x) = \sum_{j=0}^{M-2} g_{M,j} x^j.$$

For $h \in \{k, l\}$, let q_h be the four-cell vector of category h under P , and write $q_h^r = \prod_{b,y} (q_h)_{by}^{r_{by}}$. For an arm $a \in \{0, 1\}$, a cell $c \in \{0, 1\}^2$, and integers $0 \leq t \leq j \leq M-2$, let $r^{a,c,j,t}$ be the multi-index determined by

$$u^{r^{a,c,j,t}} = u_c u_{a1} u_{a0}^t u_{a1}^{j-t}.$$

The sparse arm contribution and its sparse mean are

$$\begin{aligned} \hat{A}_{a,h} &= \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} B^{-j} \binom{j}{t} \frac{\prod_{b,y} (N_{bhy}^{(1)})_{r_{by}^{a,c,j,t}}}{(m)_{|r^{a,c,j,t}|}}, \\ \bar{A}_{a,h} &= \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} B^{-1} g_{M,j} B^{-j} \binom{j}{t} q_h^{r^{a,c,j,t}}. \end{aligned}$$

Thus

$$\hat{P}_h = \hat{A}_{1,h} - \hat{A}_{0,h}, \quad \bar{P}_h = \bar{A}_{1,h} - \bar{A}_{0,h}.$$

The size condition $4M^2 \leq m$, together with $M > 0$, gives $M \leq m$. Since each displayed sparse multi-index has total degree $j+2 \leq M$, the factorial-moment identity gives, for $h = k, l$,

$$\int \hat{A}_{a,h} d\mu = \bar{A}_{a,h} \quad (a = 0, 1), \quad \int \hat{P}_h d\mu = \bar{P}_h.$$

Consequently, by expanding the centered product and integrating term by term,

$$\int (\hat{P}_k - \bar{P}_k)(\hat{P}_l - \bar{P}_l) d\mu = \int \hat{P}_k \hat{P}_l d\mu - \bar{P}_k \bar{P}_l.$$

For $h \in \{k, l\}$ and $a \in \{0, 1\}$, define the absolute arm envelope

$$\text{Env}_{a,h} = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} \left| B^{-1} g_{M,j} B^{-j} \binom{j}{t} \right| q_h^{r^{a,c,j,t}}.$$

Fix two sparse monomials with multi-indices r and s from categories k and l , and put

$$\hat{F}_{h,r} = \frac{\prod_{b,y} (N_{bhy}^{(1)})_{r_{by}}}{(m)_{|r|}}.$$

The product of the two normalized factorial counts is an average over ordered injective selections for the k -monomial and ordered injective selections for the l -monomial. Because $k \neq l$, an observation cannot satisfy both category patterns; hence exactly the disjoint pairs of selections contribute. There are $(m)_{|r|+|s|}$ such ordered pairs, and independence gives the same cell-mass product $q_k^r q_l^s$ for each pair. Therefore

$$\int \widehat{F}_{k,r} \widehat{F}_{l,s} d\mu = \Gamma_{r,s} q_k^r q_l^s, \quad \Gamma_{r,s} = \frac{(m)_{|r|+|s|}}{(m)_{|r|}(m)_{|s|}}.$$

For sparse terms, $|r| \leq M$ and $|s| \leq M$. We next bound the normalization factor. Suppose first that $|r| \leq |s|$. If $|s| = 0$, then $|r| = 0$ and $\Gamma_{r,s} = 1$. If $|s| > 0$, the condition $4|s|^2 \leq m$ follows from $|s| \leq M$ and $4M^2 \leq m$, and it implies $2|s| \leq m$. Using

$$\Gamma_{r,s} = \frac{(m - |r|)_{|s|}}{(m)_{|s|}},$$

we have $\Gamma_{r,s} \leq 1$. Also

$$\Gamma_{r,s} \geq \frac{(m - 2|s|)_{|s|}}{m_{|s|}} = \left(1 - \frac{2|s|}{m}\right)^{|s|} \geq 1 - \frac{2|s|^2}{m},$$

where the last step is Bernoulli's inequality, using $-1 \leq -2|s|/m$. Hence

$$|\Gamma_{r,s} - 1| \leq \frac{2|s|^2}{m} \leq \eta.$$

The case $|s| \leq |r|$ is identical after exchanging r and s , using the symmetry of $\Gamma_{r,s}$. Since the four-cell masses are nonnegative,

$$\left| \int \widehat{F}_{k,r} \widehat{F}_{l,s} d\mu - q_k^r q_l^s \right| \leq \eta q_k^r q_l^s, \quad |r|, |s| \leq M.$$

Multiplying by the two sparse coefficients, summing over the two finite arm expansions, and using the triangle inequality yields

$$\left| \int \widehat{A}_{a,k} \widehat{A}_{b,l} d\mu - \bar{A}_{a,k} \bar{A}_{b,l} \right| \leq \eta \text{Env}_{a,k} \text{Env}_{b,l}.$$

For $h = k, l$, the coordinates of q_h are nonnegative and sum to p_h . For fixed j , positivity of B and of the binomial coefficients gives

$$\begin{aligned} & \sum_{t=0}^j \sum_{c \in \{0,1\}^2} \left| B^{-1} g_{M,j} B^{-j} \binom{j}{t} \right| q_h^{a,c,j,t} \\ &= B^{-1} |g_{M,j}| B^{-j} \sum_{t=0}^j \sum_{c \in \{0,1\}^2} \binom{j}{t} (q_h)_c (q_h)_{a1} (q_h)_{a0}^t (q_h)_{a1}^{j-t} \\ &= B^{-1} |g_{M,j}| B^{-j} s(q_h) (q_h)_{a1} ((q_h)_{a0} + (q_h)_{a1})^j. \end{aligned}$$

Summing in j , and using $s_a(q_h) = (q_h)_{a0} + (q_h)_{a1}$, yields

$$\text{Env}_{a,h} = B^{-1} s(q_h) (q_h)_{a1} \sum_{j=0}^{M-2} |g_{M,j}| \left\{ \frac{s_a(q_h)}{B} \right\}^j.$$

The light-cell hypotheses give $p_h = s(q_h) \leq B$, while $(q_h)_{a1} \leq s(q_h)$ and $B > 0$. Hence

$$B^{-1}s(q_h)(q_h)_{a1} \leq B^{-1}s(q_h)^2 \leq B, \quad 0 \leq \frac{s_a(q_h)}{B} \leq 1.$$

It remains to record the coefficient envelope used in this display. For every $z \geq 0$,

$$\sum_{j=0}^{M-2} |g_{M,j}| z^j \leq 6^M \max\{1, z^{M-2}\}.$$

Indeed, the sign pattern of the explicit coefficients gives $\sum_{j=0}^{M-2} |g_{M,j}| = G_M(-1)$. The defining Chebyshev identity for the continuation, evaluated at -1 , gives

$$T_M(3) = 1 + 2M^2 + 2M^2 \sum_{j=0}^{M-2} |g_{M,j}|,$$

so nonnegativity of the coefficient sum implies $\sum_{j=0}^{M-2} |g_{M,j}| \leq T_M(3)$. The Chebyshev recurrence gives $T_M(3) \leq 6^M$. If $0 \leq z \leq 1$, termwise monotonicity gives $\sum |g_{M,j}| z^j \leq \sum |g_{M,j}| \leq 6^M$. If $z \geq 1$, then $j \leq M-2$ on the displayed range, so $\sum |g_{M,j}| z^j \leq z^{M-2} \sum |g_{M,j}| \leq 6^M z^{M-2}$. This proves the stated envelope. Taking $z = s_a(q_h)/B$ makes the maximum equal to 1, and therefore

$$\text{Env}_{a,h} \leq B 6^M = R.$$

Hence, for all $a, b \in \{0, 1\}$,

$$\left| \int \hat{A}_{a,k} \hat{A}_{b,l} d\mu - \bar{A}_{a,k} \bar{A}_{b,l} \right| \leq \eta R^2.$$

Define

$$C_{ab} = \int \hat{A}_{a,k} \hat{A}_{b,l} d\mu - \bar{A}_{a,k} \bar{A}_{b,l}.$$

Expanding the two arm contrasts gives

$$\int \hat{P}_k \hat{P}_l d\mu - \bar{P}_k \bar{P}_l = C_{11} - C_{10} - C_{01} + C_{00}.$$

Therefore

$$\left| \int \hat{P}_k \hat{P}_l d\mu - \bar{P}_k \bar{P}_l \right| \leq |C_{11}| + |C_{10}| + |C_{01}| + |C_{00}| \leq 4\eta R^2 = \frac{8M^2}{m} \{B 6^M\}^2.$$

Combining this bound with the centering identity and using $x \leq |x|$ yields

$$\int (\hat{P}_k - \bar{P}_k)(\hat{P}_l - \bar{P}_l) d\mu \leq \frac{8M(n)^2}{m_1} \{B(n)6^{M(n)}\}^2.$$

□

Lemma 18. `[lem:false-light-envelope]` (False-light error envelope) *Let P be a probability law on $\{1, \dots, d\} \times \{0, 1\}^2$. Suppose that:*

- **(Overlap.)** For every category k with $p_k > 0$, the treatment probability satisfies

$$\epsilon \leq \pi_k \leq 1 - \epsilon.$$

- **(Calibration cutoff.)** The sample size satisfies $n \geq N_0$, where N_0 is the least numerical cutoff satisfying the calibration inequalities used in the estimator.
- **(Logarithmic scale.)** With $L = \log(en)$, one has $L \geq 240$.

For the pilot-light indicator $S_k = \mathbf{1}\{k \in \hat{\mathcal{L}}_n\}$, the selected false-light error over categories whose population mass exceeds the light-cell threshold satisfies

$$E_{P^{\otimes n}} \left[\left\{ \sum_{\substack{k \in \{1, \dots, d\} \\ p_k > B(n)/4}} S_k (\hat{P}_k - \phi(q_k)) \right\}^2 \right] \leq 8(e+1)8192^2 \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

Proof of Lemma 18. All expectations below are under the iid infinite product law generating the observation sequence, with each statistic evaluated on its first n coordinates. Let

$$L = \log(en), \quad \ell = \log n, \quad m_j = |I_j| \ (j = 0, 1), \quad B = B(n), \quad M = M(n), \quad G = 6^{2M}.$$

The deterministic split construction of Definition 16 gives

$$m_0 = \lfloor n/2 \rfloor, \quad m_1 = n - \lfloor n/2 \rfloor.$$

By Lemma 7,

$$0 < \ell, \quad \ell \leq L, \quad L \leq 2\ell, \quad m_1 > 0, \quad n \leq 2m_1, \quad M \leq L,$$

$$B \leq \frac{8192L}{n}, \quad G^2 L^6 \leq n.$$

In particular $n > 0$. Since $B = 4096L/m_1$, $L > 0$, and $m_1 > 0$, also $B > 0$, and hence

$$B^2 \leq \frac{8192^2 L^2}{n^2}.$$

Because $n > 0$, $L = 1 + \ell$. Since $L \geq 240$,

$$\exp(-49L) \leq \exp(-L) = \frac{1}{en} \leq \frac{1}{n}.$$

Moreover $G \geq 1$, so $L^4 \leq L^6 \leq G^2 L^6 \leq n$. Together with $\ell \leq L$, this gives

$$L^2 \ell^2 \leq L^4 \leq n.$$

Therefore

$$\frac{d^2 L^2}{n^3} \leq \frac{d^2}{n^2 \ell^2} \leq \frac{1}{n} + \frac{d^2}{n^2 \ell^2}.$$

Let

$$F = \{k : p_k > B/4\}, \quad S_k = \mathbf{1}\{k \in \hat{\mathcal{L}}_n\}, \quad e_k = S_k(\hat{P}_k - \phi(q_k)).$$

By the definition of the false-light error, the selected false-light error is $\sum_{k \in F} e_k$. For each sample, Cauchy's inequality gives

$$\left(\sum_{k \in F} e_k \right)^2 \leq |F| \sum_{k \in F} e_k^2.$$

Fix $k \in F$. The cutoff defining N_0 in [Definition 35](#) and the assumption $n \geq N_0$ give

$$\alpha_0 L \geq 2, \quad 4M^2 \leq m_1, \quad \frac{4M}{m_1} \leq \frac{3B}{4}.$$

Since $M \geq 2$, the bound $4M^2 \leq m_1$ gives $16 \leq m_1$. With the split identities above, this implies $m_0 > 0$ and $m_1 \leq 2m_0$.

Write

$$G_M(x) = \sum_{j=0}^{M-2} g_{M,j} x^j, \quad g_{M,j} = (-1)^j \frac{2^{2j+3}}{M(M+j+2)} \binom{M+j+2}{2j+4}.$$

Let $\mathbf{e}_{by}$ denote the unit multi-index at the four-cell coordinate $(b, y) \in \{0, 1\}^2$. In the next display, ay is a single four-cell label, say $ay = (b, y')$, and $\mathbf{e}_{ay}$ is the corresponding unit multi-index. For $a \in \{0, 1\}$, $ay \in \{0, 1\}^2$, and $0 \leq t \leq j \leq M-2$, set

$$r(a, ay, j, t) = \mathbf{e}_{ay} + \mathbf{e}_{a1} + t\mathbf{e}_{a0} + (j-t)\mathbf{e}_{a1}, \quad |r(a, ay, j, t)| = j+2.$$

For a multi-index r , the factorial-moment monomial is

$$\mathbf{m}_{k,r} = \frac{\prod_{(b,y) \in \{0,1\}^2} (N_{bky}^{(1)})_{r_{by}}}{(m_1)_{|r|}},$$

where $(z)_s$ denotes the falling factorial. The a -arm sparse factorial-moment contribution is

$$\hat{P}_{k,a}^{\text{sp}} = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{ay \in \{0,1\}^2} B^{-1} g_{M,j} B^{-j} \binom{j}{t} \mathbf{m}_{k,r(a,ay,j,t)}.$$

By [Definition 35](#), $\hat{P}_k = \sum_{|r| \leq M} c_r \mathbf{m}_{k,r}$ in collected coefficient form, where each c_r is obtained by expanding the two arms of $\hat{P}_{M,B}$ through the coefficients $g_{M,j}$ and the binomial expansion of the arm masses, and summing the resulting sparse coefficients $c_u = B^{-1} g_{M,j} B^{-j} \binom{j}{t}$ over the sparse indices $u = (j, t, ay)$ with $r_u = r$. Since $r \mapsto \mathbf{m}_{k,r}$ enters linearly in each coefficient, regrouping this finite double sum arm by arm yields

$$\hat{P}_k = \hat{P}_{k,1}^{\text{sp}} - \hat{P}_{k,0}^{\text{sp}}.$$

For a nonnegative four-cell vector v , define the absolute sparse arm envelope

$$\mathcal{A}_a(v) = \sum_{j=0}^{M-2} \sum_{t=0}^j \sum_{ay \in \{0,1\}^2} \left| B^{-1} g_{M,j} B^{-j} \binom{j}{t} \right| v^{r(a,ay,j,t)}, \quad v^r = \prod_{(b,y) \in \{0,1\}^2} v_{by}^{r_{by}}.$$

Set

$$v_{ay}^{(k)} = q_{k,ay} + \frac{M}{m_1}.$$

This vector is nonnegative. Fix two sparse indices $u = (j, t, ay)$ and $u' = (j', t', ay')$ with $0 \leq t \leq j \leq M-2$ and $0 \leq t' \leq j' \leq M-2$. Both multi-indices $r_u, r_{u'}$ have degree at most M , so $4 \max\{|r_u|, |r_{u'}|\}^2 \leq 4M^2 \leq m_1$. Apply [Lemma 13](#) to the degree-ordered pair. Its unshifted monomial factor is bounded coordinatewise by $(v^{(k)})^r$, since $q_{k,cy} \leq q_{k,cy} + M/m_1$. Its shifted monomial factor is also bounded coordinatewise by the corresponding power of $v^{(k)}$: if s is the larger-degree multi-index in the ordered pair, then $|s| \leq M$, and hence

$$q_{k,cy} + \frac{|s|}{m_1} \leq q_{k,cy} + \frac{M}{m_1} = v_{cy}^{(k)} \quad \text{for every coordinate } cy.$$

Using these two coordinatewise bounds, and swapping u, u' before applying [Lemma 13](#) when the opposite degree order holds, gives

$$E[(c_u \mathbf{m}_{k,r_u})(c_{u'} \mathbf{m}_{k,r_{u'}})] \leq e |c_u| (v^{(k)})^{r_u} |c_{u'}| (v^{(k)})^{r_{u'}},$$

where $c_u = B^{-1} g_{M,j} B^{-j} \binom{j}{t}$ and $r_u = r(a, ay, j, t)$. Expanding the square of $\widehat{P}_{k,a}^{\text{sp}}$, summing this bound over all ordered pairs of sparse indices, and collecting the two sums into the envelope gives

$$E[(\widehat{P}_{k,a}^{\text{sp}})^2] \leq e \mathcal{A}_a(v^{(k)})^2, \quad a = 0, 1.$$

The shifted vector has total mass

$$\sum_{ay} v_{ay}^{(k)} = p_k + \frac{4M}{m_1} \leq p_k + \frac{3B}{4} \leq B \left(1 + \frac{p_k}{B}\right).$$

Let $R = \sum_{ay} v_{ay}$, $A_a(v) = v_{a0} + v_{a1}$, and

$$g_M^+(z) = \sum_{j=0}^{M-2} |g_{M,j}| z^j.$$

For every nonnegative v with $R \leq Bw$ and $w \geq 1$, the binomial summation in the sparse envelope gives

$$\mathcal{A}_a(v) = B^{-1} R v_{a1} g_M^+ \left(\frac{A_a(v)}{B} \right),$$

and, since $0 \leq A_a(v)/B \leq R/B \leq w$ with $w \geq 1$, [Lemma 14](#) gives

$$g_M^+ \left(\frac{A_a(v)}{B} \right) \leq 6^M \max \left\{ 1, \left(\frac{A_a(v)}{B} \right)^{M-2} \right\} \leq 6^M w^{M-2}.$$

Since $v_{a1} \leq R \leq Bw$, these displays imply

$$\mathcal{A}_a(v) \leq B 6^M w^M.$$

Applying this with $v = v^{(k)}$ and $w = 1 + p_k/B$ yields

$$\mathcal{A}_a(v^{(k)}) \leq B 6^M \left(1 + \frac{p_k}{B} \right)^M, \quad a = 0, 1.$$

Since $\widehat{P}_k$ is the difference of the two sparse-arm contributions, $(x - y)^2 \leq 2x^2 + 2y^2$ yields

$$E[\widehat{P}_k^2] \leq 4e \left\{ B 6^M \left(1 + \frac{p_k}{B} \right)^M \right\}^2.$$

Because $k \in F$ and $B > 0$, $p_k > 0$. The overlap condition and [Definition 6](#) give $|\phi(q_k)| \leq p_k$. Since $p_k/B \geq 0$ and $M \geq 2$,

$$\frac{p_k}{B} \leq 6^M \left(1 + \frac{p_k}{B}\right)^M,$$

and therefore

$$\phi(q_k)^2 \leq \left\{ B 6^M \left(1 + \frac{p_k}{B}\right)^M \right\}^2.$$

Using $(x - t)^2 \leq 2x^2 + 2t^2$,

$$E\left[(\hat{P}_k - \phi(q_k))^2\right] \leq 8(e + 1)B^2 \left\{ 6^M \left(1 + \frac{p_k}{B}\right)^M \right\}^2.$$

The indicator S_k is determined by the pilot split, while $\hat{P}_k - \phi(q_k)$ is computed from the estimation split and a fixed target. The split factorization gives

$$E\left[S_k^2(\hat{P}_k - \phi(q_k))^2\right] = E[S_k^2]E\left[(\hat{P}_k - \phi(q_k))^2\right].$$

Since $S_k \in \{0, 1\}$, $E[S_k^2] = E[S_k]$. The indicator integral is the probability of the pilot-light event:

$$E[S_k] = P^{\otimes n}\{k \in \hat{\mathcal{L}}_n\}.$$

For $n \geq N_0$, the event $\{k \in \hat{\mathcal{L}}_n\}$ is contained in

$$\left\{ N_k^{(0)} \leq 256L \right\}.$$

The false-light condition $p_k > B/4$, the identity $B = 4096L/m_1$, and $m_1 \leq 2m_0$ imply

$$2(256L) < m_0 p_k.$$

The pilot lower-tail bound for this threshold gives

$$E[S_k] \leq \exp\left(-\frac{m_0 p_k}{8}\right).$$

Since $p_k/B > 1/4$,

$$1 + \frac{p_k}{B} \leq 5\frac{p_k}{B}, \quad 6\left(1 + \frac{p_k}{B}\right) \leq 30\frac{p_k}{B}.$$

Also $6(1 + p_k/B) \geq 1$, so $\log(6(1 + p_k/B)) \geq 0$; and from $\log x \leq x - 1$ for $x > 0$,

$$\log\left(6\left(1 + \frac{p_k}{B}\right)\right) \leq 30\frac{p_k}{B}.$$

Together with $M \leq L$, $B = 4096L/m_1$, and $m_1 \leq 2m_0$, this yields

$$2M \log\left(6\left(1 + \frac{p_k}{B}\right)\right) \leq 60L \frac{p_k}{B}, \quad 256L \frac{p_k}{B} \leq \frac{m_0 p_k}{8}.$$

Consequently

$$-\frac{m_0 p_k}{8} + 2M \log\left(6\left(1 + \frac{p_k}{B}\right)\right) \leq -196L \frac{p_k}{B} \leq -49L,$$

and hence

$$\exp\left(-\frac{m_0 p_k}{8}\right) \left\{ 6^M \left(1 + \frac{p_k}{B}\right)^M \right\}^2 \leq \exp(-49L).$$

Combining the preceding displays gives the single-cell bound

$$E[e_k^2] \leq 8(e+1)B^2 \exp(-49L).$$

Since $|F| \leq d$, integrating the pointwise Cauchy bound and applying the single-cell estimate gives

$$E \left[\left(\sum_{k \in F} e_k \right)^2 \right] \leq |F| \sum_{k \in F} 8(e+1)B^2 \exp(-49L) \leq d^2 8(e+1)B^2 \exp(-49L).$$

Using the bounds on B^2 , $\exp(-49L)$, and the rate algebra above,

$$d^2 8(e+1)B^2 \exp(-49L) \leq 8(e+1)8192^2 \frac{d^2 L^2}{n^3} \leq 8(e+1)8192^2 \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

□

Lemma 19. [lem:light-cell-polynomial] (Light-cell polynomial rate) *Fix the overlap parameter ϵ with $0 < \epsilon \leq 1/2$. The following conditions hold:*

- **(Overlap level.)** *The fixed overlap parameter satisfies $0 < \epsilon$ and $\epsilon \leq 1/2$.*
- **(Computability.)** *The hybrid estimator admits a real-arithmetic implementation: there is a universal operation-count constant $K \in \mathbb{N}$, $K > 0$, such that for every n and d the program reads the split-count vector, returns $\hat{\tau}_n^{\text{hyb}}$ exactly, and uses at most $K d M(n)^4$ arithmetic and comparison operations.*
- **(Constants.)** *There are constants $C_\epsilon, c_\epsilon > 0$.*
- **(Experiment and dimension.)** *For every n, d , every single-observation law $P \in \text{Prob}(\{1, \dots, d\} \times \{0, 1\}^2)$, and every sample law μ_n satisfying $\mu_n \in \mathcal{E}_{n,d,\epsilon}$ as in [Definition 4](#), the dimension satisfies*

$$d \leq c_\epsilon n \log n.$$

Under these conditions, the light-cell contribution $\hat{T}_{\mathcal{L}}$ satisfies

$$E_{\mu_n} \left[\left\{ \hat{T}_{\mathcal{L}} - \sum_{k \in \hat{\mathcal{L}}_n} \phi(q_k) \right\}^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

Proof of Lemma 19. The computability assertion uses the real-arithmetic program model in which an instruction is an input read, a constant, one of the arithmetic operations $+$, $-$, $\times$, $/$, or a nonpositive comparison branch. Input and constant instructions have cost zero, and each arithmetic or branch instruction has cost one. The arithmetic is total: division by zero is interpreted as the real value 0. A branch node is compiled as an expression-tree node containing its test and both alternatives, so its operation count is the count of the test, plus the counts of both alternatives, plus one for the branch instruction. A finite list sum $x_1 + \dots + x_s$ is compiled

recursively as $x_1 + (x_2 + \dots + (x_s + 0) \dots)$; hence its cost is the sum of the costs of the listed expressions plus one addition for each list entry. The corresponding list product is compiled with a terminal factor 1 and contributes one multiplication for each list entry.

The falling-factorial factors use the guarded natural-subtraction expression

$$x \dot{-} i = \begin{cases} 0, & x - i \leq 0, \\ x - i, & x - i > 0, \end{cases}$$

implemented by one nonpositive branch with test $x - i$ and alternatives 0 and $x - i$. On natural-valued count inputs this evaluates to the truncated difference, and its operation count is $2 \text{ ops}(x) + 3$. Consequently the recursive expression $(x)_0 = 1$, $(x)_{r+1} = (x \dot{-} r)(x)_r$ has cost $r\{2 \text{ ops}(x) + 4\}$. A split-count read has cost zero, so a falling-factorial expression of a split count of order r costs $4r$. Thus a factorial monomial of multi-degree $|r|$, including the final division by $(m_1)_{|r|}$, costs

$$4|r| + 5.$$

For a light-polynomial term indexed by (j, t, a, y) , the multi-degree is $j + 2$, and multiplication by its coefficient gives cost $4j + 14$. Using the list-sum convention above, summing the four cell terms costs at most $50M(n)$, summing over $t = 0, \dots, j$ costs at most $51M(n)^2$, and summing over $j = 0, \dots, M(n) - 2$ gives one treatment-arm factorial-polynomial expression with cost at most $52M(n)^3$. The two treatment arms and their subtraction therefore give the light-cell expression cost

$$\text{ops}\{\text{one light category}\} \leq 105M(n)^4.$$

The heavy ratio expression has cost 11. The selected-category expression is defined by cases. If $n < N_0$, it is the heavy ratio expression; in this range $\hat{\mathcal{H}}_n$ is the full category set and $\hat{\mathcal{L}}_n = \emptyset$, so this expression is exact for the selected contribution. Its cost is $11 \leq 110M(n)^4$. If $n \geq N_0$, the expression branches on the first-split category count minus the threshold constant. The category-count expression is the list sum of the four split-count reads and costs four additions, the subtraction of the threshold contributes one more operation, the branch contains both the light expression and the heavy ratio expression, and the branch instruction itself contributes one operation. Since $M(n) \geq 2$, this selected-category branch also satisfies

$$\text{ops}\{\text{one selected category}\} \leq 110M(n)^4.$$

Summing these selected-category expressions over d categories uses the same recursive list sum and adds d summation nodes. The resulting expression computes $\hat{T}_{\mathcal{H}} + \hat{T}_{\mathcal{L}}$ exactly, hence

$$\text{ops}\{\hat{T}_{\mathcal{H}} + \hat{T}_{\mathcal{L}}\} \leq 111dM(n)^4.$$

For $d > 0$, the final clamp to $[-1, 1]$, as displayed in [Definition 35](#), has cost

$$4 \text{ ops}\{\hat{T}_{\mathcal{H}} + \hat{T}_{\mathcal{L}}\} + 6 \leq 450dM(n)^4.$$

For $d = 0$, the compiled program is the zero-cost constant zero, which agrees with the empty-category hybrid estimator. Thus, for every n, d , there is a real-arithmetic program reading the hybrid split-count vector, returning $\hat{\tau}_n^{\text{hyb}}$ exactly on every sample, and satisfying

$$\text{ops}\{\hat{\tau}_n^{\text{hyb}}\} \leq 450dM(n)^4.$$

The computability clause therefore holds with the universal constant $K = 450$.

For the light-cell risk bound, fix n, d, P and a sample law μ_n satisfying the hypotheses of the lemma. By the experiment-class hypothesis in [Definition 4](#), $\mu_n = P^{\otimes n}$ and P satisfies ϵ -overlap. In the endpoint case $n = 0$, the displayed rate is interpreted with the same total real convention as the arithmetic model: real division by zero is 0 and $\log 0 = 0$. In the calibrated range used below, $n > 0$ and $\log n > 0$, so the displayed rate has its usual analytic meaning. Set

$$R_{n,d} := \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

Choose

$$K_c = 4(e+1)8192^2 + 16\,8192^2, \quad K_b = \left(\frac{65536}{\epsilon\alpha_0^2}\right)^2, \quad K_f = 8(e+1)8192^2,$$

and let

$$C_\epsilon = 3(K_c + K_b + K_f), \quad c_\epsilon = 1.$$

The inequalities $\epsilon > 0$ and $\alpha_0 > 0$ make these constants positive. The estimates below are uniform over all d , so the dimension-range hypothesis is accommodated by this choice of c_ϵ .

First suppose $n \geq N_0$. Write

$$L = \log(en), \quad \ell = \log n, \quad M = M(n), \quad B = B(n), \quad m_0 = |I_0|, \quad m_1 = |I_1|,$$

and define the genuinely light and false-light population sets, together with the pilot light indicator,

$$G = \{k : p_k \leq B/4\}, \quad F = \{k : p_k > B/4\}, \quad S_k = \mathbf{1}\{k \in \hat{\mathcal{L}}_n\}.$$

Let $\bar{P}_k = P_{M,B}(q_k)$ be the sparse factorial-polynomial mean of $\hat{P}_k$. Every multi-index r appearing in the expansion has $|r| \leq M$, and the cutoff inequalities give $|r| \leq M \leq m_1$. Hence the ordered-selection factorial-moment identity applies with the required denominator order:

$$\mathbb{E}_{P^{\otimes \infty}} \left[\frac{\prod_{a,y} (N_{aky}^{(1)})_{r_{ay}}}{(m_1)_{|r|}} \right] = q_k^r.$$

Substituting this identity into the polynomial expansion of $\hat{P}_k$, equivalently applying [Lemma 8](#), gives

$$\mathbb{E}_{P^{\otimes \infty}}[\hat{P}_k] = \bar{P}_k = P_{M,B}(q_k).$$

Put

$$W_{\text{var}} = \sum_{k \in G} S_k(\hat{P}_k - \bar{P}_k), \quad W_{\text{app}} = \sum_{k \in G} S_k(\bar{P}_k - \phi(q_k)), \quad W_{\text{sel}} = \sum_{k \in F} S_k(\hat{P}_k - \phi(q_k)).$$

Partitioning the selected light categories into G and F , and adding and subtracting $\bar{P}_k$ on G , gives the pointwise identity

$$\hat{T}_{\mathcal{L}} - \sum_{k \in \hat{\mathcal{L}}_n} \phi(q_k) = W_{\text{var}} + W_{\text{app}} + W_{\text{sel}}.$$

The cutoff property in [Definition 35](#) gives $2 \leq \alpha_0 L$, $4M^2 \leq m_1$, and $4M/m_1 \leq 3B/4$. The numerical definition of α_0 gives $\alpha_0 \leq 1/120$, since $D_A = 8 \log(27/4) > 8$ and $(256D_A)^{-1} \leq 1/120$. Combining this with $2 \leq \alpha_0 L$ yields $L \geq 240$. Therefore [Lemma 7](#) supplies

$$0 < \ell, \quad \ell \leq L, \quad n \leq 2m_1, \quad M \leq L, \quad B \leq \frac{8192L}{n}, \quad \Gamma^2 L^6 \leq n,$$

where $\Gamma = 6^{2M}$, and also $\Gamma \geq 1$. For every $k \in G$, the cutoff shift gives

$$p_k + \frac{4M}{m_1} \leq B.$$

Thus [Lemma 16](#) gives, for $k \in G$,

$$\int (\hat{P}_k - \bar{P}_k)^2 dP^{\otimes \infty} \leq 8(e+1)(B6^M)^2.$$

For distinct $k, l \in G$, [Lemma 17](#) gives

$$\int (\hat{P}_k - \bar{P}_k)(\hat{P}_l - \bar{P}_l) dP^{\otimes \infty} \leq \frac{8M^2}{m_1}(B6^M)^2.$$

The pilot indicators depend on the pilot split, while the centered polynomial products depend on the estimation split; their pair expectation factors from the centered product integral and lies in $[0, 1]$. Expanding the square over any deterministic $\mathcal{S} \subseteq G$ yields

$$\int \left\{ \sum_{k \in \mathcal{S}} S_k (\hat{P}_k - \bar{P}_k) \right\}^2 dP^{\otimes \infty} \leq |\mathcal{S}| 8(e+1)(B6^M)^2 + |\mathcal{S}|^2 \frac{8M^2}{m_1} (B6^M)^2.$$

Taking $\mathcal{S} = G$ and using $|G| \leq d$, the diagonal rate conversion follows from

$$(L^2 \Gamma \ell^2)^2 = L^4 \Gamma^2 \ell^4 \leq n \ell^2,$$

where the inequality uses $\ell \leq L$ and $\Gamma^2 L^6 \leq n$. Thus $2uv \leq u^2 + v^2$, with $u = d$ and $v = L^2 \Gamma \ell^2$, gives

$$\frac{dL^2 \Gamma}{n^2} \leq \frac{1}{2} \left\{ \frac{1}{n} + \frac{d^2}{n^2 \ell^2} \right\} = \frac{1}{2} R_{n,d}.$$

The cross-rate conversion uses

$$L^4 \Gamma \ell^2 \leq L^4 \Gamma^2 L^2 = \Gamma^2 L^6 \leq n,$$

and hence

$$\frac{d^2 L^4 \Gamma}{n^3} \leq \frac{d^2}{n^2 \ell^2} \leq R_{n,d}.$$

Combining these algebraic inequalities with the preceding raw second-moment bound yields

$$\int W_{\text{var}}^2 dP^{\otimes \infty} \leq K_c R_{n,d}.$$

For the deterministic approximation term, overlap gives $q_k \in \mathcal{C}_\epsilon$, and $k \in G$ gives $s(q_k) = p_k \leq B/4 \leq B$. Applying [Lemma 10](#) to the four-cell vector q_k gives

$$|\bar{P}_k - \phi(q_k)| = |P_{M,B}(q_k) - \phi(q_k)| \leq \frac{2B}{\epsilon M^2} \quad (k \in G).$$

Consequently

$$|W_{\text{app}}| \leq d \frac{2B}{\epsilon M^2}.$$

The cutoff inequality $2 \leq \alpha_0 L$ gives $M \geq \alpha_0 L/2$, while [Lemma 7](#) gives $B \leq 8192L/n$ and $\ell \leq L$. Therefore

$$\frac{2B}{\epsilon M^2} \leq \frac{65536}{\epsilon \alpha_0^2 n L} \leq \frac{65536}{\epsilon \alpha_0^2 n \ell},$$

and hence, pointwise in the sample,

$$W_{\text{app}}^2 \leq \left(\frac{65536}{\epsilon \alpha_0^2} \right)^2 \frac{d^2}{n^2 \ell^2} \leq K_b R_{n,d}.$$

For the false-light term, observe that $W_{\text{sel}} = \sum_{k \in F} S_k(\hat{P}_k - \phi(q_k))$ is exactly the selected false-light error of [Lemma 18](#): the summation there ranges over the categories with $p_k > B/4$, which is the set F . Its hypotheses hold in the present case: the overlap condition $\epsilon \leq \pi_k \leq 1 - \epsilon$ holds for every category of positive mass because $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, the calibration cutoff $n \geq N_0$ is the standing assumption of this case, and $L \geq 240$ was derived above from the cutoff property and the numerical value of α_0 . Since W_{sel} is a function of the first n observations, its second moment under $P^{\otimes \infty}$ equals its second moment under $P^{\otimes n}$, and [Lemma 18](#) gives directly

$$\int W_{\text{sel}}^2 dP^{\otimes \infty} = E_{P^{\otimes n}}[W_{\text{sel}}^2] \leq 8(e+1)8192^2 \left\{ \frac{1}{n} + \frac{d^2}{n^2 \ell^2} \right\} = K_f R_{n,d}.$$

The bound for W_{var} is obtained on the infinite-product representation of the iid sample and transported to $P^{\otimes n}$ through equality of integrals for functions of the first n observations. Using $(x+y+z)^2 \leq 3(x^2+y^2+z^2)$, the pointwise bound on W_{app}^2 , and the two second-moment bounds, we obtain

$$\begin{aligned} \int \left\{ \hat{T}_{\mathcal{L}} - \sum_{k \in \hat{\mathcal{L}}_n} \phi(q_k) \right\}^2 dP^{\otimes n} &\leq 3 \int (W_{\text{var}}^2 + W_{\text{app}}^2 + W_{\text{sel}}^2) dP^{\otimes n} \\ &\leq 3(K_c + K_b + K_f) R_{n,d} = C_\epsilon R_{n,d}. \end{aligned}$$

Since $\mu_n = P^{\otimes n}$, this is the desired bound under μ_n .

If $n < N_0$, then $\hat{\mathcal{L}}_n = \emptyset$ by [Definition 35](#). The light contribution and its selected target are both zero, so the left-hand side is zero. Since $R_{n,d} \geq 0$, the same bound holds in this case. $\square$

[Lemmas 16](#) and [17](#) control the second moments of the sparse factorial-polynomial terms within and across categories. [Lemma 18](#) handles categories that are selected as light by the pilot split while having population mass above the light-cell scale. Combining these ingredients with the approximation certificate yields [Lemma 19](#), which gives the aggregate light-branch rate and the real-arithmetic computability claim used in [Theorem 2](#).

The fixed-interior upper bound in [Theorem 1](#) is obtained by summing the heavy and light errors from [Lemmas 15](#) and [19](#) around the pilot partition and then applying the truncation in [Definition 35](#). The lower bound is supplied by [Lemma 2](#), and the endpoint and selected-estimator conclusions follow from [Lemmas 3](#), [5](#) and [6](#).

B Verification note

The formal component of the paper covers the finite-alphabet observational model used throughout the analysis. It includes the iid product-law representation for the observed sample, the equivalent finite single-observation law on $\{1, \dots, d\} \times \{0, 1\}^2$, and the bridge between the one-observation law and the n -fold product experiment. Within that representation, [Assumptions 1](#) and [2](#) supply the sampling and fixed-overlap conditions, while [Definition 4](#) records the experiment class over which worst-case risks are evaluated.

The machine-checked development also covers the local four-cell description of the observational law. The overlap cone in [Definition 5](#), the homogeneous cell contribution in [Definition 6](#), and the ATE functional in [Definition 7](#) provide the algebraic layer used by both the lower and upper bounds. The potential-outcome overlay used for the illustrative two-category construction is represented through the consistency and conditional-exchangeability conditions in [Assumptions 3](#) and [4](#); the verified statements connect those conditions to the observed-data witness in [Definition 13](#) and [Proposition 1](#).

For the estimator and risk objects, the formalization includes the deterministic sample split, the split cell counts, the heavy–light pilot classification, the ratio branch, the Chebyshev polynomial branch, the factorial-moment light-cell estimator, and the final truncation displayed in [Definition 35](#). It also includes the minimax mean-squared-error risk in [Definition 8](#), so the upper bounds for the displayed estimator and the lower bounds for arbitrary measurable estimators are stated against the same decision problem.

The proof layer covers the internally used auxiliary bounds that feed the main statements: the light-cell polynomial control, the pilot sandwich, the heavy-cell aggregation bound, the lower-bound transfer, the centered near-randomization estimator, the one-category endpoint lower bound, the randomized endpoint bracket, and the combined upper envelope. These ingredients support the sharp fixed-interior minimax characterization in [Theorem 1](#) and the universal hybrid envelope in [Theorem 2](#). The theorem-local verification footnotes identify any published mathematical conclusions invoked as external dependencies at their point of use.

All theorem statements and proofs described above are machine-checked in Lean 4.

C Proofs of the main results

Proof of [Proposition 1](#). Fix $0 < \epsilon \leq 1/2$. Write the two categories as $x \in \{1, 2\}$, and define

$$\zeta_1 = 1, \quad \zeta_2 = 0,$$

so that the witness has $Y(0) = 0$ and $Y(1) = \zeta_x$ in category x . For an arm value $\mathfrak{a} \in \{0, 1\}$, set

$$\eta_{1,1} = \epsilon, \quad \eta_{1,0} = 1 - \epsilon, \quad \eta_{2,1} = 1 - \epsilon, \quad \eta_{2,0} = \epsilon.$$

Then the full-data law Q in [Definition 13](#) is

$$Q\{X = x, A = \mathfrak{a}, Y = y, Y(0) = y_0, Y(1) = y_1\} = \begin{cases} \frac{1}{2} \eta_{x,\mathfrak{a}}, & y_0 = 0, y_1 = \zeta_x, y = \mathfrak{a}\zeta_x, \\ 0, & \text{otherwise.} \end{cases}$$

The displayed masses are nonnegative, since $0 < \epsilon \leq 1/2$, and their total mass is

$$\frac{1}{2}\{\epsilon + (1 - \epsilon)\} + \frac{1}{2}\{(1 - \epsilon) + \epsilon\} = 1.$$

Marginalizing Q over $Y(0), Y(1)$, the observed law P has nonzero cells

$$P(X = 1, A = 1, Y = 1) = \frac{\epsilon}{2}, \quad P(X = 1, A = 0, Y = 0) = \frac{1 - \epsilon}{2},$$

$$P(X = 2, A = 1, Y = 0) = \frac{1 - \epsilon}{2}, \quad P(X = 2, A = 0, Y = 0) = \frac{\epsilon}{2}.$$

The category masses $p_x = P(X = x)$ and propensities $\pi_x = P(A = 1 \mid X = x)$ are therefore

$$p_1 = p_2 = \frac{1}{2}, \quad \pi_1 = \frac{\epsilon/2}{1/2} = \epsilon, \quad \pi_2 = \frac{(1 - \epsilon)/2}{1/2} = 1 - \epsilon.$$

Because $0 < \epsilon \leq 1/2$, each positive-mass category satisfies

$$\epsilon \leq \pi_x \leq 1 - \epsilon, \quad x \in \{1, 2\}.$$

Thus P satisfies ϵ -overlap in the sense of [Assumption 2](#).

With overlap established, evaluate the ATE functional through the weighted-regression identity

$$\tau(P) = \sum_x p_x \{\mu_{1x} - \mu_{0x}\}.$$

For completeness, this identity follows directly from [Definition 7](#): for the four-cell vector q_x , the totalized convention sets all quotients with zero denominator to 0; if $q_x = 0$, both $\phi(q_x)$ and $p_x\{\mu_{1x} - \mu_{0x}\}$ are zero, while otherwise expanding

$$\phi(q_x) = p_x \left\{ \frac{q_{x,11}}{q_{x,10} + q_{x,11}} - \frac{q_{x,01}}{q_{x,00} + q_{x,01}} \right\}$$

gives $p_x\{\mu_{1x} - \mu_{0x}\}$. In the present law,

$$\mu_{11} = 1, \quad \mu_{01} = 0, \quad \mu_{12} = 0, \quad \mu_{02} = 0,$$

and hence

$$\tau(P) = \frac{1}{2}(1 - 0) + \frac{1}{2}(0 - 0) = \frac{1}{2}.$$

The consistency assertion is immediate from the support display for Q . Every atom with positive Q -mass has $Y(0) = 0, Y(1) = \zeta_x$, and

$$Y = \begin{cases} Y(1), & A = 1, \\ Y(0), & A = 0. \end{cases}$$

Equivalently, every atom for which Y differs from the potential outcome selected by A receives mass 0, which is the condition in [Assumption 3](#).

For conditional exchangeability, define, for $x \in \{1, 2\}$, $\mathbf{a} \in \{0, 1\}$, and $\omega_0, \omega_1 \in \{0, 1\}$,

$$\Lambda_x(\mathbf{a}, \omega_0, \omega_1) = \sum_{y \in \{0, 1\}} Q\{X = x, A = \mathbf{a}, Y = y, Y(0) = \omega_0, Y(1) = \omega_1\}.$$

The finite conditional-independence identity in [Assumption 4](#) is

$$\Lambda_x(\mathbf{a}, \omega_0, \omega_1) \sum_{\mathbf{a}', \omega'_0, \omega'_1} \Lambda_x(\mathbf{a}', \omega'_0, \omega'_1) = \left\{ \sum_{\mathbf{a}'} \Lambda_x(\mathbf{a}', \omega_0, \omega_1) \right\} \left\{ \sum_{\omega'_0, \omega'_1} \Lambda_x(\mathbf{a}, \omega'_0, \omega'_1) \right\}.$$

From the definition of Q ,

$$\Lambda_x(\mathbf{a}, \omega_0, \omega_1) = \begin{cases} \frac{1}{2} \eta_{x,\mathbf{a}}, & (\omega_0, \omega_1) = (0, \zeta_x), \\ 0, & \text{otherwise.} \end{cases}$$

Also $\sum_{\mathbf{a}} \eta_{x,\mathbf{a}} = 1$, so the total mass in category x is $1/2$. If $(\omega_0, \omega_1) = (0, \zeta_x)$, the two sides of the displayed identity are both

$$\left(\frac{1}{2} \eta_{x,\mathbf{a}} \right) \left(\frac{1}{2} \right).$$

For every other pair (ω_0, ω_1) , the factor involving that potential-outcome pair is zero on both sides. Thus Q satisfies [Assumption 4](#).

Finally, the naive treated-minus-control contrast under P has

$$\sum_x P(X = x, A = 1, Y = 1) = \frac{\epsilon}{2}, \quad \sum_x P(X = x, A = 0, Y = 1) = 0,$$

and

$$\sum_x P(X = x, A = 1) = \frac{1}{2}, \quad \sum_x P(X = x, A = 0) = \frac{1}{2}.$$

Therefore

$$E_P[Y \mid A = 1] - E_P[Y \mid A = 0] = \frac{\epsilon/2}{1/2} - \frac{0}{1/2} = \epsilon.$$

Together with the preceding paragraphs, this proves all asserted conclusions. $\square$

Proof of [Theorem 1](#). Fix $0 < \epsilon < 1/2$, and write

$$r_{n,d} = \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

1. First derive the fixed-interior upper envelope for the hybrid estimator. From [Lemma 19](#) choose constants $\mathfrak{C}_{\mathcal{L}}, \mathfrak{c}_{\mathcal{L}} > 0$ such that, whenever $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ and $d \leq \mathfrak{c}_{\mathcal{L}} n \log n$,

$$E_{P^{\otimes n}} \left[\left\{ \hat{T}_{\mathcal{L}} - \sum_{k \in \hat{\mathcal{L}}_n} \phi(q_k) \right\}^2 \right] \leq \mathfrak{C}_{\mathcal{L}} r_{n,d}.$$

From [Lemma 15](#) choose $\mathfrak{C}_{\mathcal{H}}, \mathfrak{c}_{\mathcal{H}} > 0$ and $\mathfrak{N}_{\mathcal{H}} < \infty$ such that, whenever $n \geq \mathfrak{N}_{\mathcal{H}}$, $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, and $d \leq \mathfrak{c}_{\mathcal{H}} n \log n$,

$$E_{P^{\otimes n}} \left[\left\{ \hat{T}_{\mathcal{H}} - \sum_{k \in \hat{\mathcal{H}}_n} \phi(q_k) \right\}^2 \right] \leq \mathfrak{C}_{\mathcal{H}} r_{n,d}.$$

Set

$$\mathfrak{C}_{\text{up}} = 2(\mathfrak{C}_{\mathcal{L}} + \mathfrak{C}_{\mathcal{H}}), \quad \rho_{\text{up}} = \min\{\mathfrak{c}_{\mathcal{L}}, \mathfrak{c}_{\mathcal{H}}\}, \quad \mathfrak{N}_{\text{up}} = \mathfrak{N}_{\mathcal{H}}.$$

For a sample array $o_{1:n}$, define the pilot targets

$$\mathcal{T}_{\mathcal{H}}^{\circ}(P, o_{1:n}) = \sum_{k \in \hat{\mathcal{H}}_n(o_{1:n})} \phi(q_k), \quad \mathcal{T}_{\mathcal{L}}^{\circ}(P, o_{1:n}) = \sum_{k \in \hat{\mathcal{L}}_n(o_{1:n})} \phi(q_k).$$

The sets $\widehat{\mathcal{H}}_n(o_{1:n})$ and $\widehat{\mathcal{L}}_n(o_{1:n})$ partition $\{1, \dots, d\}$, hence

$$\mathcal{T}_{\mathcal{H}}^\circ(P, o_{1:n}) + \mathcal{T}_{\mathcal{L}}^\circ(P, o_{1:n}) = \sum_{k=1}^d \phi(q_k) = \tau(P),$$

where the last equality is [Definition 7](#). The interval bound needed for truncation follows from the cell formula. For each category, write

$$p_k = \sum_{a,y \in \{0,1\}} q_{aky}, \quad s_{ak} = q_{ak0} + q_{ak1}, \quad \mu_{ak} = q_{ak1}/s_{ak},$$

with the empty-arm value interpreted by the totalized convention $\mu_{ak} = 0$ when $s_{ak} = 0$. By [Definition 6](#), the zero vector has $\phi(0) = 0$, so a zero-mass category contributes $0 = p_k\{\mu_{1k} - \mu_{0k}\}$. On a positive-mass category satisfying overlap, the two arm masses are positive, and the displayed formula in [Definition 6](#) gives

$$\phi(q_k) = p_k \left\{ \frac{q_{1k1}}{s_{1k}} - \frac{q_{0k1}}{s_{0k}} \right\} = p_k \{\mu_{1k} - \mu_{0k}\}.$$

Summing this identity and using [Definition 7](#) gives

$$\tau(P) = \sum_{k=1}^d p_k \{\mu_{1k} - \mu_{0k}\}.$$

Here $p_k \geq 0$, $\sum_{k=1}^d p_k = 1$, and the totalized binary-outcome means satisfy $0 \leq \mu_{ak} \leq 1$. Therefore $|\mu_{1k} - \mu_{0k}| \leq 1$ for every k , and

$$|\tau(P)| \leq \sum_{k=1}^d p_k |\mu_{1k} - \mu_{0k}| \leq \sum_{k=1}^d p_k = 1.$$

Thus $\tau(P) \in [-1, 1]$. Let

$$\Pi_{[-1,1]}(x) = \max\{-1, \min\{1, x\}\}.$$

The elementary projection inequality

$$\{\Pi_{[-1,1]}(x) - \tau(P)\}^2 \leq \{x - \tau(P)\}^2$$

holds for every real x . Applying it to $x = \widehat{T}_{\mathcal{H}}(o_{1:n}) + \widehat{T}_{\mathcal{L}}(o_{1:n})$, and then using $(u+v)^2 \leq 2u^2 + 2v^2$, gives the pointwise bound

$$\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P)\right)^2 \leq 2 \left\{ \widehat{T}_{\mathcal{H}} - \mathcal{T}_{\mathcal{H}}^\circ(P, \cdot) \right\}^2 + 2 \left\{ \widehat{T}_{\mathcal{L}} - \mathcal{T}_{\mathcal{L}}^\circ(P, \cdot) \right\}^2.$$

Integrating with respect to $P^{\otimes n}$ and using the two component bounds yields, for $n \geq \mathfrak{N}_{\text{up}}$ and $d \leq \rho_{\text{up}} n \log n$,

$$E_{P^{\otimes n}} \left[\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] \leq \mathfrak{C}_{\text{up}} r_{n,d}.$$

Taking the supremum over $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ gives the same bound for the worst-case risk of $\widehat{\tau}_n^{\text{hyb}}$. Since this estimator is a measurable estimator on the finite sample space and hence is one admissible estimator in the infimum defining $R_{n,d,\epsilon}$ in [Definition 8](#),

$$R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_{P^{\otimes n}} \left[\left(\widehat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] \leq \mathfrak{C}_{\text{up}} r_{n,d}.$$

2. The lower-transfer step uses the fixed-sample binary-outcome lower bound of [Zeng et al., 2024, Theorem 2; Appendix C.5; Lemma 4; Appendix D.2] through Lemma 2. Thus there are $\mathfrak{a}_{\text{low}}, \rho_{\text{low}} > 0$ and $\mathfrak{N}_{\text{low}} < \infty$ such that, for $0 < d$, $n \geq \mathfrak{N}_{\text{low}}$, and $d \leq \rho_{\text{low}} n \log n$,

$$\mathfrak{a}_{\text{low}} r_{n,d} \leq R_{n,d,\epsilon}.$$

Define

$$a_\epsilon = \mathfrak{a}_{\text{low}}, \quad \rho_\epsilon = \min\{\rho_{\text{up}}, \rho_{\text{low}}\}, \quad C_\epsilon = \mathfrak{C}_{\text{up}}, \quad N_\epsilon = \max\{\mathfrak{N}_{\text{up}}, \mathfrak{N}_{\text{low}}\}.$$

These constants are positive, with $N_\epsilon < \infty$. If $0 < d$, $n \geq N_\epsilon$, and $d \leq \rho_\epsilon n \log n$, then both the upper and lower ranges apply, so

$$a_\epsilon r_{n,d} \leq R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_{P^{\otimes n}} \left[\left(\hat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] \leq C_\epsilon r_{n,d}.$$

This is the sharp sandwich assertion.

3. Fix $K > 0$. Set

$$C_K = C_\epsilon(1 + K^2).$$

Then $C_K > 0$. Suppose $0 < d$, $n \geq N_\epsilon$, $d \leq K\sqrt{n} \log n$, and $d \leq \rho_\epsilon n \log n$. The sample size is positive by the displayed range: if $n = 0$, then $K\sqrt{n} \log n = 0$, contradicting $0 < d \leq K\sqrt{n} \log n$. If $\log n \leq 0$, then $K\sqrt{n} \geq 0$ and therefore $K\sqrt{n} \log n \leq 0$, again contradicting $0 < d \leq K\sqrt{n} \log n$. Thus $\log n > 0$. With both sides of $d \leq K\sqrt{n} \log n$ nonnegative, squaring gives

$$d^2 \leq K^2 n (\log n)^2.$$

Dividing by the positive denominator $n^2(\log n)^2$ gives

$$\frac{d^2}{n^2(\log n)^2} \leq \frac{K^2}{n}.$$

Hence

$$r_{n,d} = \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \leq \frac{1 + K^2}{n}.$$

The lower side of the sandwich gives

$$\frac{a_\epsilon}{n} \leq a_\epsilon r_{n,d} \leq R_{n,d,\epsilon},$$

and the upper side gives

$$R_{n,d,\epsilon} \leq C_\epsilon r_{n,d} \leq \frac{C_\epsilon(1 + K^2)}{n} = \frac{C_K}{n}.$$

This proves the parametric interior assertion.

4. Let n_j, d_j be sequences with $n_j \rightarrow \infty$, with $d_j > 0$ eventually, and with

$$d_j \leq \rho_\epsilon n_j \log n_j$$

eventually. Finite initial indices leave convergence unchanged, so work on the tail where $n_j \geq N_\epsilon$, $n_j \geq 2$, $d_j > 0$, and the displayed range condition holds. Define

$$\eta_j = \frac{d_j}{n_j \log n_j}, \quad r_j = \frac{1}{n_j} + \eta_j^2.$$

On this tail, $r_j = r_{n_j, d_j}$ and $\eta_j \geq 0$. The overlap class is nonempty for such d_j : since $d_j > 0$, the endpoint parametric law on $\{1, \dots, d_j\}$ with uniform category masses, treatment probability $1/2$, and both binary response means $1/2$ has propensity $1/2$ in every positive-mass category, and therefore satisfies the ϵ -overlap condition because $0 < \epsilon < 1/2$. Applying the sandwich on the tail gives

$$a_\epsilon r_j \leq R_{n_j, d_j, \epsilon} \leq C_\epsilon r_j.$$

If $R_{n_j, d_j, \epsilon} \rightarrow 0$, then

$$0 \leq r_j \leq a_\epsilon^{-1} R_{n_j, d_j, \epsilon}$$

eventually, so $r_j \rightarrow 0$. Since

$$0 \leq \eta_j^2 \leq r_j,$$

we have $\eta_j^2 \rightarrow 0$, and the eventual nonnegativity of η_j gives $\eta_j \rightarrow 0$. Conversely, if $\eta_j \rightarrow 0$, then $1/n_j \rightarrow 0$ and $\eta_j^2 \rightarrow 0$, so $r_j \rightarrow 0$. Since minimax risks are nonnegative, the eventual bound

$$0 \leq R_{n_j, d_j, \epsilon} \leq C_\epsilon r_j$$

then yields $R_{n_j, d_j, \epsilon} \rightarrow 0$. Therefore

$$R_{n_j, d_j, \epsilon} \rightarrow 0 \iff \frac{d_j}{n_j \log n_j} \rightarrow 0.$$

□

Proof of Theorem 2. 1. **Computability.** Use the real-arithmetic program model in which a program is a finite register list with input reads, real constants, additions, subtractions, multiplications, divisions, and branch comparisons. Input reads and constants have cost zero, while each arithmetic operation and comparison has cost one. Division is the total real-field operation, so division by a zero denominator returns zero. A branch comparison has a test expression and two branch expressions and returns the first branch when the test value is nonpositive and the second branch otherwise.

For $n, d \in \mathbb{N}$, the input alphabet for the program is

$$\{0, 1\} \times \{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}.$$

At input coordinate (j, k, a, y) the program reads the real number

$$N_{aky}^{(j)} = \#\{i \in I_j : (X_i, A_i, Y_i) = (k, a, y)\}.$$

Equivalently, the input vector is the collection of the $8d$ split cell counts $N_{aky}^{(j)}$. Sums and products below are compiled as expression trees folded onto the initial constants 0 and 1: a list sum has cost equal to the sum of the costs of its entries plus one addition for

each entry, and a list product has cost equal to the sum of the costs of its entries plus one multiplication for each entry. Thus a four-term category count has cost 4, not 3.

If $d = 0$, the program is the constant-zero program. Its operation count is zero, and its value agrees with $\hat{\tau}_n^{\text{hyb}}$, since the heavy and light sums range over the empty alphabet and the truncation of zero is zero.

Now suppose $d > 0$, and write $M = M(n)$. A split-count input has cost zero. For a natural-valued expression x , the truncated subtraction $(x - i)_+$ is compiled as a branch on $x - i$, returning 0 on the nonpositive branch and $x - i$ on the positive branch. This expression has cost $2 \text{op}(x) + 3$. The falling factorial $x(x-1)_+ \cdots (x-r+1)_+$ is compiled by the corresponding folded product, so its cost is $r\{2 \text{op}(x) + 4\}$. Therefore, for a split-count input x , the falling-factorial expression of order r has cost $4r$.

For a multi-index r , the factorial monomial expression is

$$\frac{\prod_{a,y} (N_{aky}^{(1)})_{r_{ay}}}{(m_1)_{|r|}},$$

with the same total division convention as the program model. The folded product over the four cells contributes $4|r| + 4$ operations, and the final division by the constant denominator contributes one more, giving

$$\text{op}\{\text{factorial monomial } r\} = 4|r| + 5.$$

For the multi-indices used in the arm expansion of $P_{M,B}$, $|r| = j + 2$. After multiplication by its coefficient, one fixed-cell summand at level (j, t) has cost

$$4j + 14.$$

Since $M \geq 2$, the nested sums in one treatment arm satisfy

$$\text{op}\{\text{sum over } (a, y) \text{ at fixed } j, t\} \leq 50M,$$

$$\text{op}\{\text{sum over } t \text{ at fixed } j\} \leq 51M^2, \quad \text{op}\{\text{one arm sum}\} \leq 52M^3.$$

Subtracting the two arm sums gives the light-cell expression, and hence

$$\text{op}\{\text{light-cell expression for category } k\} \leq 105M^4.$$

The heavy expression for category k is

$$\frac{N_k^{(1)}}{m_1} \left\{ \frac{N_{1k1}^{(1)}}{N_{1k0}^{(1)} + N_{1k1}^{(1)}} - \frac{N_{0k1}^{(1)}}{N_{0k0}^{(1)} + N_{0k1}^{(1)}} \right\},$$

where $N_k^{(1)} = \sum_{a,y} N_{aky}^{(1)}$, again using total division at zero denominators. This expression has cost 11 and agrees exactly with $\hat{h}_k$. The pilot count expression $\sum_{a,y} N_{aky}^{(0)}$ has cost 4, so the comparison expression $\sum_{a,y} N_{aky}^{(0)} - \lambda_0 L$ has cost 5. For $n \geq N_0$, the selected category expression branches to the light expression when the pilot count is at most $\lambda_0 L$ and to the heavy expression when it is larger. Its cost is at most

$$5 + 105M^4 + 11 + 1 \leq 110M^4,$$

where the final 1 is the branch comparison itself. For $n < N_0$, the selected category expression is the heavy expression, whose cost 11 is also at most $110M^4$. Thus each selected category expression has operation count at most $110M(n)^4$.

Summing the d selected category expressions uses one addition node per category in the folded sum, so the untruncated expression has cost at most

$$111 d M(n)^4.$$

The truncation

$$x \mapsto \max\{-1, \min\{1, x\}\}$$

is compiled by two branch comparisons, with the relevant expression duplicated in the branch subtrees. Consequently

$$\text{op}\{\text{clamp}(x)\} = 4 \text{op}(x) + 6.$$

For $d > 0$, this gives

$$\text{op}(\mathcal{A}_{n,d}) \leq 4 \cdot 111 d M(n)^4 + 6 \leq 450 d M(n)^4,$$

because $1 \leq dM(n)^4$. Together with the zero-alphabet branch, the compiled program $\mathcal{A}_{n,d}$ satisfies

$$\text{op}(\mathcal{A}_{n,d}) \leq 450 d M(n)^4$$

and, for every sample,

$$\mathcal{A}_{n,d}((N_{aky}^{(j)})_{j,k,a,y}) = \hat{\tau}_n^{\text{hyb}}(O_{1:n}).$$

Taking $K = 450$ proves the computability clause.

2. **Fixed-interior hybrid bound.** Fix $0 < \epsilon < 1/2$. Let $C_L, c_L > 0$ be the constants supplied by [Lemma 19](#), and let $C_H, \rho_H > 0$ and $N_H \in \mathbb{N}$ be the constants supplied by [Lemma 15](#). Set

$$C_\epsilon = 2(C_L + C_H), \quad \rho_\epsilon = \min\{c_L, \rho_H\}, \quad N_\epsilon = N_H.$$

These constants are positive in the required real coordinates.

For a sample $z = O_{1:n}$, define

$$T_{\mathcal{H}}^\circ(P, z) = \sum_{k \in \hat{\mathcal{H}}_n(z)} \phi(q_k), \quad T_{\mathcal{L}}^\circ(P, z) = \sum_{k \in \hat{\mathcal{L}}_n(z)} \phi(q_k).$$

Since $\hat{\mathcal{L}}_n(z) = \hat{\mathcal{H}}_n(z)^c$,

$$T_{\mathcal{H}}^\circ(P, z) + T_{\mathcal{L}}^\circ(P, z) = \tau(P).$$

Also $\tau(P) \in [-1, 1]$ for every law in the overlap class. Hence the truncation map is contractive around $\tau(P)$, and

$$\begin{aligned} \left(\hat{\tau}_n^{\text{hyb}}(z) - \tau(P) \right)^2 &\leq \left[\{ \hat{T}_{\mathcal{H}}(z) - T_{\mathcal{H}}^\circ(P, z) \} + \{ \hat{T}_{\mathcal{L}}(z) - T_{\mathcal{L}}^\circ(P, z) \} \right]^2 \\ &\leq 2 \{ \hat{T}_{\mathcal{H}}(z) - T_{\mathcal{H}}^\circ(P, z) \}^2 + 2 \{ \hat{T}_{\mathcal{L}}(z) - T_{\mathcal{L}}^\circ(P, z) \}^2. \end{aligned}$$

Integrating under any $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ gives

$$\begin{aligned} E_P \left[\left(\hat{\tau}_n^{\text{hyb}} - \tau(P) \right)^2 \right] &\leq 2C_H \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} + 2C_L \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \\ &= C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}, \end{aligned}$$

whenever $0 < n$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$. The light and heavy bounds apply because this dimension condition implies both

$$d \leq c_L n \log n \quad \text{and} \quad d \leq \rho_H n \log n.$$

Taking the supremum over $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ proves the fixed-interior calibration bound for $\hat{\tau}_n^{\text{hyb}}$. In the auxiliary zero-alphabet case used by the selector lemma, the experiment class is empty and the displayed upper bound follows from the empty-supremum convention and nonnegativity of the rate.

3. **Selector envelope.** Apply [Lemma 6](#) with the constants just constructed and with the hybrid bound from the previous step. For every $n, d \in \mathbb{N}$ with $0 < n$, $0 < d$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$, the selected estimator that chooses $\hat{\tau}_n^{\text{hyb}}$ when

$$C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and chooses $\hat{\tau}_{\text{ctr}}$ otherwise satisfies

$$R_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[\left(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P) \right)^2 \right]$$

and

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[\left(\hat{\tau}_{C_\epsilon, \epsilon}^{\text{sel}} - \tau(P) \right)^2 \right] \leq \max\{C_\epsilon, 4\} \left[\frac{1}{n} + \min \left\{ \frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2 \right\} \right].$$

4. **Centered randomization bound.** Fix $0 < \epsilon \leq 1/2$, $0 < n$, and $0 < d$. For every $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, [Lemma 3](#) gives

$$E_P \left[\left(\hat{\tau}_{\text{ctr}} - \tau(P) \right)^2 \right] \leq \frac{1}{n} + 4(1/2 - \epsilon)^2.$$

The right-hand side is independent of P , so taking the supremum over the experiment class gives

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P \left[\left(\hat{\tau}_{\text{ctr}} - \tau(P) \right)^2 \right] \leq \frac{1}{n} + 4(1/2 - \epsilon)^2.$$

The same conclusion covers the empty-class branch through the same empty-supremum convention.

5. **Endpoint bracket.** For $0 < n$ and $0 < d$, [Lemma 5](#) gives directly

$$\frac{1}{100n} \leq R_{n,d,1/2} \leq \frac{1}{n}.$$

This is the endpoint bracket. □

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